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Failure Approaches

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March 30, 2026

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1 Introduction

1.1 Motivation

Robust control design for uncertain dynamical systems remains a central problem in modern control theory, especially in applications where disturbances are stochastic, partially unknown, or only imperfectly characterised. In many practical settings, the designer is required not only to stabilize the closed-loop system but also to guarantee satisfactory performance under a family of plausible disturbance distributions rather than under a single nominal model.

This challenge naturally motivates the use of *distributionally robust* formulations, in which uncertainty is described through ambiguity sets of probability distributions. Among the available tools, Wasserstein-based ambiguity sets have recently attracted considerable attention because they provide a flexible and mathematically tractable way to account for distributional mismatch while remaining closely connected to data-driven settings.

At the same time, many practical problems do not provide full access to an exact state-space model. Instead, only input-state-output trajectories may be available, which raises the question of whether model-based robust synthesis methods can be extended to settings where the plant description must be inferred from data. This creates a natural tension between two desirable objectives: preserving the convexity and guarantees of LMI-based controller synthesis, and enabling the method to operate in a genuinely data-driven setting.

1.2 Research Problem

This thesis studies the synthesis of dynamic output-feedback controllers for discrete-time linear time-invariant systems under stochastic disturbances, with particular emphasis on distributional uncertainty. The starting point is a model-based distributionally robust LMI framework in which the plant is assumed to be known and the disturbance distribution belongs to a Wasserstein ambiguity set centered around a nominal covariance description.

A central question then arises: can this framework be extended to data-driven settings by replacing the unknown system matrices with estimates obtained from measured trajectories, while preserving the convex structure of the synthesis problem?

As shown in this work, a direct two-stage approach based on regularised regression followed by LMI synthesis fails to preserve convexity. This negative result is not merely technical; it plays a key conceptual role in motivating the need for a different formulation.

1.3 Objectives and Contributions

The main objectives of this thesis are threefold.

I. First, a model-based distributionally robust LMI synthesis framework is formulated for linear time-invariant systems with dynamic output feedback. This provides the theoretical baseline and reference point for all subsequent developments.

II. Second, the thesis investigates a natural data-driven extension in which the plant matrices are identified from trajectories and then substituted into the model-based synthesis pipeline.

The analysis shows that this naive extension destroys the affine dependence required for convex semidefinite programming, thereby yielding a fundamentally nonconvex problem.

III. Third, the proposed methods are evaluated experimentally through a controlled comparison between model-based and data-driven configurations under a common simulation and performance-assessment pipeline.

The contribution of the thesis is therefore not only the derivation of a robust LMI synthesis framework, but also the clarification of a structural obstacle that arises when regularised identification is combined with convex robust controller design.

1.4 Thesis Structure

The remainder of the thesis is organised as follows.

Chapter 2 introduces the model-based distributionally robust LMI synthesis framework for linear time-invariant systems under stochastic disturbances, establishing the reference formulation and notation used throughout the work.

Chapter 3 presents a simulation-based baseline method, which serves both as a benchmark for performance comparison and as a reference for understanding the limitations of direct optimisation approaches.

Chapters 4–7 develop a sequence of data-driven extensions of the distributionally robust problem. Starting from a direct data-driven parametrisation, the discussion progressively introduces relaxation techniques and structural modifications, including Young-type bounds, Schur-complement enhancements, and Willems-inspired formulations. Each approach is analysed in terms of tractability, robustness, and inherent limitations.

Chapter 8 describes the experimental framework, including dataset generation, identification procedures, evaluation metrics, and comparison protocols used to ensure a consistent assessment across all methods.

Chapter 9 presents the numerical results and provides a comparative analysis of the proposed approaches, highlighting the trade-offs between performance, robustness, and computational complexity.

The appendices are divided into two parts. Part I collects general methodological components, including disturbance modelling, dataset construction, and evaluation tools. Part II contains detailed derivations and algorithm-specific analyses supporting the developments presented in the main chapters.

Method	**Main Formulation**	**Derivations**	**Results**	**Algorithm**
Baseline	Chapter 3	Appendix G	Section 9.1	Algorithm F.1
DeePC	Chapter 4	Appendix H	—	Algorithm F.2
Young	Chapter 5	Appendix I	Section 9.2	Algorithm F.3
Young-Schur	Chapter 6	Appendix J	Section 9.3	Algorithm F.4
WFL	Chapter 7	Appendix K	—	Algorithm F.5

Table 1.1: Guide to the organisation of the proposed methods across the thesis.

2 Distributionally Robust LMI Synthesis for LTI Systems

2.1 Problem Setting

This chapter addresses the synthesis of output-feedback controllers for linear time-invariant (LTI) systems under stochastic disturbances, with a particular focus on distributionally robust formulations. The objective is to design controllers that ensure closed-loop stability while optimising performance under uncertainty in the disturbance distribution.

We consider the standard discrete-time LTI system described by:

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix},$$

where $x_t \in \mathbb{R}^{n_x}$ is the state, $u_t \in \mathbb{R}^{n_u}$ is the control input, and $w_t \in \mathbb{R}^{n_w}$ represents an exogenous disturbance. The outputs $y_t \in \mathbb{R}^{n_y}$ and $z_t \in \mathbb{R}^{n_z}$ correspond to the measured and performance variables, respectively.

We adopt a dynamic output-feedback controller of the form:

$$\begin{bmatrix} x_{c,t+1} \\ u_t \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} x_{c,t} \\ y_t \end{bmatrix}.$$

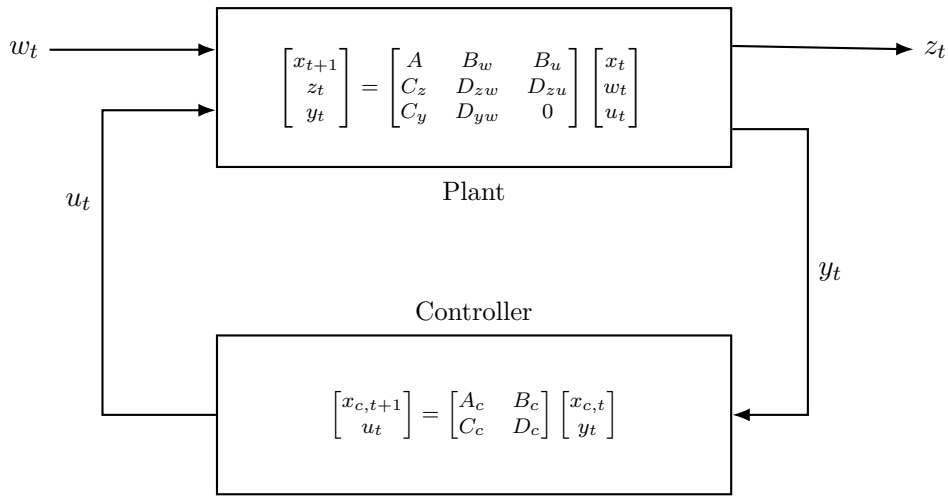


Figure 2.1: Closed-loop interconnection between plant and controller via output feedback.

Closed-Loop Representation. To facilitate analysis, we introduce the augmented state vector:

$$\chi_t = \begin{bmatrix} x_t^\top & x_{c,t}^\top \end{bmatrix}^\top.$$

The resulting closed-loop system can be compactly expressed as:

$$\begin{aligned} \chi_{t+1} &= \mathcal{A}\,\chi_t + \mathcal{B}\,w_t, \\ z_t &= \mathcal{C}\,\chi_t + \mathcal{D}\,w_t, \end{aligned}$$

where the lifted matrices are defined as:

$$\mathcal{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix},$$
$$\mathcal{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad \mathcal{D} = D_{zw} + D_{zu} D_c D_{yw}.$$

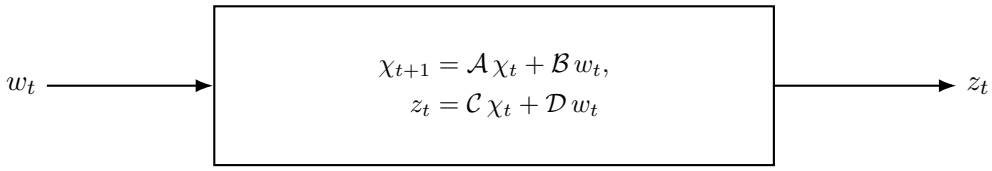


Figure 2.2: Lifted closed-loop system representation with augmented state χ_t.

This lifted formulation will be instrumental in deriving both simulation-based and LMI-based optimization procedures.

2.2 Benchmark Model Description

We adopt a benchmark configuration inspired by [1], representing a flexible turbine–shaft–pitch system of the form:

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix},$$

where $x_t \in \mathbb{R}^{n_x}$ denotes the state, $u_t \in \mathbb{R}^{n_u}$ the control input, and $w_t \in \mathbb{R}^{n_w}$ an exogenous disturbance. The vectors $y_t \in \mathbb{R}^{n_y}$ and $z_t \in \mathbb{R}^{n_z}$ represent the measured and performance outputs, respectively.

The continuous-time version of this model describes a flexible turbine–shaft–pitch system:

$$\dot{x}(t) = A_{CT}x(t) + B_{CT}u(t) + E_{CT}w(t),$$

with

$$A_{CT} = \begin{bmatrix} \frac{k_\omega}{J} & \frac{k_h}{J} & 0 & \frac{k_\phi}{J} & 0 & \frac{k_\beta}{J} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{f_\omega}{m} & -\omega_f^2 & -2\zeta_f\omega_f & 0 & 0 & \frac{f_\beta}{m} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{m_\omega}{I_t} & 0 & 0 & -\omega_t^2 & -2\zeta_t\omega_t & \frac{m_\beta}{I_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega_p^2 & -2\zeta_p\omega_p \end{bmatrix},$$

$$B_{CT} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega_p^2 \end{bmatrix}, \quad E_{CT} = \begin{bmatrix} \frac{k_v}{J} & 0 \\ 0 & 0 \\ \frac{f_v}{m} & \frac{f_v}{m} \\ 0 & 0 \\ \frac{m_v}{I_t} & \frac{m_v}{I_t} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

which is discretised with sampling time T_s via

$$A = e^{A_{CT}T_s}, \qquad B_u = \int_0^{T_s} e^{A_{CT}\tau} B_{CT}\, d\tau, \qquad B_w = \int_0^{T_s} e^{A_{CT}\tau} E_{CT}\, d\tau.$$

The output matrices $(C_z, D_{zw}, D_{zu}, C_y, D_{yw})$ are selected to reflect a standard $\mathcal{H}_2$-type performance objective:

$$z_t = \begin{bmatrix} Q_x^{1/2} & 0 \\ 0 & R_u^{1/2} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \qquad Q_x \succeq 0, R_u \succ 0 \qquad y_t = C_y x_t + D_{yw} w_t,$$

with $Q_x \succeq 0, R_u \succ 0$ and $D_{zw} = 0$.

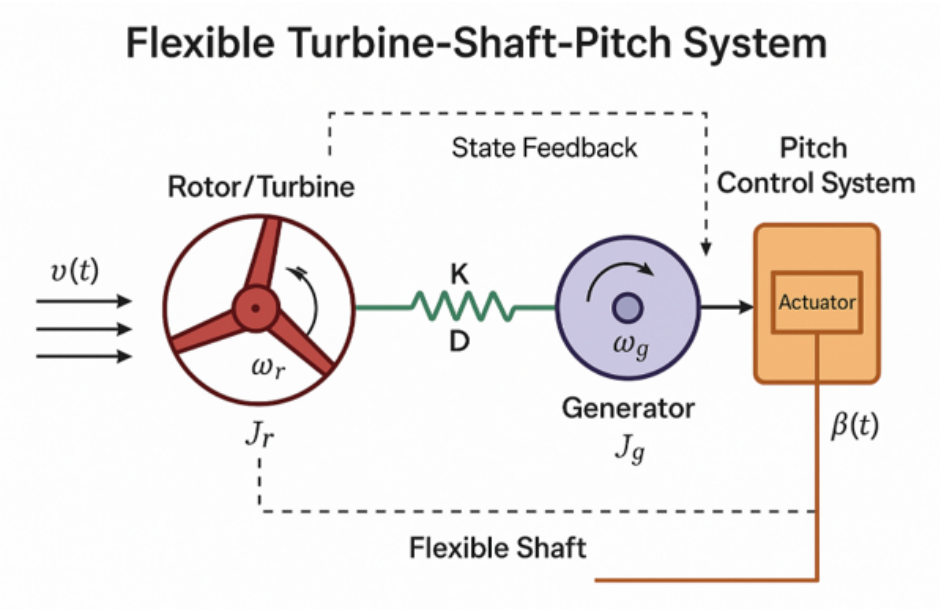


Figure 2.3: Representation of the benchmark model

2.3 Disturbance Uncertainty Model

We model the exogenous disturbance w_t as a stochastic process with uncertain second-order statistics. Specifically, we assume that its marginal distribution belongs to a 2-Wasserstein ambiguity set centered at a nominal Gaussian model.

We fix

$$P_{\text{nom}} = \mathcal{N}(0, \Sigma_{\text{nom}}),$$

and define the ambiguity set

$$\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) = \{\mathcal{N}(0, \Sigma) \mid W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \leq \gamma\}.$$

This set captures uncertainty in the disturbance covariance, allowing deviations from Σ_{nom} within a Wasserstein radius γ.

Two disturbance models are considered.

Independent disturbances

$$w_t \sim \mathcal{N}(0, \Sigma_{\text{use}}), \quad t = 0, \ldots, T-1,$$

independent across time.

Correlated disturbances

$$w_{t+1} = \rho w_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$

with

$$Q = (1 - \rho^2)\Sigma_{\text{use}}, \quad |\rho| < 1,$$

so that the stationary marginal distribution remains $\mathcal{N}(0, \Sigma_{\text{use}})$.

The covariance Σ_{use} is obtained by projecting a candidate covariance onto the Wasserstein ball, ensuring consistency with the ambiguity set.

Detailed construction, sampling procedures, and empirical visualizations are provided in Appendix B.

2.4 Optimization Problems

We now formulate the controller synthesis problem under uncertainty.

Baseline (Simulation-Based) Formulation. The baseline approach seeks to directly minimize the long-run average cost under stochastic disturbances:

$$\min_{\theta} J(\theta) \quad \text{s.t.} \quad \rho(\mathcal{A}(\theta)) < 1,$$

where $\theta \leftrightarrow (A_c, B_c, C_c, D_c)$ parameterizes the controller.

The objective is defined as:

$$J(\theta) = \sup_{w \in \mathcal{W}} \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|z_t\|^2.$$

The complete formulation of this method can be found in Chapter 3, and its results in Chapter 9.1.

Distributionally Robust LMI Formulation. To address uncertainty in the disturbance distribution, we adopt a distributionally robust optimization (DRO) framework based on Wasserstein ambiguity sets [2].

We introduce decision variables:

$$\lambda \geq 0, \quad Q, \quad X = X^\top, \quad Y = Y^\top, \quad K, L, M, N,$$

and define the lifted matrices $\mathbb{P}, \mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ accordingly.

The optimization problem becomes:

$$\min \ \text{trace}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2,$$

subject to a set of Linear Matrix Inequalities (LMIs) ensuring stability and robust performance.

Two ambiguity models are considered:

Correlated Disturbances The joint disturbance structure is preserved, leading to a single large LMI constraint.

Independent Disturbances The disturbance components are decoupled, yielding two separate LMI conditions.

More details of this method can be found in [2] and its extensions in Chapters 4-7.

Algorithm 1 DRO-LMI Controller Synthesis

```
Define ambiguity set radius γ
Construct LMI constraints (correlated or independent)
Solve resulting semidefinite program (SDP)
Recover controller matrices
Validate via simulation
```

2.5 Discussion

The baseline approach offers a flexible yet potentially fragile solution because it relies on a nominal disturbance model. In contrast, the DRO-LMI formulation explicitly accounts for distributional uncertainty, offering robustness guarantees at the cost of increased computational complexity.

This trade-off between robustness and tractability will be explored empirically in subsequent chapters.

Although the preceding formulation assumes that the plant matrices are known, this assumption is often unrealistic in practical applications. In many data-driven settings, only input-output or state trajectory measurements are available, and the system matrices must be inferred from data. This naturally motivates the question of whether the same distributionally robust LMI framework can be extended to an identified model without losing its convex synthesis structure.

3 Simulation-Based Baseline

This chapter introduces the simulation-based baseline corresponding to the formulation presented in Chapter 2. While the previous chapter defined the baseline as an optimisation problem over the controller parameters, the present chapter summarises the resulting practical surrogate formulation and its interpretation.

The detailed numerical implementation of this approach—including simulation, Monte Carlo approximation, stability handling, and numerical optimisation—is reported in Appendix G.

The baseline method serves two purposes. First, it provides a direct benchmark against which the subsequent LMI-based and data-driven formulations can be compared. Second, it illustrates the main computational difficulties of direct controller optimisation, namely the need to approximate a stochastic infinite-horizon objective, the non-convexity of the optimisation landscape, and the lack of an explicit convex stability certificate.

3.1 Main Idea

The baseline approach seeks a dynamic output-feedback controller

$$\begin{bmatrix} x_{c,t+1} \\ u_t \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} x_{c,t} \\ y_t \end{bmatrix}$$

that minimizes the asymptotic closed-loop performance under stochastic excitation.

Letting

$$\chi_t = \begin{bmatrix} x_t^\top & x_{c,t}^\top \end{bmatrix}^\top, \qquad \theta = \mathrm{vec}(A_c, B_c, C_c, D_c),$$

the closed-loop system is

$$\chi_{t+1} = \mathcal{A}(\theta)\chi_t + \mathcal{B}(\theta)w_t, \tag{3.1}$$

$$z_t = \mathcal{C}(\theta)\chi_t + \mathcal{D}(\theta)w_t. \tag{3.2}$$

The ideal optimisation problem can be written as

$$\min_\theta J(\theta) \qquad \text{subject to} \qquad \rho(\mathcal{A}(\theta)) < 1,$$

where $J(\theta)$ denotes the steady-state average quadratic cost.

This problem is not directly tractable because:

- the cost is defined over an infinite horizon;
- the expectation is taken with respect to a stochastic disturbance process;
- the dependence of the cost on θ is nonlinear and non-convex;
- the stability constraint $\rho(\mathcal{A}(\theta)) < 1$ is itself non-convex.

The implemented baseline, therefore, replaces the ideal problem with a finite-horizon Monte Carlo surrogate combined with a soft stability penalty.

3.2 Final Program

Practical surrogate problem. The implemented baseline solves the following simulation-based optimisation problem:

$$\begin{aligned} \min_{\theta=\mathrm{vec}(A_c,B_c,C_c,D_c)} \quad & \bar{J}_T(\theta) + \mu_\rho\Big(\max\{0, \rho(\mathcal{A}(\theta)) - \rho_{\max}\}\Big)^2 \\ \text{s.t.} \quad & \underline{\theta}_i \le \theta_i \le \overline{\theta}_i, \qquad i = 1, \dots, n_\theta, \end{aligned}$$

where the Monte Carlo objective is

$$\bar{J}_T(\theta) = \frac{1}{N_s} \sum_{s=1}^{N_s} \frac{1}{T - T_{\text{burn}}} \sum_{t=T_{\text{burn}}}^{T-1} \|z_t^{(s)}\|_2^2,$$

and trajectories are generated according to

$$\chi_{t+1}^{(s)} = \mathcal{A}(\theta)\chi_t^{(s)} + \mathcal{B}(\theta)w_t^{(s)}, \qquad z_t^{(s)} = \mathcal{C}(\theta)\chi_t^{(s)} + \mathcal{D}(\theta)w_t^{(s)},$$

with

$$\chi_0^{(s)} = 0, \qquad w_t^{(s)} \in \mathcal{W}_{Wass}.$$

The complete numerical procedure underlying this formulation is described in Appendix G.

3.3 Interpretation of the Baseline Method

The implemented baseline should be interpreted as a direct simulation-driven controller tuning method. Its main strength lies in its conceptual simplicity:

- it directly optimises the controller parameters;
- it does not require reformulating the problem as an SDP;
- it is flexible with respect to the objective, simulation setup, and controller structure.

At the same time, the method exhibits important limitations:

1. **Non-convexity:** the mapping $\theta \mapsto J(\theta)$ is highly nonlinear and generally admits multiple local minima.
2. **Simulation dependence:** the objective is only available through repeated stochastic simulation, which makes optimisation computationally expensive.
3. **Weak stability handling:** stability is not guaranteed by construction, but only encouraged through spectral-radius penalization and heuristic projection.
4. **Nominal uncertainty model:** the method relies on a prescribed covariance Σ_w and therefore does not explicitly protect against distributional mismatch.

These observations clarify why the baseline is best viewed as a reference implementation rather than a final synthesis framework. It provides a useful benchmark and a direct optimisation perspective, but it lacks the structural guarantees and robustness properties that motivate the later LMI-based formulations.

3.4 Discussion and Role in the Overall Report

The baseline implementation occupies an important position in the methodological progression of this work. It represents the most direct approach to controller design in the considered setting: parameterise the controller, simulate the closed loop under stochastic excitation, estimate a cost, and optimise numerically.

This approach is valuable because it establishes a clear benchmark. Any subsequent method, whether model-based LMI or data-driven, should be interpreted relative to this reference in terms of:

- achieved closed-loop performance;
- computational burden;
- sensitivity to uncertainty;
- strength of the associated stability and robustness guarantees.

At the same time, the baseline also exposes the core limitations that motivate the rest of the report. In particular:

- the optimisation remains non-convex;
- the handling of uncertainty is nominal rather than distributionally robust;
- stability is verified numerically rather than enforced through a convex certificate;
- the procedure is difficult to generalize into a principled, direct data-driven synthesis framework.

For these reasons, the baseline chapter should not be regarded as a mere implementation appendix. Rather, it provides the computational and conceptual starting point from which the model-based LMI formulation, and later the data-driven reformulation attempts, naturally emerge.

In summary, the baseline method converts the abstract stochastic control problem into a practical numerical routine based on simulation, Monte Carlo averaging, and local nonlinear optimisation. This makes it an appropriate benchmark for the more structured synthesis procedures developed in the following chapters.

The corresponding numerical results for the independent disturbance setting are reported in Section 9.1.

4 Data-Driven DRO-LMI Synthesis via Auxiliary Inverse Variables

In Chapter 2, the distributionally robust LMI synthesis problem was developed under the assumption that the plant matrices are known. In the present setting, however, only trajectory data are available. A natural idea is therefore to reconstruct the unknown state-space operators directly from measured data and embed them into the DRO-LMI design.

Rather than explicitly computing regularised inverse Gram matrices offline, this chapter adopts the implementation-oriented parametrisation based on auxiliary positive semidefinite matrices

$$G_x \succeq 0, \qquad G_y \succeq 0, \qquad G_z \succeq 0,$$

which are intended to represent inverse regularised data Gramians. This yields a compact data-driven reconstruction of the plant matrices while preserving a structured optimisation model.

4.1 Data-Driven Parametrisation

Let the measured data be stacked as

$$X, \quad X_+, \quad U, \quad W, \quad Y, \quad Z,$$

and define the regressor matrices

$$D_x := \begin{bmatrix} X \\ U \\ W \end{bmatrix}, \qquad D_y := \begin{bmatrix} X \\ W \end{bmatrix}, \qquad D_z := \begin{bmatrix} X \\ U \\ W \end{bmatrix}.$$

The data-driven operators are then defined by

$$O_x := X_+ D_x^\top G_x, \tag{4.1}$$

$$O_y := Y D_y^\top G_y, \tag{4.2}$$

$$O_z := Z D_z^\top G_z, \tag{4.3}$$

from which the matrices

$$A,\ B_u,\ B_w,\ C_y,\ D_{yw},\ C_z,\ D_{zu},\ D_{zw}$$

are extracted block-by-block. In particular,

$$\begin{bmatrix} A & B_u & B_w \end{bmatrix} = O_x, \qquad \begin{bmatrix} C_y & D_{yw} \end{bmatrix} = O_y, \qquad \begin{bmatrix} C_z & D_{zu} & D_{zw} \end{bmatrix} = O_z.$$

The auxiliary matrices G_x, G_y, G_z are not arbitrary: they are driven toward the inverse regularised Gramians through the convex penalties

$$\begin{aligned} J_{\text{est}} = \mu_x \Big(&\operatorname{tr}\big((D_x D_x^\top + r_x I) G_x\big) - \log\det G_x \Big) \\ &+ \mu_y \Big(\operatorname{tr}\big((D_y D_y^\top + r_y I) G_y\big) - \log\det G_y \Big) \\ &+ \mu_z \Big(\operatorname{tr}\big((D_z D_z^\top + r_z I) G_z\big) - \log\det G_z \Big). \end{aligned} \tag{4.4}$$

The derivation and interpretation of this construction are given in Appendix H.1.

4.2 Lifted Closed-Loop Realisation

Using the synthesis variables

$$X = X^\top, \qquad Y = Y^\top, \qquad K,\ L,\ M,\ N,$$

define

$$P := \begin{bmatrix} Y & I \\ I & X \end{bmatrix}.$$

The lifted closed-loop matrices used in the implementation are

$$\mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix}, \tag{4.5}$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \tag{4.6}$$

$$\mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \tag{4.7}$$

$$\mathbf{D} = D_{zw} + D_{zu} N D_{yw}. \tag{4.8}$$

4.3 Final Program

Let

$$\lambda \geq 0, \qquad Q \succeq 0,$$

and let Σ_{nom} denote the nominal disturbance covariance. The DRO contribution to the objective is

$$J_{\text{dro}} = \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2.$$

Final program. The data-driven DRO-LMI synthesis problem with auxiliary inverse variables is:

$$
\begin{aligned}
\min_{\substack{\lambda\geq 0,\ Q\succeq 0,\\ G_x\succ 0,\ G_y\succ 0,\ G_z\succ 0,\\ X=X^\top,\ Y=Y^\top,\ K,\ L,\ M,\ N}} \quad & \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \alpha\, J_{\text{est}} \\
\text{s.t.} \quad & P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \\
& \begin{cases} \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q-\lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, & w \in \mathcal{W}^{\text{corr}}_{\text{Wasserstein}}, \\ \begin{cases} \begin{bmatrix} -P & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -P & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \\ \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q-\lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \end{cases} & w \in \mathcal{W}^{\text{ind}}_{\text{Wasserstein}}. \end{cases}
\end{aligned}
$$

with J_{est} given by (4.4), and where $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ are defined by (4.5)–(4.8).

4.4 Interpretation and Limitation

This formulation is attractive because it uses trajectory data directly and embeds the inverse-Gram regularisation inside the optimisation problem. However, this does *not* produce a genuinely convex direct data-driven LMI synthesis problem.

Indeed, while the estimation term J_{est} is convex in G_x, G_y, G_z, the lifted matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ contain products between the reconstructed matrices and the controller variables X, Y, K, L, M, N. Consequently, the semidefinite constraints are no longer affine in the decision variables. The detailed derivation is given in Appendix H.2.

One may attempt to restore convexity by replacing the exact inverse parametrisation with convex outer bounds or by freezing G_x, G_y, G_z in a preliminary identification stage. However, such modifications introduce two drawbacks:

1. the resulting constraints become overly conservative, hence often too restrictive in practice;
2. the method ceases to be purely direct data-driven, since it again relies on an intermediate identified model.

A precise discussion is provided in Appendix H.3.

Takeaway. The auxiliary-inverse formulation is a useful data-driven modelling device, but it does not by itself yield a convex direct data-driven DRO-LMI synthesis method.

5 Data-Driven DRO-LMI Relaxations Based on Identification Residuals

5.1 Motivation

Chapter 4 showed that directly combining data-driven model identification with the model-based DRO-LMI synthesis framework is not straightforward. In particular, once the identified matrices are substituted explicitly into the lifted closed-loop formulation and the identification mismatch is retained as an optimization variable, the synthesis problem generally loses convexity. This creates a methodological tension: on the one hand, one would like to preserve the data-driven nature of the approach; on the other hand, one would also like to retain the tractability and structural guarantees offered by convex semidefinite formulations.

This chapter addresses that tension through a different strategy. Rather than substituting the identification error directly into the plant matrices and optimizing over the resulting non-convex expressions, the mismatch between the nominal one-step predictor and the measured data is interpreted as a bounded uncertainty budget. This uncertainty is then injected into the lifted DRO-LMI by means of Young-type inequalities and auxiliary slack variables. The resulting formulation remains convex and therefore computationally tractable, while still accounting for the uncertainty induced by data-driven identification.

The approach developed here is thus a convex residual-based extension of the model-based DRO-LMI framework. It preserves the main structure of the output-feedback synthesis problem, but modifies the Lyapunov-related block to absorb the effect of identification mismatch through conservative upper bounds.

5.2 General Idea of the Method

High-level principle. The proposed methodology starts from a nominal model identified directly from data, together with a residual-based disturbance representation. This preliminary construction is described in Appendix E. Denote the identified matrices by

$$(\hat{A}, \hat{B}_u, \hat{B}_w, \hat{C}_y, \hat{D}_{yw}, \hat{C}_z, \hat{D}_{zu}, \hat{D}_{zw}),$$

and let

$$\beta_A, \qquad \beta_B$$

be the uncertainty surrogates associated with the state matrix and input matrix parts of the one-step regression error.

The central idea is to keep the standard lifted output-feedback parametrization based on

$$P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix},$$

and to write the uncertain lifted state block as a nominal part plus two perturbation components:

$$\mathbf{A}_{\text{true}} = \mathbf{A} + E_A + E_B.$$

Here:

- E_A collects the part of the lifted uncertainty induced by the error on A;
- E_B collects the part induced by the error on B_u.

These perturbations create mixed bilinear terms of the form

$$2\langle E_A, P\mathbf{A}\rangle, \qquad 2\langle E_B, P\mathbf{A}\rangle,$$

which would be non-convex if kept explicitly. The key step is therefore to upper-bound them using Young's inequality:

$$2\langle U, V\rangle \leq s\|U\|_F^2 + \frac{1}{s}\|V\|_F^2, \qquad s > 0.$$

This transforms the original non-convex uncertainty coupling into a convex relaxation involving:

- slack variables associated with the uncertainty radii;
- selector matrices that localize the uncertainty contribution in the Lyapunov corner of the lifted kernel;
- a regularized state block replacing the original $-P$.

Two relaxation variants. The implementation supports two variants of this idea:

1. an *isotropic Young-type relaxation*, where both uncertainty contributions are handled through scalar slack variables;
2. a *residual-directional relaxation*, where the uncertainty associated with A is distributed across the principal directions of the residual covariance.

Two DRO disturbance structures. For either relaxation, the code also supports two DRO formulations:

1. a *correlated* case, where the Lyapunov, performance, and disturbance ambiguity blocks are coupled in a single large LMI;
2. an *independent* case, where the disturbance ambiguity block is separated into a second LMI.

Hence, the overall methodology comprises four convex synthesis programs:

$$\text{(isotropic or directional)} \times \text{(correlated or independent)}.$$

5.3 Nominal Lifted Closed-Loop Parametrisation

Given the identified nominal model, the lifted closed-loop matrices used in the synthesis stage are

$$\mathbf{A} = \begin{bmatrix} \hat{A}Y + \hat{B}_u M & \hat{A} + \hat{B}_u N\hat{C}_y \\ K & X\hat{A} + L\hat{C}_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} \hat{B}_w + \hat{B}_u N\hat{D}_{yw} \\ X\hat{B}_w + L\hat{D}_{yw} \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} \hat{C}_z Y + \hat{D}_{zu} M & \hat{C}_z + \hat{D}_{zu} N\hat{C}_y \end{bmatrix},$$

$$\mathbf{D} = \hat{D}_{zw} + \hat{D}_{zu} N\hat{D}_{yw}.$$

The optimization variables are

$$X = X^\top, \qquad Y = Y^\top, \qquad K,\ L,\ M,\ N, \qquad Q \succeq 0, \qquad \lambda \geq 0.$$

The nominal DRO objective remains

$$J_{\mathrm{DRO}} = \mathrm{tr}(Q\Sigma_{\mathrm{nom}}) + \lambda\gamma^2,$$

where Σ_{nom} and γ are provided by the identification and noise-model stage.

5.4 Uncertainty Injection into the Lifted State Block

Let the true state and input matrices be written as

$$A = \hat{A} + \Delta_A, \qquad B_u = \hat{B}_u + \Delta_B.$$

Then the corresponding lifted state block can be decomposed as

$$\mathbf{A}_{\mathrm{true}} = \mathbf{A} + E_A + E_B,$$

with

$$E_A = \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix}, \qquad E_B = \begin{bmatrix} \Delta_B M & \Delta_B N\hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

The detailed derivation of this decomposition is given in Appendix I. The important point for the present chapter is that the uncertainty enters the synthesis problem only through the lifted state block $\mathbf{A}$, and therefore affects the Lyapunov corner of the DRO-LMI.

The residual-based uncertainty surrogates β_A and β_B, derived in Appendix E, provide computable scales for these two perturbation components. The proposed convex relaxations differ in how these scales are embedded into the LMI.

5.5 Isotropic Young-Type Relaxation

5.5.i Basic construction. The first variant uses scalar slack variables for both uncertainty contributions:

$$s_A \geq 0, \qquad \tau_A \geq 0, \qquad s_B \geq 0, \qquad \tau_B \geq 0.$$

The effective uncertainty magnitudes used in the implementation are denoted by

$$\beta_A^\star, \qquad \beta_B^\star,$$

and are related to the data-driven radii β_A, β_B through conservative scaling rules. The Young bounds are encoded by the PSD constraints

$$\begin{bmatrix} s_A & \beta_A^\star \\ \beta_A^\star & \tau_A \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0.$$

The corresponding regularized Lyapunov corner is

$$\mathrm{state_blk} = -P + (\tau_A + \tau_B)I_{2n_x}.$$

The selector matrices are chosen isotropically as

$$S_A = S_B = \begin{bmatrix} I_{n_x} \\ 0_{n_x \times n_x} \end{bmatrix},$$

and the slack block associated with the A-part is

$$\Pi_A = -s_A I_{n_x}.$$

5.5.ii Interpretation. This isotropic formulation treats the two uncertainty contributions in a uniform manner. The A- and B-induced errors both inflate the Lyapunov corner by scalar amounts τ_A and τ_B, while the corresponding Schur-complement rows and columns inject the quadratic Young penalties associated with $1/s_A$ and $1/s_B$.

The advantage of this formulation is simplicity. It introduces only scalar uncertainty slacks and therefore leads to a relatively compact SDP. Its drawback is conservatism: all state directions are treated equally, even though the residual covariance may be highly anisotropic.

5.6 Residual-Directional Relaxation

5.6.i Motivation. The residual covariance

$$S_R = \frac{1}{T} R R^\top$$

contains directional information on how the one-step nominal model fails to explain the measured successor states. This information can be exploited to refine the Young-type relaxation.

Instead of treating the uncertainty associated with A isotropically, the second variant distributes it along the dominant eigendirections of the residual covariance.

5.6.ii Directional weights and slacks. Let

$$S_R = U_A \operatorname{diag}(\sigma_1, \dots, \sigma_{n_x})\, U_A^\top, \qquad \sigma_1 \geq \cdots \geq \sigma_{n_x} \geq 0.$$

Define normalized directional weights

$$w_A^{(i)} = \frac{\sqrt{\sigma_i}}{\max_j \sqrt{\sigma_j}}, \qquad i = 1, \dots, n_x.$$

Then the effective uncertainty scales are chosen as

$$\beta_{A,i}^\star = \beta_A\, w_A^{(i)}, \qquad i = 1, \dots, n_x.$$

The A-part now uses vector slacks

$$s_A \in \mathbb{R}_{\geq 0}^{n_x}, \qquad \tau_A \in \mathbb{R}_{\geq 0}^{n_x},$$

with PSD constraints

$$\begin{bmatrix} s_{A,i} & \beta_{A,i}^\star \\ \beta_{A,i}^\star & \tau_{A,i} \end{bmatrix} \succeq 0, \qquad i = 1, \dots, n_x.$$

The B-part remains scalar:

$$\begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0.$$

The selector matrices are

$$S_A = \begin{bmatrix} I_{n_x} \\ 0_{n_x \times n_x} \end{bmatrix} U_A, \qquad S_B = \begin{bmatrix} I_{n_x} \\ 0_{n_x \times n_x} \end{bmatrix},$$

and the A-slack block becomes

$$\Pi_A = -\operatorname{Diag}(s_A).$$

The regularized Lyapunov corner is

$$\text{state_blk} = -P + \Big(\sum_{i=1}^{n_x} \tau_{A,i} + \tau_B\Big) I_{2n_x}.$$

5.6.iii Interpretation. This formulation remains convex, but allocates the uncertainty associated with A according to the principal directions of the residual covariance. In this sense, it is more data-aware than the isotropic formulation. Directions associated with larger residual variance receive a larger uncertainty budget, while low-energy directions are less penalized.

The formulation is still conservative, since the true perturbation is not modeled exactly, but it is typically less isotropic and more structured than the scalar Young variant.

5.7 Correlated and Independent DRO-LMI Structures

The uncertainty relaxations described above modify only the Lyapunov-related part of the synthesis problem. Independently of whether the isotropic or directional relaxation is used, the code supports two different DRO-LMI structures.

5.7.i Correlated case. In the correlated case, the full robust synthesis condition is written as one large LMI:

$$\mathcal{M}_{\text{corr}} = \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top & 0 & 0 \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top & 0 & 0 \\ 0 & \lambda I_w & -Q-\lambda I_w & 0 & 0 & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & \text{state_blk} & 0 & S_A & S_B \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z & 0 & 0 \\ 0 & 0 & 0 & S_A^\top & 0 & \Pi_A & 0 \\ 0 & 0 & 0 & S_B^\top & 0 & 0 & -s_B I_{n_x} \end{bmatrix} \prec 0, \tag{5.1}$$

together with

$$\text{state_blk} \prec 0.$$

In the isotropic case,

$$\Pi_A = -s_A I_{n_x},$$

whereas in the directional case,

$$\Pi_A = -\operatorname{Diag}(s_A).$$

5.7.ii Independent case. In the independent case, the robust certificate is split into two LMIs. The first one contains the Lyapunov and Young-relaxation structure:

$$\mathcal{M}_1 = \begin{bmatrix} -P & \mathbf{A}^\top & \mathbf{C}^\top & 0 & 0 \\ \mathbf{A} & \text{state_blk} & 0 & S_A & S_B \\ \mathbf{C} & 0 & -I_z & 0 & 0 \\ 0 & S_A^\top & 0 & \Pi_A & 0 \\ 0 & S_B^\top & 0 & 0 & -s_B I_{n_x} \end{bmatrix} \prec 0, \tag{5.2}$$

with

$$\text{state_blk} \prec 0,$$

while the second one contains the disturbance ambiguity block:

$$\mathcal{M}_2 = \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q-\lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0. \tag{5.3}$$

Thus, the difference between the correlated and independent models lies in how the disturbance ambiguity is coupled with the Lyapunov structure: jointly in the first case, separately in the second.

5.8 Regularization and Numerical Stabilization

In addition to the main DRO objective, the code includes a regularization term to keep the controller variables and uncertainty slacks well behaved.

To control the size of the controller matrices, spectral norm epigraph variables

$$t_K,\ t_L,\ t_M,\ t_N \geq 0$$

are introduced through the LMIs

$$\begin{bmatrix} t_K I_{n_x} & K \\ K^\top & t_K I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_L I_{n_x} & L \\ L^\top & t_L I_{n_y} \end{bmatrix} \succeq 0,$$

$$\begin{bmatrix} t_M I_{n_u} & M \\ M^\top & t_M I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_N I_{n_u} & N \\ N^\top & t_N I_{n_y} \end{bmatrix} \succeq 0.$$

The regularization term then has the form

$$J_{\text{reg}} = J_{\text{slack}} + J_{\text{spec}},$$

where:

$$J_{\text{spec}} = \mu(t_K + t_L + t_M + t_N),$$

and J_{slack} depends on the chosen relaxation.

For the isotropic case:

$$J_{\text{slack}} = \mu \left(s_A + \frac{\tau_A}{\beta_A^{\star 2} + \varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2} + \varepsilon} \right).$$

For the directional case:

$$J_{\text{slack}} = \mu \left(\sum_{i=1}^{n_x} s_{A,i} + \sum_{i=1}^{n_x} \frac{\tau_{A,i}}{\beta_{A,i}^{\star 2} + \varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2} + \varepsilon} \right).$$

The full objective is therefore

$$\min\ \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + J_{\text{reg}}.$$

5.9 Final Optimization Programs

For clarity, the four implemented programs are summarized below.

5.9.i Isotropic program. The isotropic Young-relaxed formulation can be written as

$$\min_{\substack{\lambda\geq 0,\ Q,\ X=X^\top,\ Y=Y^\top,\\ K,\ L,\ M,\ N,\ s_A,s_B,\tau_A,\tau_B,\\ t_K,t_L,t_M,t_N}} \quad \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \mu\left(s_A + \frac{\tau_A}{\beta_A^{\star 2}+\varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2}+\varepsilon} + t_K + t_L + t_M + t_N\right)$$

$$\text{s.t.} \quad P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succeq 0, \qquad Q \succeq 0, \qquad \lambda \geq 0,$$

$$\begin{bmatrix} s_A & \beta_A^\star \\ \beta_A^\star & \tau_A \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0,$$

spectral-norm epigraph (Appendix I.13) constraints for K, L, M, N,

$$S_A = S_B = \begin{bmatrix} I_{n_x} \\ 0 \end{bmatrix}, \qquad \Pi_A = -s_A I_{n_x},$$

$$\text{state_blk} = -P + (\tau_A + \tau_B) I_{2n_x},$$

$$\begin{cases} \mathcal{M}_{\text{corr}} \prec 0, & w \in \mathcal{W}_{\text{Wass}}^{\text{corr}}, \\ \mathcal{M}_1 \prec 0, \quad \mathcal{M}_2 \prec 0, & w \in \mathcal{W}_{\text{Wass}}^{\text{ind}}, \end{cases} \qquad \text{state_blk} \prec 0.$$

5.9.ii Directional program. The directional Young-relaxed formulation can be written as

$$\min_{\substack{\lambda\geq 0,\ Q,\ X=X^\top,\ Y=Y^\top,\\ K,\ L,\ M,\ N,\ \{s_{A,i},\tau_{A,i}\}_{i=1}^{n_x},\ s_B,\tau_B,\\ t_K,t_L,t_M,t_N}} \quad \begin{aligned} &\operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \\ &+ \mu\left(\sum_{i=1}^{n_x} s_{A,i} + \sum_{i=1}^{n_x} \frac{\tau_{A,i}}{\beta_{A,i}^{\star 2}+\varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2}+\varepsilon} + t_K + t_L + t_M + t_N\right) \end{aligned}$$

$$\text{s.t.} \quad P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succeq 0, \qquad Q \succeq 0, \qquad \lambda \geq 0,$$

$$\begin{bmatrix} s_{A,i} & \beta_{A,i}^\star \\ \beta_{A,i}^\star & \tau_{A,i} \end{bmatrix} \succeq 0, \qquad i = 1, \ldots, n_x,$$

$$\begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0,$$

spectral-norm epigraph (Appendix I.13) constraints for K, L, M, N,

$$S_A = \begin{bmatrix} I_{n_x} \\ 0 \end{bmatrix} U_A, \qquad S_B = \begin{bmatrix} I_{n_x} \\ 0 \end{bmatrix}, \qquad \Pi_A = -\operatorname{Diag}(s_A),$$

$$\text{state_blk} = -P + \Big(\sum_{i=1}^{n_x} \tau_{A,i} + \tau_B\Big) I_{2n_x},$$

$$\begin{cases} \mathcal{M}_{\text{corr}} \prec 0, & w \in \mathcal{W}_{\text{Wass}}^{\text{corr}}, \\ \mathcal{M}_1 \prec 0, \quad \mathcal{M}_2 \prec 0, & w \in \mathcal{W}_{\text{Wass}}^{\text{ind}}, \end{cases} \qquad \text{state_blk} \prec 0.$$

5.10 Algorithmic Summaries

Algorithm 2 Isotropic Young-Type Data-Driven DRO-LMI Synthesis

Input: data trajectory, noise parameters $(\Sigma_{\text{nom}}, \gamma)$, regularization parameters
Compute the nominal model and disturbance channel as in Appendix E
Build the nominal lifted matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$
Compute effective uncertainty scales $\beta_A^\star, \beta_B^\star$
Introduce scalar slacks s_A, τ_A, s_B, τ_B
Build state_blk $= -P + (\tau_A + \tau_B) I_{2n_x}$
Add selector blocks $S_A = S_B = [I \;\; 0]^\top$ and $\Pi_A = -s_A I$
Add spectral norm epigraph constraints for K, L, M, N
if correlated model **then**
    solve the single augmented LMI $\mathcal{M}_{\text{corr}} \prec 0$
else
    solve the pair of LMIs $\mathcal{M}_1 \prec 0$, $\mathcal{M}_2 \prec 0$
end if
Output: controller variables and DRO certificate

Algorithm 3 Residual-Directional Data-Driven DRO-LMI Synthesis

Input: data trajectory, noise parameters $(\Sigma_{\text{nom}}, \gamma)$, regularization parameters
Compute the nominal model and residual covariance as in Appendix E
Build the nominal lifted matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$
Eigendecompose the residual covariance $S_R = U_A \operatorname{diag}(\sigma_i) U_A^\top$
Compute directional weights $w_A^{(i)}$ and radii $\beta_{A,i}^\star$
Introduce vector slacks s_A, τ_A and scalar slacks s_B, τ_B
Build rotated selector $S_A = [I \;\; 0]^\top U_A$ and $\Pi_A = -\operatorname{Diag}(s_A)$
Build state_blk $= -P + (\sum_i \tau_{A,i} + \tau_B) I_{2n_x}$
Add spectral norm epigraph constraints for K, L, M, N
if correlated model **then**
    solve the single augmented LMI $\mathcal{M}_{\text{corr}} \prec 0$
else
    solve the pair of LMIs $\mathcal{M}_1 \prec 0$, $\mathcal{M}_2 \prec 0$
end if
Output: controller variables and DRO certificate

5.11 Limitations and Discussion

The methodology developed in this chapter provides a convex way to incorporate identification uncertainty into data-driven DRO-LMI synthesis. Its main advantage is that it avoids the non-convexity that would arise from directly optimizing over the unknown regression errors. At the same time, it retains the main output-feedback and distributionally robust structure of the model-based synthesis framework.

However, this convexity is obtained through conservative upper bounds. The uncertain terms are not modeled exactly, but are instead relaxed through Young-type inequalities and auxiliary slack variables. This introduces several limitations.

i. First, the method may be conservative when the identification residual is small but highly structured, since the uncertainty is still embedded through upper bounds rather than exact algebraic identities. This is particularly evident in the isotropic formulation, where all state directions are treated uniformly.

ii. Second, even in the directional variant, the uncertainty associated with the input matrix B_u is still handled through a scalar slack, and the state-corner inflation remains isotropic through a multiple of the identity. Thus, the directional refinement only partially captures the geometry of the residual mismatch.

iii. Third, the quality of the resulting controller depends on the reliability of the residual-based uncertainty surrogates β_A, β_B. These quantities are data-consistent and computable, but they do not constitute exact worst-case identification bounds.

iv. Fourth, the method inherits the numerical sensitivity of semidefinite synthesis problems. In practice, additional regularization on the controller matrices and the uncertainty slacks is needed to avoid excessively large solutions or poor conditioning.

Despite these limitations, the chapter introduces a useful intermediate step between purely nominal data-driven design and more structured robust formulations. It shows that residual-based uncertainty can be incorporated directly into the DRO-LMI synthesis stage while preserving convexity, and it provides two concrete realizations of this principle:

- a simpler isotropic Young-type relaxation;
- a more structured residual-directional relaxation.

The detailed derivations supporting these formulations are deferred to Appendix I, while the residual-based identification backbone is presented in Appendix E. Numerical results are discussed in Chapter 9.2.

6 Schur-Enhanced Young-Type Data-Driven DRO-LMI Relaxation

6.1 Motivation and limitations of the first Young-type relaxation

The Young-type relaxation introduced in Chapter 5 provides a convex data-driven formulation for distributionally robust output-feedback synthesis. Its main advantage is that it allows one to work directly from trajectory data, while accounting for identification mismatch through an explicit uncertainty budget derived from the one-step regression residuals.

However, that first formulation remains conservative. In particular, the perturbation terms generated by the identification errors are absorbed through comparatively coarse scalar bounds, and the Lyapunov inequality is regularized in a way that does not fully exploit the matrix structure of the uncertainty. As a consequence, the resulting feasible set may be unnecessarily restricted, especially when the estimated model mismatch is moderate but directionally structured.

This motivates a refined formulation in which the same data-driven uncertainty description is retained, but the robustification of the Lyapunov inequality is performed through an additional Schur-complement lifting. The purpose is to obtain a tighter convex surrogate, while preserving semidefinite representability and compatibility with dynamic output-feedback synthesis.

6.2 Main contribution

The main contribution of this chapter is a *Schur-enhanced Young-type data-driven DRO-LMI relaxation* for dynamic output-feedback control. The proposed formulation preserves the same identification pipeline as the previous chapter, but modifies the robust synthesis step in three key ways:

i. the inverse Lyapunov metric is upper bounded through an explicit semidefinite lifting;

ii. the mixed perturbation terms are absorbed by means of an auxiliary diagonal slack matrix and additional Schur-complement variables;

iii. the nominal Lyapunov block is replaced by an uncertainty-shrunk matrix $\bar{P}$, and the lower-right Lyapunov block is replaced by an auxiliary matrix P_s, both of which are linked to the uncertainty bound through convex matrix inequalities.

The resulting problem remains a convex semidefinite program and can therefore be solved with standard conic solvers.

6.3 Lifted controller parameterisation

Once having collected data and estimate the model plant (Appendix E), we follow the same procedure in Chapter 5.3 to obtain $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ from $\hat{A}_x, \hat{B}_u, \hat{B}_w, \hat{C}_y, \hat{D}_{yw}, \hat{C}_z, \hat{D}_{zu}, \hat{D}_{zw}$.

The disturbance ambiguity is described through a nominal covariance Σ_{nom} and a radius parameter γ. The optimization introduces

$$\lambda \geq 0, \qquad Q = Q^\top \succeq 0,$$

and minimizes the dual distributionally robust objective

$$J_{\text{DRO}} = \text{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2.$$

6.4 Residual-based uncertainty budget

In the Schur-enhanced formulation, the uncertainty entering the lifted state block is summarized by an isotropic matrix bound

$$\beta_E = \gamma_E I_{2n_x},$$

where γ_E is built from residual-based uncertainty magnitudes and norm bounds on the lifted controller variables. The detailed derivation of β_E is deferred to Appendix J.

6.5 Schur-enhanced robustification

The main refinement over the previous Young-type formulation is obtained by introducing:

$$S = \text{diag}(s_A, s_B) \in \mathbb{S}^{2n_x}, \qquad \sigma_s \geq 0, \qquad \sigma_p \geq 0,$$

together with two auxiliary matrices

$$H \in \mathbb{S}^{2n_x}, \qquad P_s \in \mathbb{S}^{2n_x}.$$

The variable σ_p is used to upper bound P^{-1} through

$$\begin{bmatrix} \sigma_p I & I \\ I & P \end{bmatrix} \succ 0,$$

while S is constrained by

$$S \succ 0, \qquad \sigma_s I - S \succ 0.$$

Defining

$$G := \sigma_p I,$$

the additional Schur lifting is enforced through

$$\begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0, \qquad \begin{bmatrix} G + H & I \\ I & P_s \end{bmatrix} \succ 0.$$

Finally, the Lyapunov block is reduced according to

$$\bar{P} := P - (\sigma_s + \sigma_p)\beta_E, \qquad \bar{P} \succ 0.$$

6.6 Final optimization programs

The final Schur-enhanced Young-type data-driven DRO-LMI synthesis problem can be stated compactly as:

$$
\begin{aligned}
\min_{\substack{\lambda\geq 0,\ Q,\ X=X^\top,\ Y=Y^\top,\\ K,\ L,\ M,\ N,\ S,\ H,\ P_s,\\ \sigma_s\geq 0,\ \sigma_p\geq 0,\\ t_K,t_L,t_X,t_Y,t_M,t_N}} \quad & \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \rho\,(t_K + t_L + t_X + t_Y + t_M + t_N) + \varepsilon \\
\text{s.t.} \quad & P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad G = \sigma_p I, \qquad P_{\text{bar}} = P - (\sigma_s + \sigma_p)\beta_E, \\
& Q \succ 0, \quad P \succ 0, \quad S \succ 0, \quad \sigma_s I - S \succ 0, \quad H \succ 0, \quad P_s \succ 0, \quad P_{\text{bar}} \succ 0, \\
& \begin{bmatrix} \sigma_p I & I \\ I & P \end{bmatrix} \succ 0, \qquad \begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0, \qquad \begin{bmatrix} G+H & I \\ I & P_s \end{bmatrix} \succ 0, \\
& \begin{cases}
\begin{bmatrix} -P_{\text{bar}} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_{n_w} & \lambda I_{n_w} & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_{n_w} & -Q - \lambda I_{n_w} & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P_s & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_{n_z} \end{bmatrix} \prec 0, & w \in \mathcal{W}^{\text{corr}}_{\text{Wasserstein}}, \\
\begin{cases}
\begin{bmatrix} -P_{\text{bar}} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -P_s & 0 \\ \mathbf{C} & 0 & -I_{n_z} \end{bmatrix} \prec 0, \\
\begin{bmatrix} -\lambda I_{n_w} & \lambda I_{n_w} & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_{n_w} & -Q - \lambda I_{n_w} & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I_{n_z} \end{bmatrix} \prec 0,
\end{cases} & w \in \mathcal{W}^{\text{ind}}_{\text{Wasserstein}}.
\end{cases}
\end{aligned}
$$

6.7 Spectral-norm regularization

To avoid excessively large controller and lifting matrices, the implementation optionally introduces epigraph variables

$$t_K, t_L, t_X, t_Y, t_M, t_N \geq 0,$$

with LMI constraints of the form

$$\begin{bmatrix} t_J I_{n^J_{\text{row}}} & J \\ J^\top & t_J I_{n^J_{\text{col}}} \end{bmatrix} \succeq 0, \qquad J \in \{K, L, X, Y, M, N\}.$$

These imply

$$\|K\|_2 \leq t_K, \quad \|L\|_2 \leq t_L, \quad \|X\|_2 \leq t_X, \quad \|Y\|_2 \leq t_Y, \quad \|M\|_2 \leq t_M, \quad \|N\|_2 \leq t_N.$$

The corresponding regularization term is

$$\mathcal{R}_{\text{spec}} = \rho\,(t_K + t_L + t_X + t_Y + t_M + t_N),$$

so that the full objective becomes

$$J_{\text{tot}} = \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \mathcal{R}_{\text{spec}} + \varepsilon.$$

6.8 Discussion

The Schur-enhanced Young relaxation refines the previous data-driven Young-type DRO-LMI without modifying the identification stage. Its role is to propagate the model-estimation uncertainty through a tighter convex lifting, replacing the direct coarse inflation of the Lyapunov block by a structured Schur-complement mechanism involving S, σ_s, σ_p, H, and P_s.

The resulting formulation remains a semidefinite program and can therefore be solved efficiently with conic optimization tools. Numerical results are reported in Chapter 9.3.

The Schur-enhanced Young relaxation improves the convex treatment of identification uncertainty, but it still relies on an intermediate state-space model identified from data. The quality of the resulting controller therefore, remains tied to the regression stage and to the way the identification mismatch is quantified.

This limitation motivates a different perspective. Instead of first estimating a nominal realization and then robustifying the synthesis problem against model error, one may attempt to work directly with the set of admissible trajectories consistent with the measured data. The next chapter develops this idea through a behavioral formulation based on Willems' Fundamental Lemma.

7 Willems-Fundamental-Lemma-Based Data-Driven DRO-LMI Formulation

7.1 Motivation

Chapters 5 and 6 introduced two convex data-driven relaxations in which a nominal state-space model is first estimated from measured trajectories and the resulting identification error is incorporated into the synthesis stage through robust LMI constraints. Although these formulations preserve convexity, they still rely on an intermediate model-reconstruction step and on explicit bounds for the associated estimation error.

This chapter investigates a different direction. Instead of selecting a nominal realization and robustifying it against perturbations, we seek to characterize directly the set of plant matrices that are consistent with the measured data. To this end, we adopt a behavioral perspective inspired by Willems' Fundamental Lemma, and use Hankel-based consistency constraints to define an admissible class of plant representations. The resulting formulation aims to reduce the dependence on explicit system identification while preserving compatibility with robust control synthesis.

We consider the discrete-time linear time-invariant plant

$$\begin{aligned} x_{k+1} &= A_x x_k + B_u u_k + B_w w_k, \\ z_k &= C_z x_k + D_{zu} u_k + \underbrace{D_{zw}}_{\to \mathbf{0}_{n_z \times n_w}} w_k, \\ y_k &= C_y x_k + \underbrace{D_{yu}}_{\to \mathbf{0}_{n_y \times n_u}} u_k + \underbrace{D_{yw}}_{\to \mathbf{0}_{n_y \times n_w}} w_k, \end{aligned}$$

or equivalently

$$\begin{aligned} \begin{bmatrix} x_{k+1} \\ y_k \end{bmatrix} &= \begin{bmatrix} A_x & B_u \\ C_y & \mathbf{0}_{n_y \times n_u} \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \begin{bmatrix} B_w \\ \mathbf{0}_{n_y \times n_w} \end{bmatrix} w_k = \mathcal{M}_P \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \bar{B}_w \, w_k, \\ z_k &= C_z x_k + D_{zu} u_k. \end{aligned}$$

Here:

- $x_k \in \mathbb{R}^{n_x}$ is the system state,
- $u_k \in \mathbb{R}^{n_u}$ is the control input,
- $w_k \in \mathbb{R}^{n_w}$ is an exogenous disturbance,
- $z_k \in \mathbb{R}^{n_z}$ is the performance output,
- $y_k \in \mathbb{R}^{n_y}$ is the measured output.

The disturbance w_k is assumed to be unmeasured but bounded, i.e.

$$w_k \in \mathcal{W}_2^{\text{set}},$$

and may be estimated or reproduced through suitable filtering or internal-model mechanisms. Moreover, C_z and D_{zu} are assumed known or designer-chosen.

7.2 Dynamic Output-Feedback Structure

The plant is controlled by a dynamic output-feedback controller of the form

$$\begin{aligned} x_{k+1}^c &= A_c x_k^c + B_c y_k, \\ u_k &= C_c x_k^c + D_c y_k, \end{aligned}$$

where $x_k^c \in \mathbb{R}^{n_c}$ denotes the controller state.

Equivalently,

$$\begin{bmatrix} x_{k+1}^c \\ u_k \end{bmatrix} = \mathcal{M}_C \begin{bmatrix} x_k^c \\ y_k \end{bmatrix}, \qquad \mathcal{M}_C := \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix}.$$

The corresponding augmented closed-loop state is

$$\chi_k := \begin{bmatrix} x_k \\ x_k^c \end{bmatrix},$$

and the closed-loop dynamics can be expressed as

$$\begin{aligned} \chi_{k+1} &= \mathcal{A}\chi_k + \mathcal{B}w_k, \\ z_k &= \mathcal{C}\chi_k + \mathcal{D}w_k, \end{aligned}$$

or compactly,

$$\begin{bmatrix} \chi_{k+1} \\ z_k \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \chi_k \\ w_k \end{bmatrix} = \mathcal{M}_{CL} \begin{bmatrix} \chi_k \\ w_k \end{bmatrix}.$$

Hence, $\mathcal{M}_{CL} = f(\mathcal{M}_P, \mathcal{M}_C)$.

The explicit derivation of the lifted closed-loop matrices, the associated factorization, and the controller reconstruction formulas are deferred to Appendix K.

7.3 Disturbance Absorption and Behavioral Reformulation

To simplify the structure, we adopt the substitution

$$\bar{w}_k \mapsto B_w w_k.$$

Under this normalization, and using $D_{zw} \to 0$ and $D_{yw} \to 0$, the direct feedthrough term satisfies

$$\mathcal{D} = D_{zw} + D_{zu} D_c D_{yw} = 0.$$

This yields a normalized closed-loop disturbance channel, which is particularly convenient for behavioral consistency constraints and subsequent robust synthesis.

The full matrix expressions resulting from this substitution are reported in Appendix K.

7.4 Behavior-Consistent Matrix Reconstruction from Willems-Type Constraints

7.4.i Conceptual Idea. The objective is to construct a set of matrix blocks $\mathcal{M}_P$ that are consistent with the observed system behavior, without identifying a unique realization and without assuming Gaussian disturbances.

In contrast to purely data-consistent formulations, the admissibility of trajectories is enforced explicitly via Willems' Fundamental Lemma through Hankel parametrisation.

7.4.ii Hankel-Based Behavioral Parametrisation. Let $\{x_k, u_k, y_k\}_{k=0}^{T}$ be measured data generated by a persistently exciting input u.

Define the one-step Hankel matrices

$$\mathcal{H}(x), \quad \mathcal{H}(u), \quad \mathcal{H}(x^+), \quad \mathcal{H}(y),$$

where $x_k^+ := x_{k+1}$.

By Willems' Fundamental Lemma, for any admissible one-step trajectory (x_k, u_k, x_{k+1}, y_k), there exists a coefficient vector g such that

$$\begin{bmatrix} x_k \\ u_k \\ x_{k+1} \\ y_k \end{bmatrix} = \begin{bmatrix} \mathcal{H}(x) \\ \mathcal{H}(u) \\ \mathcal{H}(x^+) \\ \mathcal{H}(y) \end{bmatrix} g. \tag{7.1}$$

7.4.iii Decision Variables. Introduce the following decision variables:

- $\mathcal{M}_P \in \mathbb{R}^{(n_x+n_y)\times(n_x+n_u)}$, the plant block,
- $g \in \mathbb{R}^N$, behavioral coefficient vectors,
- $w \in \mathbb{R}^{n_x+n_y}$, disturbance realizations.

7.4.iv Behavior-Consistency Constraint. Impose that for every admissible trajectory parametrised by g, the plant block $\mathcal{M}_P$ maps past signals to future signals up to disturbance:

$$\begin{bmatrix} \mathcal{H}(x^+) \\ \mathcal{H}(y) \end{bmatrix} g = \mathcal{M}_P \begin{bmatrix} \mathcal{H}(x) \\ \mathcal{H}(u) \end{bmatrix} g + w. \tag{7.2}$$

This constraint enforces that $\mathcal{M}_P$ is consistent with the entire behavioral set spanned by the data.

7.4.v Disturbance Set. Assume the disturbance satisfies

$$w \in \mathcal{W},$$

where $\mathcal{W}$ is a known uncertainty set, e.g.

$$\mathcal{W} := \{w \mid \|w\|_2 \le \varepsilon\} \qquad \text{or} \qquad \mathcal{W} := \{w \mid w^\top Q^{-1} w \le 1\}.$$

No distributional assumptions on w are required at this stage.

7.4.vi Set of Behavior-Consistent Plant Blocks. Define the admissible plant block set

$$\mathcal{M}_{\text{WFL}} := \{\mathcal{M}_P \mid \forall g \ \exists w \in \mathcal{W} \text{ such that (7.2) holds}\}. \tag{7.3}$$

Each element of $\mathcal{M}_{\text{WFL}}$ represents a realization of the observed behavior consistent with Willems' lemma and disturbance bounds.

7.4.vii Interpretation. The set $\mathcal{M}_{\text{WFL}}$:

- is invariant under state-space coordinate changes,
- does not identify a unique plant,
- captures all matrix realizations compatible with the behavioral data,
- can be propagated into robust or distributionally robust control synthesis.

This construction is fully behavioral and relies explicitly on Willems' Fundamental Lemma through Hankel parametrisation.

7.4.viii Remark. If a single representative $\mathcal{M}_P$ is required, an additional selection criterion (e.g. minimum Frobenius norm) may be imposed. Such a choice does not correspond to system identification, but to the selection of a realization within the behavior-consistent equivalence class.

7.5 Optimization Viewpoints over the Willems-Consistent Set

7.5.i Joint Willems-Consistent Plant and Controller Optimisation. Consider the optimisation problem

$$\begin{aligned} \min_{\theta,\mathcal{M}_P,g,w} \quad & J_{\text{ctrl}}(\theta,\mathcal{M}_P) \\ \text{s.t.} \quad & \begin{bmatrix}\mathcal{H}(x^+)\\ \mathcal{H}(y)\end{bmatrix} g = \mathcal{M}_P \begin{bmatrix}\mathcal{H}(x)\\ \mathcal{H}(u)\end{bmatrix} g + w, \\ & w \in \mathcal{W}, \\ & \Phi_{\text{cl}}(\mathcal{M}_P,\theta) \preceq 0. \end{aligned} \tag{7.4}$$

The matrix $\mathcal{M}_P$ is selected implicitly by the optimisation objective, subject to behavioral consistency enforced via Willems' lemma. No unique plant identification is assumed or required.

7.5.ii Robust Controller Synthesis over the Willems-Consistent Set. Define again

$$\mathcal{M}_{\text{WFL}} := \left\{ \mathcal{M}_P \;\middle|\; \forall g\ \exists w \in \mathcal{W} : \begin{bmatrix}\mathcal{H}(x^+)\\ \mathcal{H}(y)\end{bmatrix} g = \mathcal{M}_P \begin{bmatrix}\mathcal{H}(x)\\ \mathcal{H}(u)\end{bmatrix} g + w \right\}.$$

The robust synthesis problem is

$$\begin{aligned} \min_{\theta} \quad & \sup_{\mathcal{M}_P\in\mathcal{M}_{\text{WFL}}} J_{\text{ctrl}}(\theta,\mathcal{M}_P) \\ \text{s.t.} \quad & \sup_{\mathcal{M}_P\in\mathcal{M}_{\text{WFL}}} \Phi_{\text{cl}}(\mathcal{M}_P,\theta) \preceq 0. \end{aligned} \tag{7.5}$$

This formulation avoids selecting a single realization and yields guarantees for all behavior-consistent plants.

7.5.iii Regularised Selection of a Representative Plant Block. If a single representative realization is required, consider

$$\begin{aligned} \min_{\mathcal{M}_P,g,w} \quad & \|\mathcal{M}_P\|_F^2 + \lambda\|w\|_2^2 \\ \text{s.t.} \quad & \begin{bmatrix}\mathcal{H}(x^+)\\ \mathcal{H}(y)\end{bmatrix} g = \mathcal{M}_P \begin{bmatrix}\mathcal{H}(x)\\ \mathcal{H}(u)\end{bmatrix} g + w, \\ & w \in \mathcal{W}. \end{aligned} \tag{7.6}$$

The resulting $\mathcal{M}_P$ is a behaviour-consistent realisation selected by a regularization criterion and does not correspond to a uniquely identifiable plant model.

7.5.iv Representative Model Selection within the Willems-Feasible Set. Given the data-consistent model set

$$\mathcal{M}_{\text{WFL}} := \left\{ \mathcal{M}_P \;\middle|\; \forall g\ \exists w \in \mathcal{W} : \begin{bmatrix}\mathcal{H}(x^+)\\ \mathcal{H}(y)\end{bmatrix} g = \mathcal{M}_P \begin{bmatrix}\mathcal{H}(x)\\ \mathcal{H}(u)\end{bmatrix} g + w \right\},$$

model identification is in general non-unique. Consequently, a representative element of $\mathcal{M}_{\text{WFL}}$ must be selected according to an additional principled criterion.

Analytic-center selection. A geometry-driven representative can be obtained by selecting the analytic center of the feasible set, defined as the solution of

$$\begin{aligned} \min_{\mathcal{M}_P} \quad & -\log\det(S(\mathcal{M}_P)) \\ \text{s.t.} \quad & \mathcal{M}_P \in \operatorname{int}(\mathcal{M}_{\mathrm{WFL}}), \end{aligned} \tag{7.7}$$

where $S(\mathcal{M}_P)$ denotes the slack matrix associated with the data-consistency constraints. This formulation selects the most interior element of $\mathcal{M}_{\mathrm{WFL}}$, maximizing robustness with respect to perturbations of the feasible set boundaries.

Performance-aligned selection. Alternatively, model selection can be directly aligned with the control objective by solving

$$\begin{aligned} \min_{\mathcal{M}_P,\,\pi} \quad & J_{\mathrm{cl}}(\mathcal{M}_P, \pi) \\ \text{s.t.} \quad & \mathcal{M}_P \in \mathcal{M}_{\mathrm{WFL}}, \\ & \pi \in \Pi_{\mathrm{adm}}, \end{aligned} \tag{7.8}$$

where J_{cl} denotes a closed-loop performance metric and π is a controller selected from the admissible class Π_{adm}. This formulation does not aim to recover the true plant, but instead selects a data-consistent model that is optimal with respect to the induced closed-loop behavior.

Discussion. These formulations highlight that regularization within $\mathcal{M}_{\mathrm{WFL}}$ does not identify a unique physical system, but rather selects a representative element according to a chosen geometric or performance-driven criterion.

7.6 Relaxations, Limitations, and Practical Deviations

The full controller reconstruction used in the convexification is exact only under the structural conditions detailed in Appendix K. In practice, several relaxations or deviations may be introduced.

7.6.i Relaxations that Preserve the Main Guarantee.

(R1) Young–Schur (or equivalent) relaxation of the factorization. A standard convex relaxation replaces the explicit inversion/factorization of S with auxiliary variables and Schur complements, eliminating the need to compute the reconstruction factors explicitly. In this case, the controller is not reconstructed by a closed-form formula, but the feasibility set remains a convex outer-approximation of the original one. The resulting guarantee is conservative but remains meaningful: any feasible solution still certifies robust stability/performance for the relaxed problem.

(R2) Structured X, Y (diagonal / block-diagonal). Restricting X and Y to diagonal or block-diagonal forms reduces degrees of freedom, often improving numerical conditioning. The original proof structure remains unchanged, but feasibility becomes more conservative and the best achievable performance can degrade.

(R3) Sequential factorization (majorization / SCP). One may fix a factorization $S \approx U_1 V_1^\top$ from a previous iterate and solve a convex subproblem for (X, Y, K, L, M, N), then update the factorization and repeat. This becomes a sequential convex programming heuristic. The original global guarantees no longer hold, but each subproblem remains well-posed and often yields practical controllers when the full nonconvex formulation is numerically unstable.

(R4) Replace strict definiteness with a margin. Enforcing $S \succeq \delta I$ with $\delta > 0$ improves numerical stability and makes the factorization well-conditioned. This does not change the theoretical structure; it only strengthens the assumptions.

7.6.ii Deliberate Deviations from the Original Construction.

(B1) Treating $\mathcal{M}_C$ as a free decision variable. If (A_c, B_c, C_c, D_c) are optimized directly while (X, Y, K, L, M, N) are also present, then the change-of-variables argument connecting the LMI solution to a realizable controller is broken. The feasible set becomes strictly larger, and feasibility no longer implies that the controller corresponds to the intended convexification. The gain is numerical simplicity and the ability to search a larger space; the loss is the proof that the LMI encodes the same controller class as the original formulation.

(B2) Dropping the constraint $I - XY \succ 0$. If the positivity of $S = I - XY$ is not enforced, then the factorization and reconstruction formulas are not well-defined. The gain is avoiding strict matrix inequalities; the loss is that the controller map becomes undefined or ill-conditioned, and the resulting controller, if extracted by an ad-hoc procedure, is not covered by the original guarantee.

(B3) Freezing the factorization (one-shot linearization). A pragmatic compromise is to compute a factorization of S from a nominal (X, Y) or from an earlier solution and keep the factors fixed while optimizing the remaining variables. This converts the reconstruction into an affine map but changes the feasible set. The gain is improved numerical robustness and speed; the loss is that the solution is only guaranteed to satisfy the modified linearized constraints, not the original ones.

(B4) Replacing the reconstruction by post-processing. Another common choice is to solve a convex problem that yields (X, Y, K, L, M, N) but then obtain (A_c, B_c, C_c, D_c) by a separate identification step. The gain is decoupling the optimization from the realization extraction; the loss is that feasibility/performance are no longer jointly certified by a single optimization problem, and post-processing can destroy the certified margins.

7.7 Final Program

Collecting the behavioral consistency condition, the Willems-feasible plant block, and the lifted closed-loop robust constraints, the final optimization program becomes

$$
\begin{aligned}
\min_{\substack{\lambda\geq 0,\ Q,\\ X=X^\top,\ Y=Y^\top,\\ P,K,L,M,N,\\ \mathcal{M}_P\in\mathcal{M}_{WFL},\ g}} \quad & \operatorname{trace}(Q\Sigma_{\text{nom}}) + \lambda\,\gamma^2 + \zeta(\mathcal{M}_P) \\
\text{s.t.} \quad & \begin{bmatrix}\mathcal{H}(x^+)\\ \mathcal{H}(y)\end{bmatrix} g = \mathcal{M}_P \begin{bmatrix}\mathcal{H}(x)\\ \mathcal{H}(u)\end{bmatrix} g + w, \\
& \mathbf{P} = \begin{bmatrix} P & \mathbf{0}_{2n_x\times n_z} \\ \mathbf{0}_{n_z\times 2n_x} & \mathbf{I}_{n_z}\end{bmatrix}, \quad P \succ 0 \\
& \begin{cases}
\begin{bmatrix} -\mathbf{P}\Big(\mathbf{I}_{2n_x+n_w} + (\lambda-1)\,\Psi_w\Psi_w^\top\Big) & \lambda\,\Psi_w & (\mathbf{P}\mathcal{M}_{CL})^\top \\ \lambda\,\Psi_w^\top & -Q-\lambda\,\mathbf{I}_{n_w} & \mathbf{0}_{n_w\times(2n_x+n_z)} \\ \mathbf{P}\mathcal{M}_{CL} & \mathbf{0}_{n_z\times n_w} & -\mathbf{P}\end{bmatrix} \prec 0, & w\in\mathcal{W}^{corr}_{\text{Wasserstein}} \\
\begin{cases}
\begin{bmatrix} -\Psi_z^\top\ \mathbf{P}\ \Psi_z & \begin{bmatrix}\mathbf{P}\ \mathcal{M}_{CL}\ \Psi_{2n_x}\end{bmatrix}^\top \\ \mathbf{P}\ \mathcal{M}_{CL}\ \Psi_{2n_x} & -\mathbf{P}\end{bmatrix} \prec 0 \\
\begin{bmatrix} -\lambda\,\mathbf{I}_{n_w} & \lambda\,\mathbf{I}_{n_w} & \begin{bmatrix}\mathbf{P}\ \mathcal{M}_{CL}\ \Psi_{n_w}\end{bmatrix}^\top \\ \lambda\,\mathbf{I}_{n_w} & -Q\ -\lambda\,\mathbf{I}_{n_w} & \mathbf{0}_{n_w\times(2n_x+n_z)} \\ \mathbf{P}\ \mathcal{M}_{CL}\ \Psi_{n_w} & \mathbf{0}_{(2n_x+n_z)\times n_w} & -\mathbf{P}\end{bmatrix} \prec 0
\end{cases}, & w\in\mathcal{W}^{ind}_{\text{Wasserstein}}
\end{cases} \\
& \mathcal{M}_{CL} = \mathcal{M}_{CL}(\mathcal{M}_P, A_c, B_c, C_c, D_c),
\end{aligned}
$$

where the explicit selector-based expansion of $\mathcal{M}_{CL}$ and the controller realization formulas are given in Appendix K.

The final program should therefore be interpreted as a distributionally robust synthesis problem over the entire Willems-consistent plant set, rather than as a nominal model-based synthesis problem with a separate identification stage.

8 Experimental Setup

8.1 Chapter Overview

This chapter describes the experimental protocol adopted to evaluate the distributionally robust output-feedback synthesis pipeline introduced in the previous chapters. The experiments compare two configurations of the same DRO_LMI framework: a model-based configuration, in which the nominal plant matrices are assumed to be known, and a data-driven configuration, in which the same synthesis pipeline is supplied with quantities identified from simulated trajectories.

The purpose of this comparison is to isolate the effect of replacing nominal model information with estimated quantities while keeping the optimization problem, ambiguity description, and solver settings fixed. In this way, any observed difference can be attributed to the quality and structural properties of the plant information provided to the synthesis stage.

8.2 Plant, Ambiguity Set, and Common Synthesis Framework

The plant is modeled as a discrete-time linear time-invariant system of the form

$$x_{k+1} = Ax_k + B_u u_k + B_w w_k, \qquad z_k = C_z x_k + D_{zu} u_k + D_{zw} w_k, \qquad y_k = C_y x_k + D_{yw} w_k,$$

where $x_k \in \mathbb{R}^{n_x}$ denotes the state, $u_k \in \mathbb{R}^{n_u}$ the control input, $w_k \in \mathbb{R}^{n_w}$ the exogenous disturbance, $z_k \in \mathbb{R}^{n_z}$ the performance output, and $y_k \in \mathbb{R}^{n_y}$ the measured output.

Uncertainty in the disturbance distribution is modelled through a 2-Wasserstein ambiguity set centred at a nominal zero-mean distribution with covariance Σ_{nom} and radius $\gamma > 0$. The ambiguity parameters $(\gamma, \Sigma_{\text{nom}})$ are fixed across all experiments.

In both configurations, the controller is synthesized using the same DRO_LMI procedure, which produces a dynamic output-feedback controller of the form

$$x^c_{k+1} = A_c x^c_k + B_c y_k, \qquad u_k = C_c x^c_k + D_c y_k.$$

Thus, the distinction between the two experimental settings lies only in the source of the plant and disturbance quantities supplied to the synthesis pipeline.

8.3 Performance Output Definition

For the performance output, we fix the weighting matrices as

$$z = \begin{bmatrix} Q_x^{1/2} x \\ R_u^{1/2} u \end{bmatrix},$$

with

$$C_z = \begin{bmatrix} \operatorname{diag}(\sqrt{Q_x}) \\ 0_{n_u \times n_x} \end{bmatrix}, \quad D_{zu} = \begin{bmatrix} 0_{n_x \times n_u} \\ \operatorname{diag}(\sqrt{R_u}) \end{bmatrix}, \quad D_{zw} = 0,$$

and

$$Q_x = I_{n_x}, \qquad R_u = I_{n_u}.$$

This choice ensures that the quadratic cost $\|z\|_2^2$ is consistent across model-based and data-driven formulations.

8.4 Experimental Configurations

Two configurations are considered throughout the experiments.

Model-Based Configuration. In the model-based configuration, the synthesis pipeline is initialized directly with the nominal state-space matrices

$$(A, B_u, B_w, C_y, D_{yw}, C_z, D_{zu}, D_{zw})$$

and with the nominal ambiguity description $(\gamma, \Sigma_{\text{nom}})$. No identification step is performed.

Data-Driven Configuration. In the data-driven configuration, the same synthesis pipeline is executed using estimated plant matrices and disturbance statistics obtained from open-loop simulated trajectories. The optimization problem itself is not modified: ambiguity radius, regularization choices, and solver settings are kept unchanged. This ensures that the comparison isolates the effect of substituting nominal quantities with data-driven estimates.

8.5 Dataset Generation and Identification Procedure

For the data-driven configuration, open-loop datasets are generated using a dedicated simulator. Each trajectory consists of samples of the form

$$\{x_k, u_k, y_k, z_k, x_{k+1}\}_{k=0}^{T-1}.$$

The initial condition x_0 is drawn from a prescribed distribution, the input sequence u_k is chosen to be persistently exciting, and the disturbance w_k is sampled from the nominal distribution associated with Σ_{nom}.

To reduce sensitivity to individual realizations, N_{sims} independent datasets are generated with different random seeds. Rather than averaging trajectories, which would alter the dynamics, a representative dataset is selected by means of a medoid criterion: among the available datasets, the one minimizing the total weighted Frobenius distance to all others is retained.

From the selected dataset, the regression matrices

$$X_- = \begin{bmatrix} x_0 & \dots & x_{T-1} \end{bmatrix}, \qquad U_- = \begin{bmatrix} u_0 & \dots & u_{T-1} \end{bmatrix}, \qquad X_+ = \begin{bmatrix} x_1 & \dots & x_T \end{bmatrix}$$

are formed, and the state transition and input matrices are estimated via least squares:

$$\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix} = X_+ \begin{bmatrix} X_- \\ U_- \end{bmatrix}^{\dagger}.$$

The residual

$$R_x = X_+ - \hat{A}X_- - \hat{B}_u U_-$$

is interpreted as the contribution of the disturbance channel and is used to build an effective estimate $\hat{B}_w$ together with an empirical covariance matrix $\widehat{\Sigma}_w$. Analogous regressions are carried out to estimate $(\hat{C}_y, \hat{D}_{yw})$ and $(\hat{C}_z, \hat{D}_{zu}, \hat{D}_{zw})$.

8.6 Closed-Loop Simulation Protocol

Once a controller has been synthesized, the plant and controller are interconnected to form the closed-loop system with augmented state

$$\xi_k = \begin{bmatrix} x_k^\top & (x_k^c)^\top \end{bmatrix}^\top.$$

The resulting dynamics are simulated over a finite horizon T_{sim} using the same sampling time adopted during identification.

Each simulation run is specified by an initial condition, a disturbance realization, and, when relevant, additional exogenous signals entering the problem formulation. To ensure a fair comparison, the model-based and data-driven controllers are evaluated under identical realizations of initial conditions and disturbance sequences.

8.7 Choice of Initial Conditions

In the main experiments, the initial condition is selected so as to excite the most critical dynamical modes of the plant and controller. Specifically, structured initial conditions are constructed from dominant eigenvectors of the plant and controller state matrices, yielding a reproducible and deliberately challenging perturbation for closed-loop evaluation.

This choice reduces the variability associated with purely random initialization and ensures that all controllers are tested under comparable transient conditions. Full details of the construction are provided in Appendix A.2.

8.8 Evaluation Metrics

The comparison between the model-based and data-driven configurations is based on a collection of closed-loop performance and computational indicators. These include:

- a finite-horizon quadratic cost J, combining state deviation, control effort, and performance output;
- the spectral radius $\rho(A_{\text{cl}})$ of the closed-loop state matrix, used as a proxy for stability and robustness;
- peak values of the performance output and control input, such as

$$z_{\max} = \max_k \|z_k\|_2, \qquad u_{\max} = \max_k \|u_k\|_2;$$

- tail statistics computed over the final portion of the horizon, used to quantify steady-state variability;
- computational indicators, including solver objective value, wall-clock solution time, and number of attempts required to obtain a feasible controller.

When repeated runs are performed, each metric is summarized through its empirical mean and standard deviation across seeds.

Table 8.1: Closed-loop performance and computational metrics used in the experimental comparison.

Category	Metric	Description
Performance	J	Finite-horizon quadratic cost over the simulation horizon.
Stability	$\rho(A_{\mathrm{cl}})$	Spectral radius of the closed-loop state matrix.
Performance	SNR	Signal-to-Noise Ratio between z_k and w_k.
Performance	$z_{\max}$	Peak performance output, $z_{\max} = \max_k \lVert z_k \rVert_2$.
Performance	$u_{\max}$	Peak control effort, $u_{\max} = \max_k \lVert u_k \rVert_2$.
Steady-state	σ_z^{tail}	Standard deviation of z_k over the last portion of the horizon.
Steady-state	σ_y^{tail}	Standard deviation of y_k over the last portion of the horizon.
Computational	obj	Optimal objective value of the DRO_LMI problem.
Computational	t_{solve}	Wall-clock solution time.
Computational	attempts	Number of solver attempts until feasibility.
Diagnostic	stress	Scalar indicator of design tightness or conservatism.
Aggregation	$\mu \pm \sigma$	Mean and standard deviation across repeated runs.

8.9 Comparison Protocol

For each experimental setting, the comparison is carried out as follows:

1. a controller is synthesized using the model-based configuration;
2. a second controller is synthesized using the data-driven configuration;
3. both controllers are simulated under the same realizations of initial condition and disturbance sequence;
4. all performance and computational metrics are recorded and aggregated across runs.

This protocol ensures that any observed difference can be attributed to the substitution of nominal plant information with identified quantities, rather than to changes in the optimization problem or simulation environment.

8.10 Implementation and Hardware

All experiments are implemented in Python, using standard numerical and optimization libraries for matrix computations, convex optimization and data handling. The DRO_LMI synthesis problems are solved by off–the–shelf conic solvers, while simulations rely on custom routines for the plant, controller and closed–loop models. The experiments are run on a single workstation with multi–core CPU and sufficient RAM to handle the moderate state dimensions and trajectory lengths considered; no specialized hardware is required.

9 Numerical Results for the proposed methods

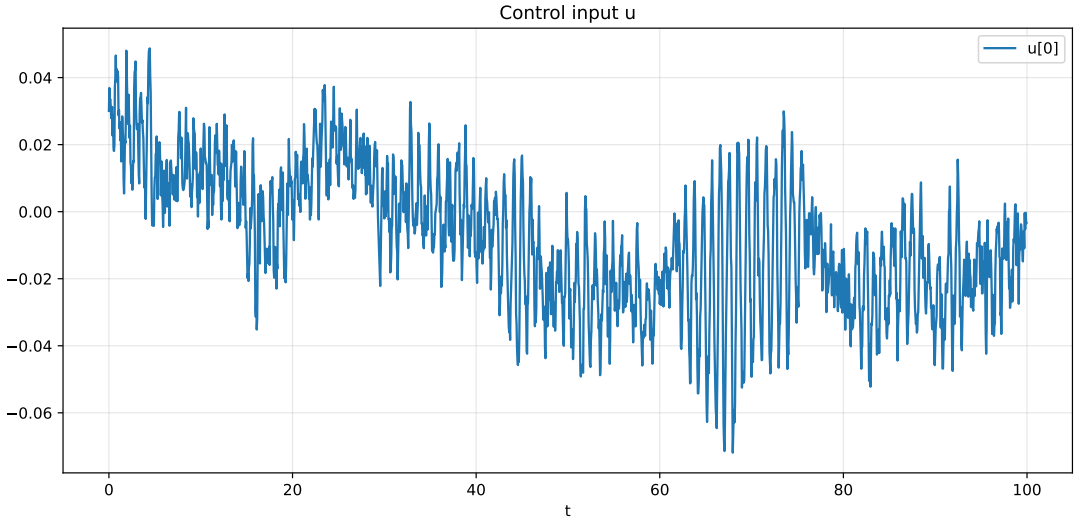
Closed-Loop SNR and Spectral Analysis To further characterize the closed-loop behaviour beyond worst-case guarantees, we evaluate steady-state signal-to-noise ratios (SNR) and frequency-domain properties under stochastic disturbances. The SNR is defined based on the steady-state covariance of the closed-loop system and quantifies the ratio between state-propagated signal energy and direct disturbance injection.

A detailed derivation, together with spectral analysis and worst-case disturbance characterisation, is reported in Appendix D.

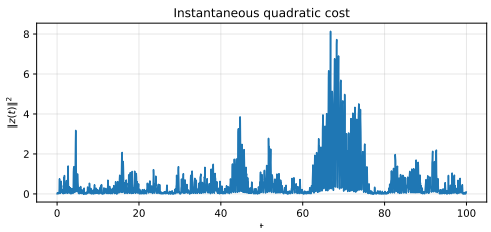
9.1 Baseline Results: Independent and Correlated Disturbance Models

This section compares the closed-loop behaviour of the baseline controller under two disturbance models: *independent* and *correlated* noise, both generated from the same Wasserstein ambiguity set. Rather than displaying all internal trajectories, the analysis focuses on the control input together with performance indicators derived from the output z, namely the quadratic cost and the signal-to-noise ratio (SNR). This provides a compact yet informative description of both transient behaviour and steady-state performance.

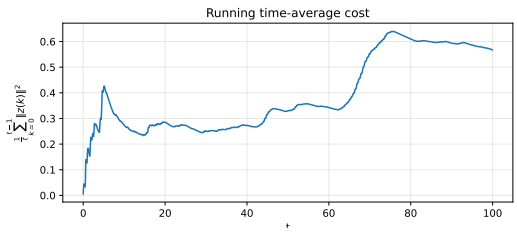
9.1.i Independent Disturbance Case. Figure 9.1 summarises the closed-loop behaviour under independent disturbances. The top panel shows the control input, while the remaining panels report the instantaneous cost, the running time-average cost, and the SNR evolution.



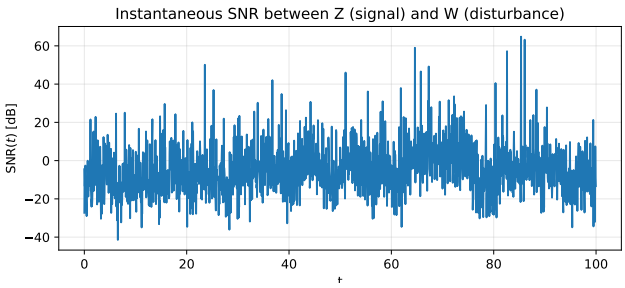
(a) Control input u



(b) Instantaneous cost



(c) Running average cost



(d) SNR evolution

Figure 9.1: Closed-loop performance under independent disturbances.

The simulation yields a final time-average cost

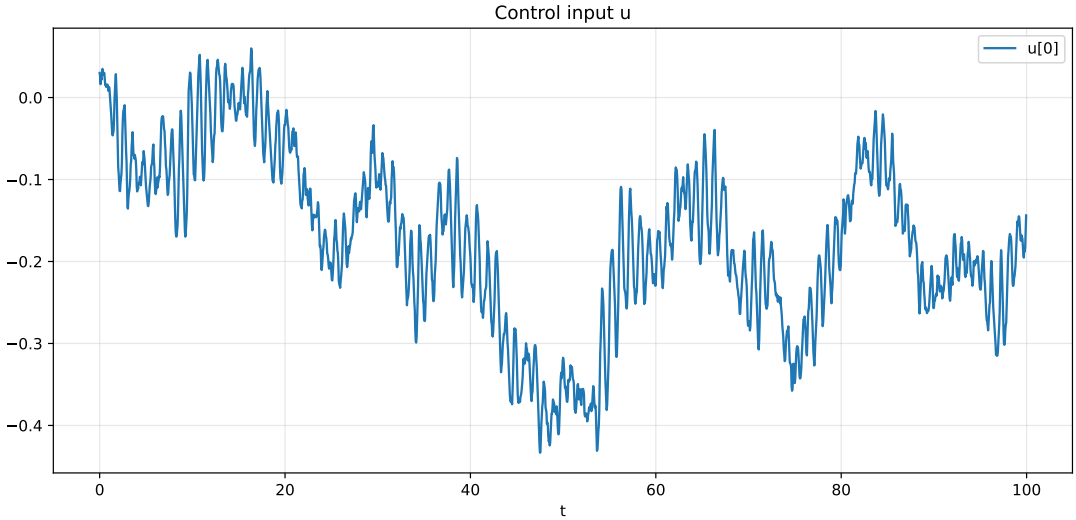
$$J_{\text{indep}} \approx 0.568,$$

together with a global signal-to-noise ratio

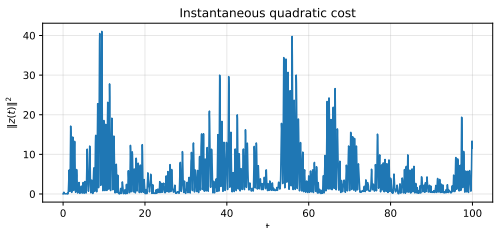
$$\text{SNR}_{\text{indep}} \approx -6.15 \text{ dB}.$$

The cost evolution shows a rapid convergence to a low steady-state value, indicating that the controller effectively attenuates the disturbance impact when the noise is temporally independent. At the same time, the negative SNR reflects that the disturbance power dominates the signal z, which is consistent with a regime where the controller aggressively suppresses output energy.

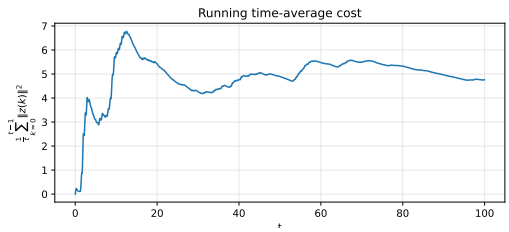
9.1.ii Correlated Disturbance Case. The same analysis is reported in Figure 9.2 for the correlated disturbance model. The structure of the figure is identical, allowing for a direct visual comparison with the independent case.



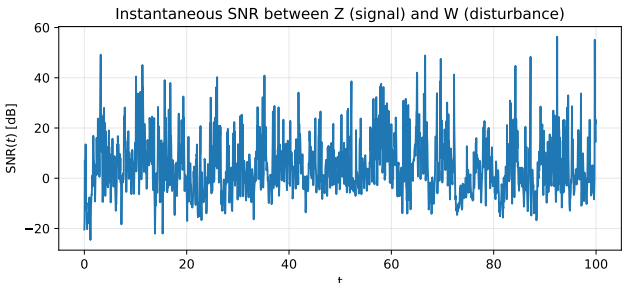
(a) Control input u



(b) Instantaneous cost



(c) Running average cost



(d) SNR evolution

Figure 9.2: Closed-loop performance under correlated disturbances.

In this case, the final performance metrics are

$$J_{\text{corr}} \approx 4.763, \qquad \text{SNR}_{\text{corr}} \approx 2.11 \text{ dB}.$$

Compared to the independent case, the cost is significantly higher, indicating a substantial degradation in closed-loop performance. This is consistent with the presence of temporal correlation in the disturbance, which introduces structured excitation that is harder for the controller to reject.

Interestingly, the SNR becomes positive, meaning that the output signal power exceeds the disturbance power. This reflects a qualitatively different regime: instead of strongly suppressing the output, the controller allows larger output excursions, which contributes to the increased cost.

9.1.iii Discussion. The comparison highlights a clear trade-off induced by the disturbance structure. Under independent noise, the controller achieves low cost and strong attenuation of

the output, at the expense of operating in a disturbance-dominated regime (negative SNR). In contrast, correlated disturbances lead to higher cost and larger output energy, but also to a higher relative signal power.

This behaviour is consistent with the theoretical differences between the two models. The independent formulation restricts the disturbance to act locally in time, which is easier to compensate for, whereas the correlated model captures more adversarial temporal patterns that propagate through the system dynamics.

Overall, these results confirm that the disturbance modeling choice has a direct and measurable impact on both performance and signal characteristics, and therefore plays a critical role in the evaluation of distributionally robust controllers.

9.2 Young-Type Relaxations: Comparative Results

This section compares the model-based design (MB) and the direct data-driven design (DD) under the two disturbance structures introduced earlier, namely the *independent* and *correlated* 2-Wasserstein settings. For each disturbance model, two data-driven relaxations are considered:

- the *isotropic* Young relaxation, based on scalar uncertainty slacks;
- the *residual-directional* relaxation, corresponding to `approach=Mats`, where the uncertainty associated with the identified state matrix is distributed along the dominant eigendirections of the residual covariance.

The goal of this section is threefold:

1. to assess the behavior of the two relaxations under independent disturbances;
2. to repeat the same analysis under correlated disturbances;
3. to identify the main trade-offs that emerge across all four data-driven configurations.

9.2.i Independent disturbances. We begin with the *independent* 2-Wasserstein disturbance setting. This corresponds to the decoupled DRO-LMI structure, where the uncertainty contribution associated with the disturbance enters through a separated ambiguity block.

Directional relaxation (Residual-based, Mats)

Setup. In this experiment, DD corresponds to the data-driven `approach=Mats` formulation, i.e., the residual-directional Young relaxation. The uncertainty associated with the identified state matrix is distributed along the dominant eigendirections of the residual covariance, thereby introducing an anisotropic uncertainty allocation.

Summary. Table 9.1 reports the main numerical indicators extracted from the simulation outputs.

Table 9.1: Residual-directional Young relaxation under independent disturbances.

Metric	**MB**	**DD (Mats)**	**DD - MB**
Cost J	9.4346	17.4583	+8.0238
Objective	14.3861	3170.5858	+3156.1997
λ	19.1761	1348.4888	+1329.3127
$\rho(A_{\mathrm{cl}})$	0.999654	1.000375	+0.000721
SNR [dB]	-1.5768	6.8520	+8.4288
Time [s]	0.3904	16.8122	+16.4218
Attempts	1	1	0
Stress	3.2021	1.2091	-1.9929
Violations [%]	0.0000	0.1765	+0.1765
Solver	MOSEK	SCS	–
Status	optimal	optimal_inaccurate	–

Performance analysis. The distributionally robust objective strongly favors the model-based design:

$$J_{\mathrm{obj}}^{\mathrm{MB}} = 14.3861, \qquad J_{\mathrm{obj}}^{\mathrm{DD}} = 3170.5858.$$

At the same time, the empirical stage cost satisfies

$$J^{\mathrm{MB}} = 9.4346, \qquad J^{\mathrm{DD}} = 17.4583.$$

Thus, both the robust objective and the realized cost are worse for the directional DD controller.

The spectral radii are

$$\rho(A_{\mathrm{cl,MB}}) = 0.999654 < 1, \qquad \rho(A_{\mathrm{cl,DD}}) = 1.000375 > 1.$$

Hence, the model-based controller is strictly stable, whereas the directional DD controller is slightly outside the unit disk.

The signal-to-noise ratios are

$$\mathrm{SNR}^{\mathrm{MB}} = -1.5768 \text{ dB}, \qquad \mathrm{SNR}^{\mathrm{DD}} = 6.8520 \text{ dB}.$$

This indicates a substantial improvement in disturbance attenuation for the data-driven controller, but this benefit is offset by the loss of robust performance and stability margin.

Numerical aspects. The model-based problem is solved by MOSEK with status `optimal` in approximately 0.39 seconds and with no reported violations. By contrast, the directional DD controller is obtained from SCS with status `optimal_inaccurate` after approximately 16.81 seconds, with six violated constraints over a total of thirty-four.

Discussion. Under independent disturbances, the directional relaxation improves the disturbance-shaping capability, as reflected by the large SNR increase, but does not translate this into a better closed-loop controller. In particular:

- the robust objective becomes dramatically larger;
- the empirical cost increases;
- the spectral radius moves slightly above one;
- numerical reliability worsens considerably.

The evidence suggests that the residual-directional relaxation is too aggressive in its current form, and that the anisotropic uncertainty allocation is not sufficiently regularized.

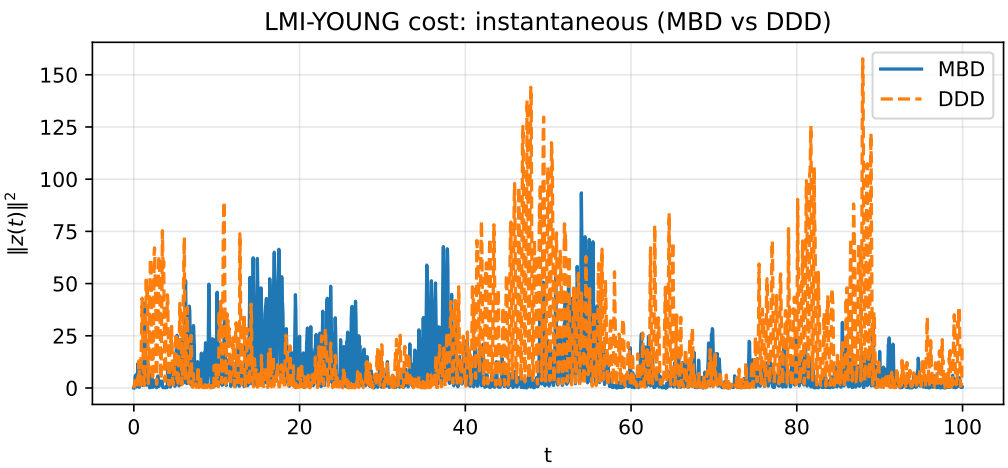


Figure 9.3: Instantaneous cost $z(t)^2$ under independent disturbances for the residual-directional relaxation.

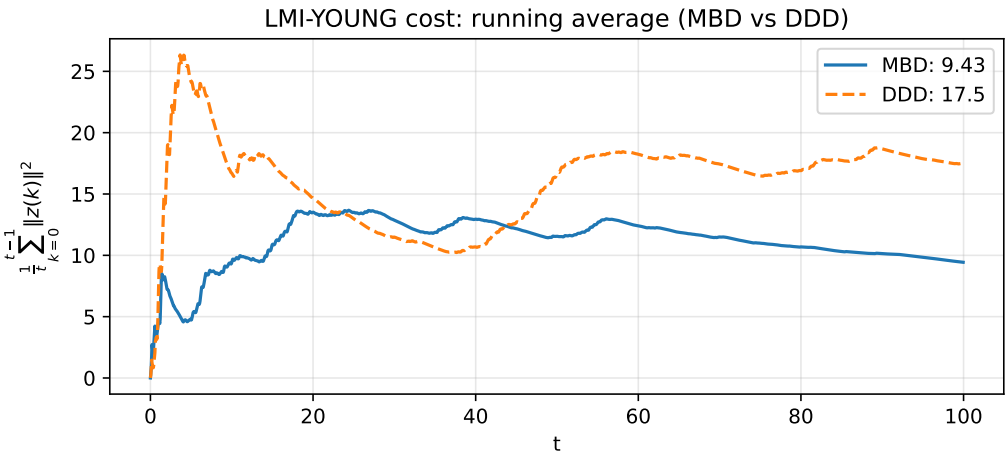


Figure 9.4: Running-average cost under independent disturbances: the directional DD controller converges to a worse empirical level than MB.

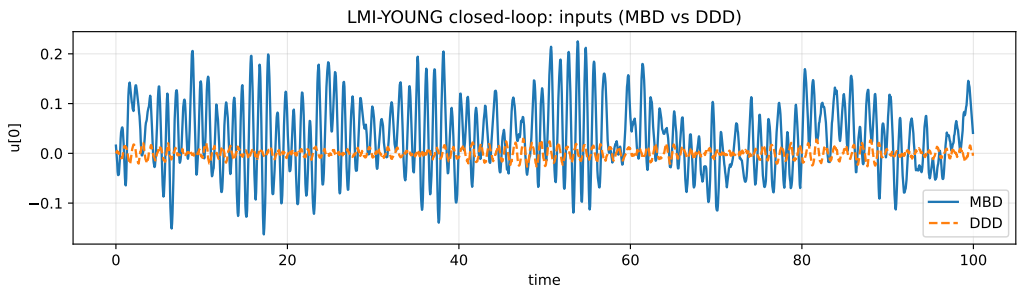


Figure 9.5: Control input comparison under independent disturbances for the residual-directional relaxation.

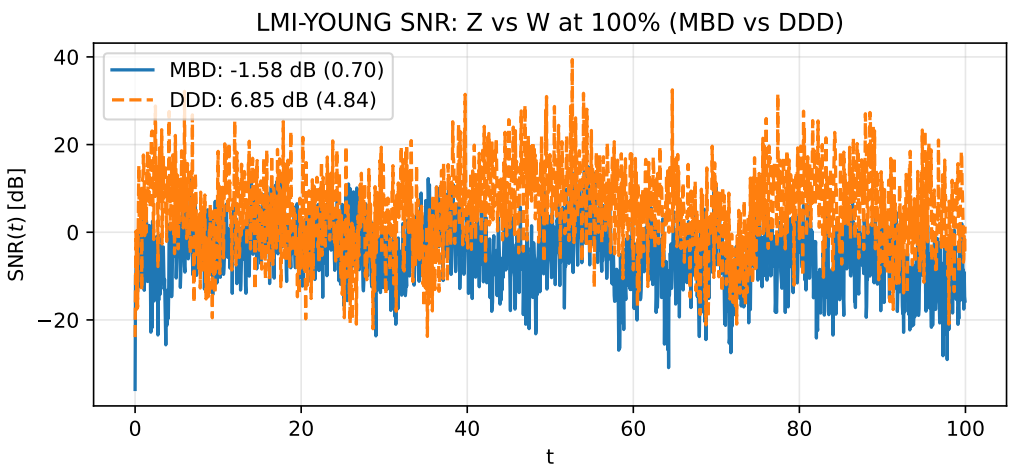


Figure 9.6: SNR comparison under independent disturbances for the residual-directional relaxation.

Isotropic relaxation

Setup. We now consider the standard isotropic Young relaxation under the same independent disturbance model. In this case, the identification uncertainty is bounded uniformly, without

directional weighting.

Summary. Table 9.2 summarizes the results.

Table 9.2: Isotropic Young relaxation under independent disturbances.

Metric	**MB**	**DD (iso)**	**DD - MB**
Cost J	5.9956	13.2509	+7.2553
Objective	14.3861	187.7242	+173.3381
λ	19.1761	82.0643	+62.8882
$\rho(A_{\mathrm{cl}})$	0.999654	0.998817	-0.000837
SNR [dB]	-3.4868	6.1144	+9.6012
Time [s]	0.3643	16.6979	+16.3336
Attempts	1	1	0
Stress	3.0881	1.1472	-1.9409
Violations [%]	0.0000	0.3529	+0.3529
Solver	MOSEK	SCS	–
Status	optimal	optimal_inaccurate	–

Performance analysis. The objective values are

$$J_{\mathrm{obj}}^{\mathrm{MB}} = 14.3861, \qquad J_{\mathrm{obj}}^{\mathrm{DD}} = 187.7242,$$

while the empirical costs are

$$J^{\mathrm{MB}} = 5.9956, \qquad J^{\mathrm{DD}} = 13.2509.$$

Thus, both criteria still favor MB, but the degradation of DD is substantially smaller than in the directional case.

The spectral radii are

$$\rho(A_{\mathrm{cl,MB}}) = 0.999654, \qquad \rho(A_{\mathrm{cl,DD}}) = 0.998817.$$

Both controllers are strictly stable, and the DD controller is slightly more contractive than the model-based one.

The SNR values are

$$\mathrm{SNR}^{\mathrm{MB}} = -3.4868 \text{ dB}, \qquad \mathrm{SNR}^{\mathrm{DD}} = 6.1144 \text{ dB}.$$

Hence, the isotropic relaxation still achieves a strong improvement in disturbance attenuation.

Numerical aspects. The MB problem is solved by MOSEK with status `optimal` in approximately 0.36 seconds and no violations. The isotropic DD problem is solved by SCS with status `optimal_inaccurate`, taking approximately 16.7 seconds and returning six violations out of seventeen constraints.

Discussion. The isotropic relaxation yields a much more balanced result than the directional one:

- stability is preserved;
- the objective remains worse than MB, but by a much smaller margin;
- the empirical cost is still worse;

- the SNR is strongly improved;
- numerical reliability remains weaker than in the model-based case.

Overall, under independent disturbances, the isotropic formulation appears to provide the best compromise among the two data-driven relaxations.

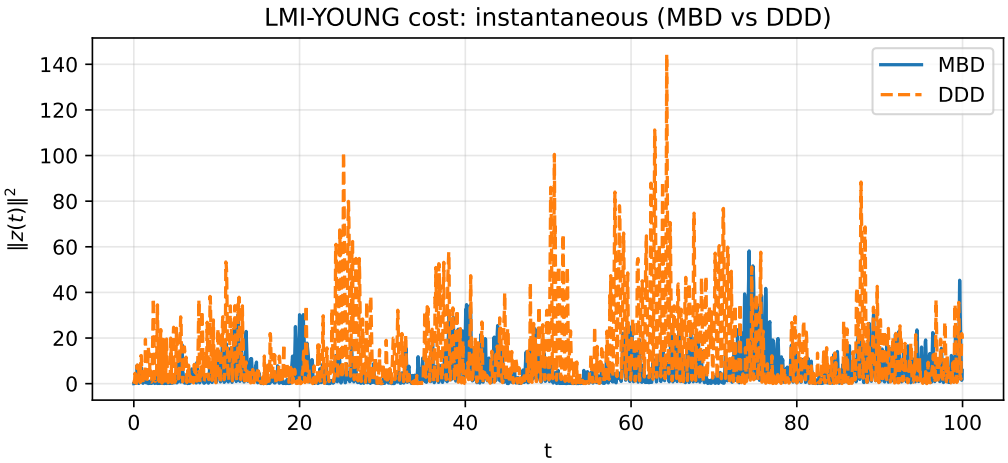


Figure 9.7: Instantaneous cost $z(t)^2$ under independent disturbances for the isotropic relaxation.

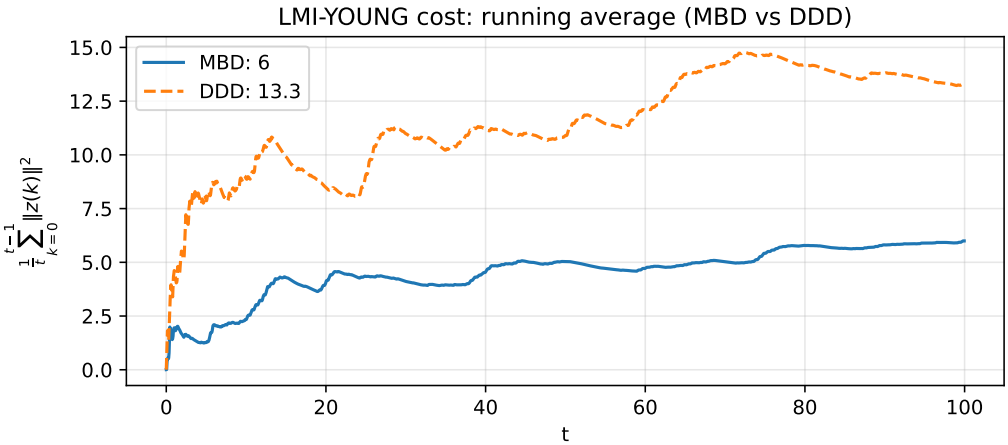


Figure 9.8: Running-average cost under independent disturbances for the isotropic relaxation.

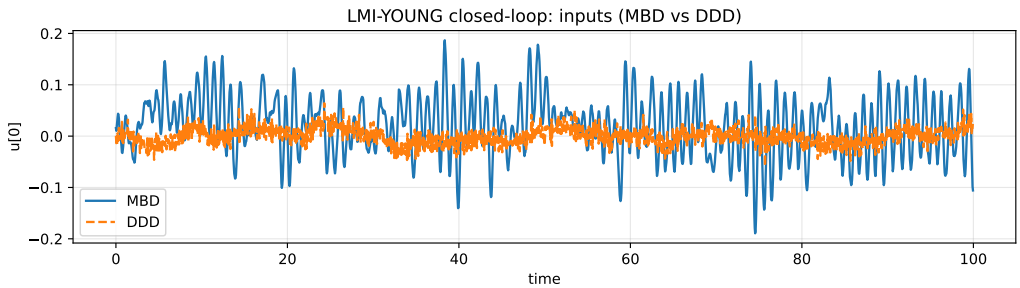


Figure 9.9: Control input comparison under independent disturbances for the isotropic relaxation.

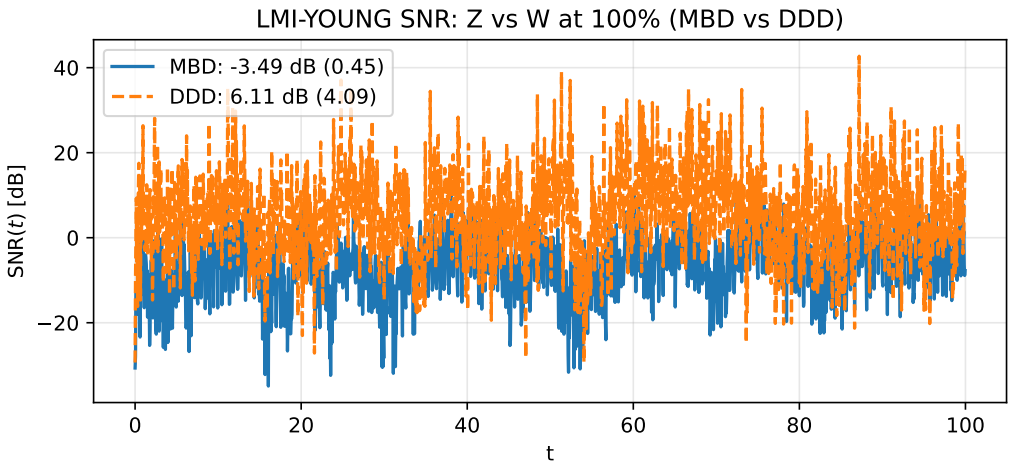


Figure 9.10: SNR comparison under independent disturbances for the isotropic relaxation.

9.2.ii Correlated disturbances. We now repeat the same analysis under the *correlated* disturbance model, corresponding to the fully coupled DRO-LMI kernel.

Directional relaxation (Residual-based, Mats)

Setup. The first correlated experiment uses the residual-directional `approach=Mats` formulation.

Summary. The main indicators are reported in Table 9.3.

Table 9.3: Residual-directional Young relaxation under correlated disturbances.

Metric	MB	DD (Mats)	DD - MB
Cost J	27.68	170.65	+142.97
Objective	878.57	6596.03	+5717.46
λ	1164.03	16588.23	+15424.20
$\rho(A_{\mathrm{cl}})$	0.9925	1.0000	+0.0075
SNR [dB]	3.14	16.52	+13.38
Time [s]	0.741	14.89	+14.15
Violations [%]	0.0000	0.12	+0.12
Solver	MOSEK	SCS	–

Performance analysis. The objective values are

$$J_{\mathrm{obj}}^{\mathrm{MB}} = 878.57, \qquad J_{\mathrm{obj}}^{\mathrm{DD}} = 6596.03,$$

and the empirical costs are

$$J^{\mathrm{MB}} = 27.68, \qquad J^{\mathrm{DD}} = 170.65.$$

Thus, under correlated disturbances, both the robust objective and the realized cost degrade severely for the directional data-driven controller.

The spectral radii are

$$\rho(A_{\mathrm{cl,MB}}) = 0.9925, \qquad \rho(A_{\mathrm{cl,DD}}) \approx 1.$$

The DD controller is therefore marginally unstable, more severely than in the independent case.

The SNR values are

$$\mathrm{SNR}^{\mathrm{MB}} = 3.14 \text{ dB}, \qquad \mathrm{SNR}^{\mathrm{DD}} = 16.52 \text{ dB}.$$

Hence, the directional relaxation produces an even stronger disturbance-shaping effect than in the independent setting.

Numerical aspects. The model-based problem is solved by MOSEK with status `optimal` in approximately 0.74 seconds and no violations. The directional DD problem is solved by SCS in approximately 14.89 seconds, again with status `optimal_inaccurate` and nonzero violations.

Discussion. The correlated setting amplifies the weaknesses already observed in the independent case:

- the robust objective becomes extremely large;
- the empirical cost deteriorates strongly;
- the stability margin is lost;
- SNR improves even further.

This is the most aggressive of all four data-driven configurations and appears to overfit the dominant uncertainty directions.

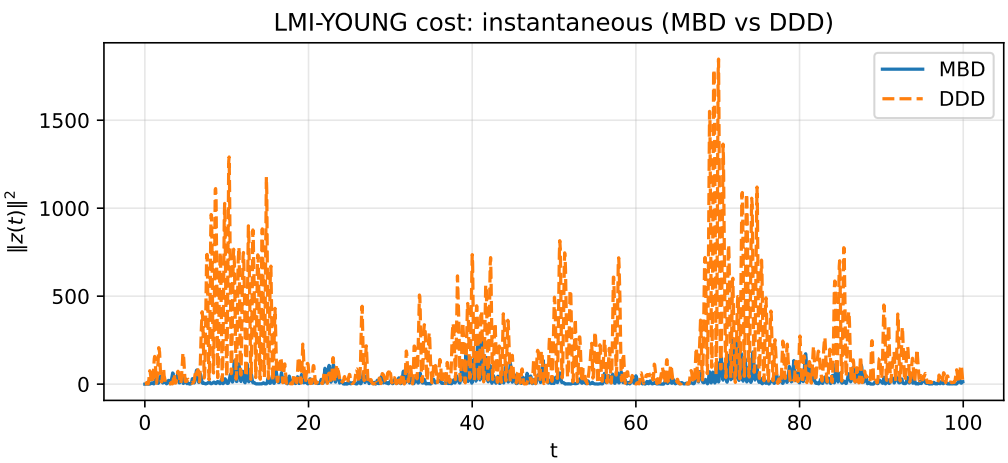


Figure 9.11: Instantaneous cost $z(t)^2$ under correlated disturbances for the residual-directional relaxation.

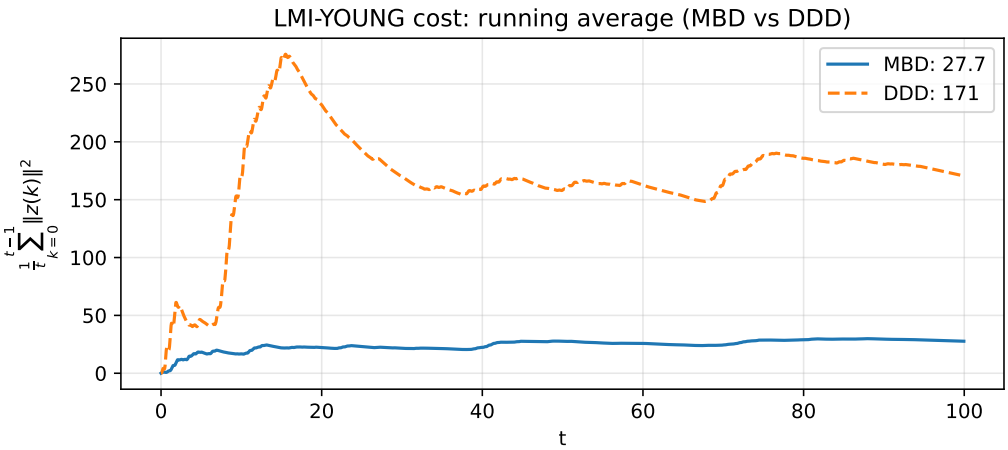


Figure 9.12: Running-average cost under correlated disturbances for the residual-directional relaxation.

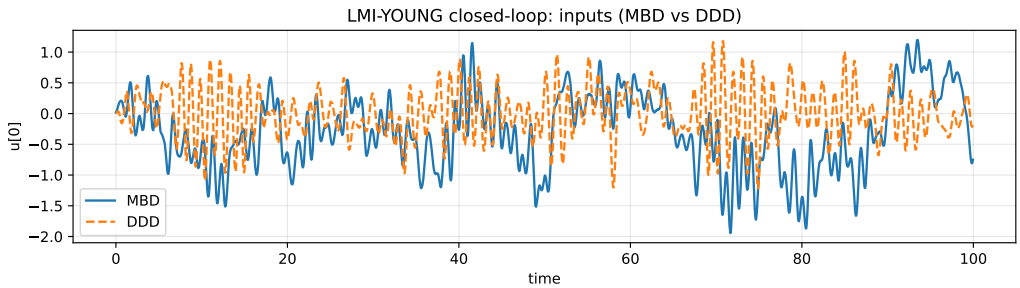


Figure 9.13: Control input comparison under correlated disturbances for the residual-directional relaxation.

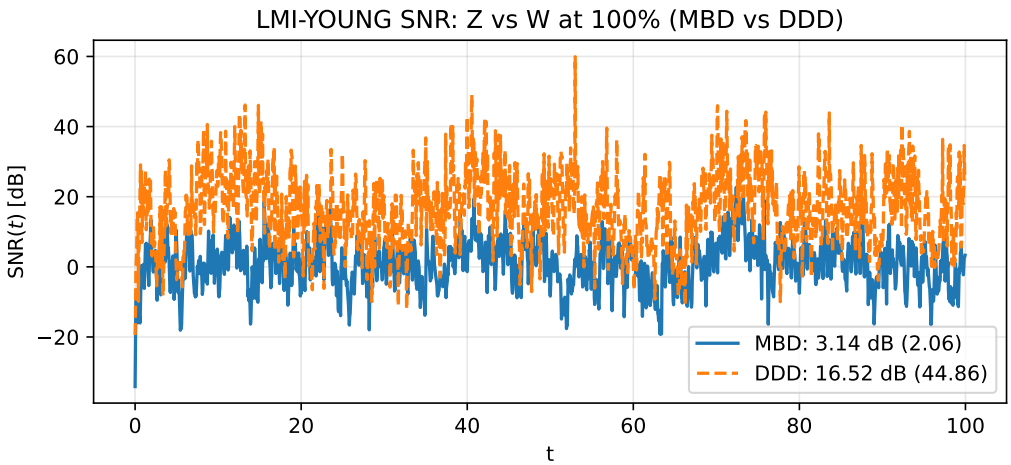


Figure 9.14: SNR comparison under correlated disturbances for the residual-directional relaxation.

Isotropic relaxation

Setup. We finally consider the isotropic Young relaxation under correlated disturbances.

Summary. The numerical indicators are reported in Table 9.4.

Table 9.4: Isotropic Young relaxation under correlated disturbances.

Metric	**MB**	**DD (iso)**	**DD - MB**
Cost J	6.73	47.72	+40.99
Objective	878.57	349.98	-528.59
λ	1164.03	290.95	-873.08
$\rho(A_{\mathrm{cl}})$	0.9925	1.0001	+0.0076
SNR [dB]	-3.09	11.35	+14.44
Time [s]	0.739	17.02	+16.28
Violations [%]	0.0000	0.375	+0.375
Solver	MOSEK	SCS	–

Performance analysis. The most striking feature of this case is that the objective values reverse their ordering:

$$J_{\mathrm{obj}}^{\mathrm{MB}} = 878.57, \qquad J_{\mathrm{obj}}^{\mathrm{DD}} = 349.98.$$

Hence, the isotropic data-driven controller achieves a *smaller* DRO objective than the model-based one.

However, the empirical cost tells the opposite story:

$$J^{\mathrm{MB}} = 6.73, \qquad J^{\mathrm{DD}} = 47.72.$$

Thus, the improved robust certificate does not translate into better realized performance.

The spectral radii are

$$\rho(A_{\mathrm{cl,MB}}) = 0.9925, \qquad \rho(A_{\mathrm{cl,DD}}) = 1.0001.$$

So the DD controller is again slightly unstable.

The SNR values are

$$\mathrm{SNR}^{\mathrm{MB}} = -3.09 \text{ dB}, \qquad \mathrm{SNR}^{\mathrm{DD}} = 11.35 \text{ dB}.$$

Hence, the isotropic relaxation still delivers a strong gain in disturbance attenuation.

Numerical aspects. The model-based design is solved by MOSEK with status `optimal`. The isotropic DD controller is obtained by SCS with status `optimal_inaccurate`, approximately 17.0 seconds of computation time, and nonzero constraint violations.

Discussion. This case is qualitatively different from all previous ones:

- the DRO objective is lower for DD;
- the empirical cost is nevertheless much worse;
- the spectral radius is slightly above one;
- the SNR is strongly improved.

Thus, under correlated disturbances, the isotropic relaxation appears less conservative from the viewpoint of the robust certificate, but this comes at the price of poorer realized regulation quality and slight loss of stability.

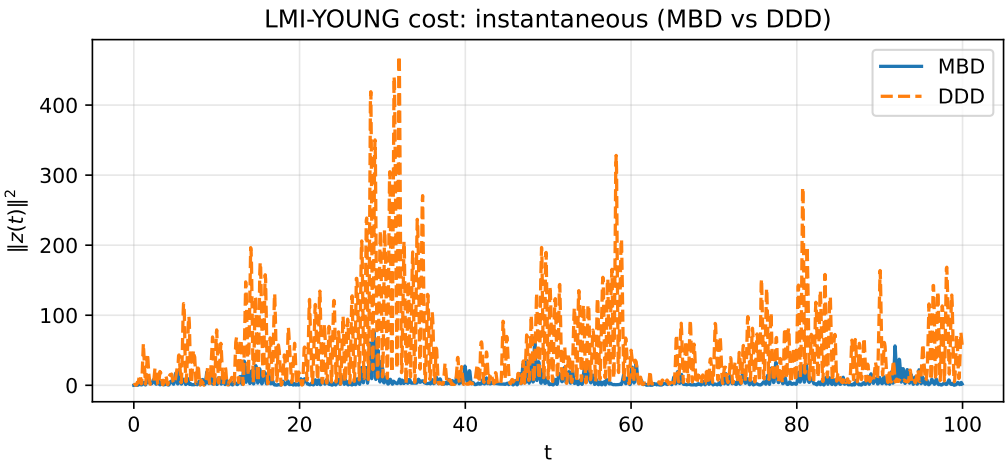


Figure 9.15: Instantaneous cost $z(t)^2$ under correlated disturbances for the isotropic relaxation.

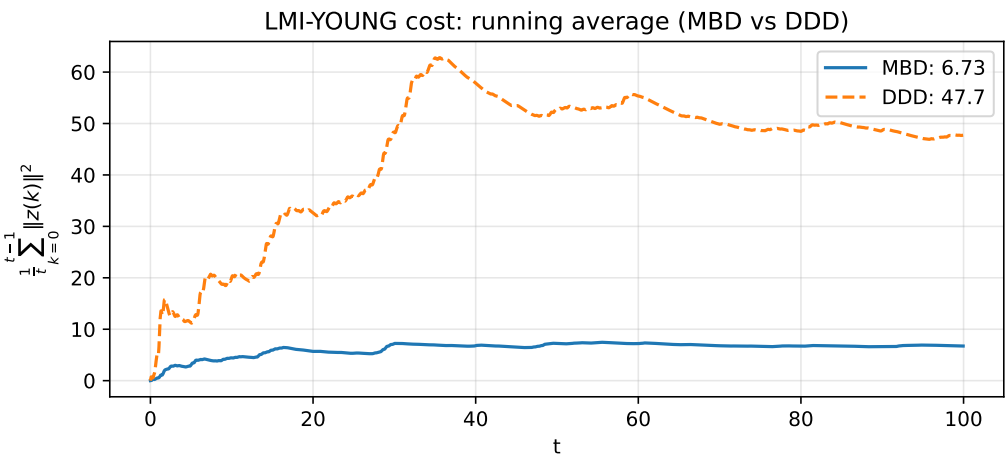


Figure 9.16: Running-average cost under correlated disturbances for the isotropic relaxation.

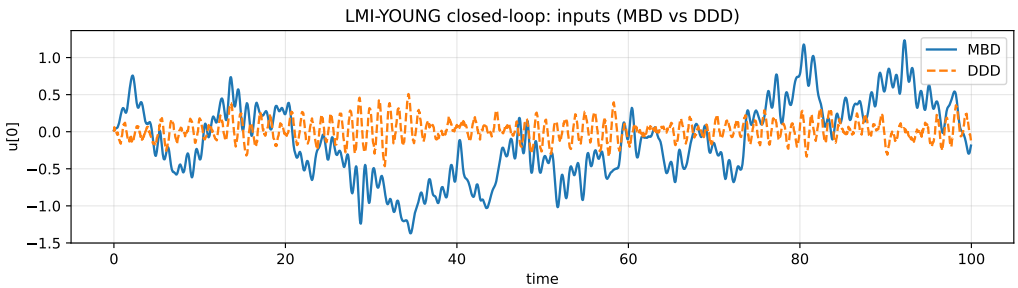


Figure 9.17: Control input comparison under correlated disturbances for the isotropic relaxation.

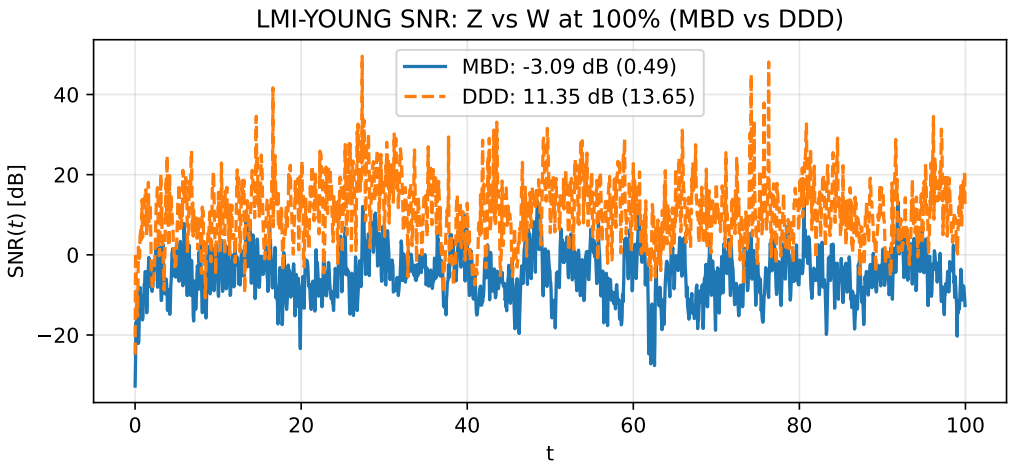


Figure 9.18: SNR comparison under correlated disturbances for the isotropic relaxation.

9.2.iii Global comparison and insights. The four data-driven configurations reveal a clear trade-off between robustness, empirical performance, stability margin, and disturbance attenuation. Table 9.5 summarizes the dominant qualitative trends.

Table 9.5: Qualitative comparison of the four Young-type data-driven relaxations.

Metric	**Indep. + Iso**	**Indep. + Dir**	**Corr. + Iso**	**Corr. + Dir**
Stability	best among DD	slightly unstable	slightly unstable	marginally unstable
Empirical cost	moderately worse than MB	worse than MB	much worse than MB	worst
DRO objective	moderately worse than MB	much worse than MB	better than MB	worst
SNR	strongly improved	strongly improved	very strongly improved	strongest improvement

Main observations. Several conclusions emerge from this comparison.

- **Directional vs isotropic.** Directional relaxations consistently achieve larger SNR gains, which indicates better shaping of the regulated output relative to the disturbance. However, this comes at the cost of much larger objective values and weaker stability margins. In both independent and correlated settings, the directional method is the most aggressive formulation.

- **Independent vs correlated disturbances.** Moving from the independent to the correlated setting amplifies the instability and numerical fragility of the data-driven controllers. The correlated case is therefore more demanding and exposes the limitations of the present relaxations more clearly.

- **Best compromise.** Among the four data-driven options, the isotropic relaxation under independent disturbances provides the most balanced behavior. It preserves closed-loop stability, improves SNR substantially, and limits the increase in the objective relative to the other data-driven configurations.

- **Most problematic case.** The directional relaxation under correlated disturbances is the most critical configuration. It yields the strongest improvement in SNR, but also the largest empirical cost, the largest robust objective, and the weakest stability margin.
- **A special case.** The isotropic relaxation under correlated disturbances is particularly interesting because it achieves a lower DRO objective than MB, yet still performs worse empirically and becomes slightly unstable. This indicates that a better robust certificate does not necessarily imply better realized regulation in the current formulation.

Interpretation. Taken together, these results suggest that uncertainty shaping along residual covariance directions is not universally beneficial. While it improves disturbance rejection in selected directions, it may also introduce excessive flexibility in sensitive dynamical modes, especially when the disturbance model is correlated.

The isotropic formulation, by contrast, is less expressive but also less aggressive. Under independent disturbances, this yields the most reliable overall compromise. Under correlated disturbances, even the isotropic relaxation becomes fragile, which indicates that additional regularization or structural constraints are needed if the data-driven Young framework is to compete consistently with the model-based baseline.

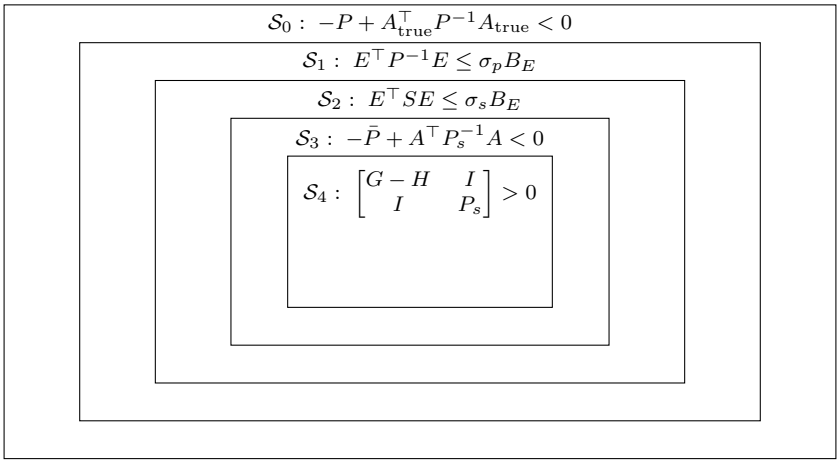
Concluding insight. The main empirical lesson of this chapter is therefore not that one data-driven relaxation uniformly dominates the others, but that the *interaction* between uncertainty structure and relaxation geometry is decisive. In particular:

- anisotropic uncertainty allocation can be beneficial for disturbance attenuation,
- but it must be controlled carefully to avoid loss of robust stability,
- and this issue becomes significantly more pronounced when the disturbance model is correlated.

These findings motivate the need for tighter regularization or more structured convex relaxations in the next developments.

9.3 Schur-Enhanced Young-Type Relaxation Results

9.3.i Geometric Interpretation of Conservatism. Before presenting the numerical results, it is useful to interpret the effect of the successive relaxations introduced in the Schur-enhanced Young formulation.



$$\mathcal{S}_4 \subset \mathcal{S}_3 \subset \mathcal{S}_2 \subset \mathcal{S}_1 \subset \mathcal{S}_0$$

Each additional bound (Young inequality, Schur lifting, inverse bounding) introduces a strict contraction of the feasible set. While convexity is preserved, the admissible controller set shrinks significantly.

This geometric contraction is the key mechanism explaining the behaviour observed in the simulations.

9.3.ii Independent Disturbance Case.

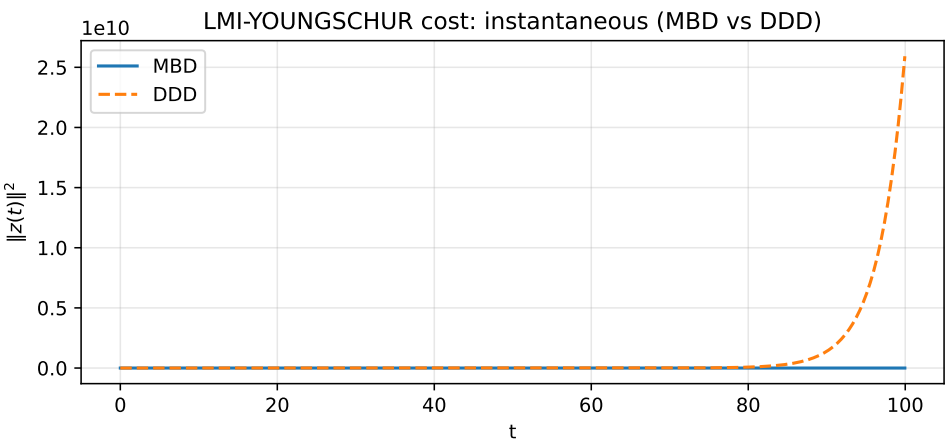


Figure 9.19: Instantaneous cost comparison (independent disturbances).

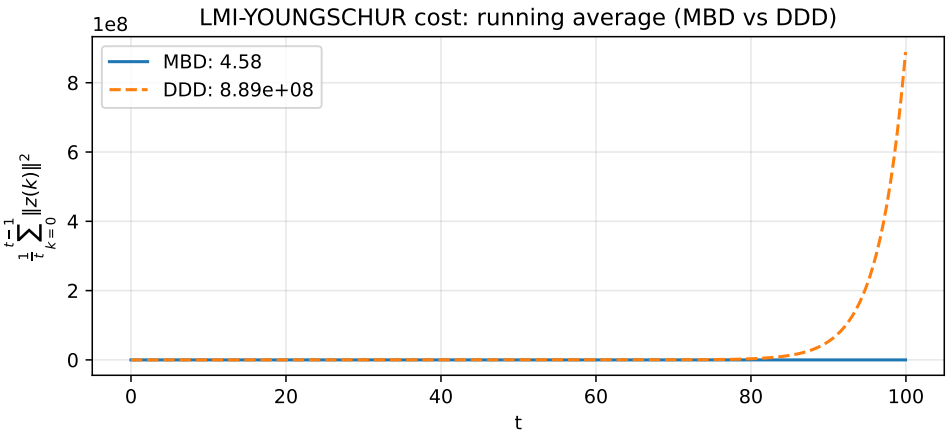


Figure 9.20: Running average cost (independent disturbances).

Closed-loop performance. The performance degradation is immediately evident. The data-driven formulation (DD) exhibits a dramatic increase in cost, reaching values on the order of 10^8–10^{10}, while the baseline controller (MBD) remains well-behaved.

The quantitative results further confirm this: the closed-loop cost explodes to extremely large values, despite the optimisation problem being solved to optimality.

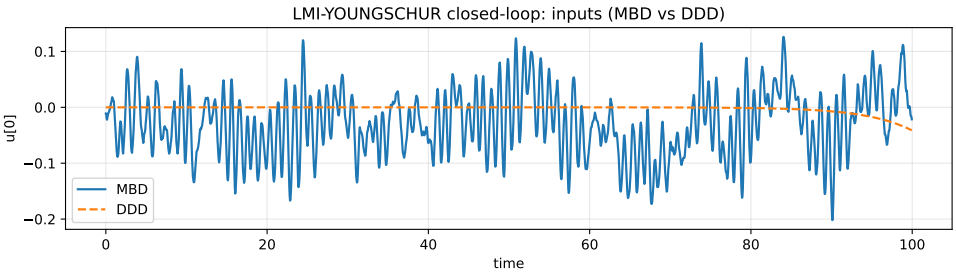


Figure 9.21: Control inputs (independent disturbances).

Control action. The control input generated by the Young-Schur solution is essentially zero over the entire horizon.

This is the crucial observation:

- the optimiser finds a *feasible* solution,
- but the solution corresponds to an almost inactive controller,
- hence the system behaves almost in open-loop.

Interpretation. The spectral radius of the closed-loop matrix exceeds one ($\rho(A_{cl}) > 1$), confirming instability.

This behaviour is not due to numerical issues or infeasibility. Instead, it is a direct consequence of the overly conservative uncertainty handling:

- the Young bounds enforce large safety margins,
- the Schur lifting further tightens admissible directions,
- the feasible controller set collapses toward low-gain solutions.

As a result, the optimiser prefers a nearly zero-control policy, which trivially satisfies the robustness constraints but fails to stabilise the system.

9.3.iii Correlated Disturbance Case.

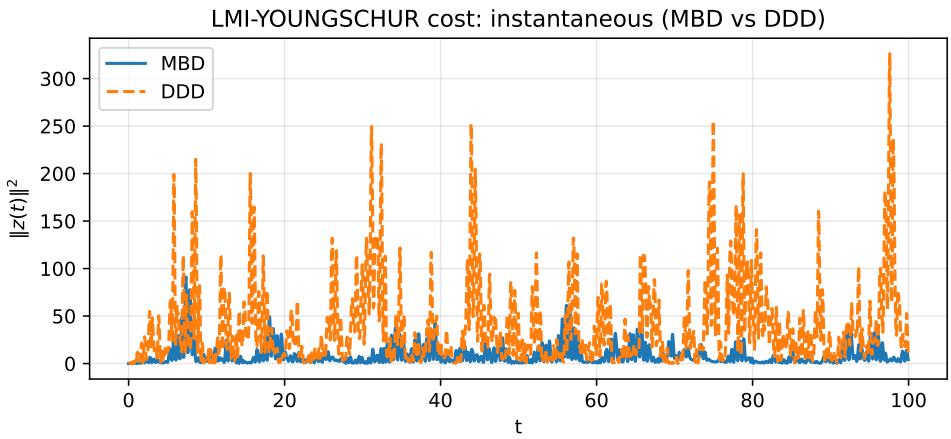


Figure 9.22: Instantaneous cost comparison (correlated disturbances).

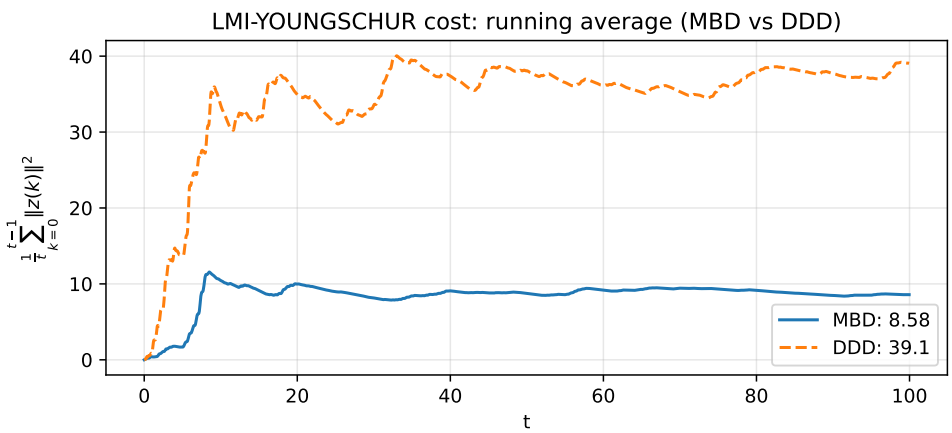


Figure 9.23: Running average cost (correlated disturbances).

Closed-loop performance. The correlated case amplifies the same phenomenon. The cost grows rapidly and remains significantly higher than the baseline solution.

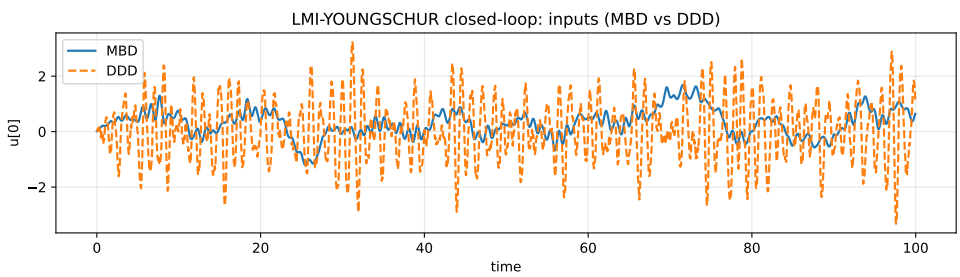


Figure 9.24: Control inputs (correlated disturbances).

Control action. Again, the control input is almost null over time, indicating that the controller is effectively inactive.

Interpretation. The instability is even more pronounced in this case. The spectral radius of the closed-loop system becomes extremely large ($\rho(A_{cl}) \gg 1$), indicating severe divergence.

This highlights a key limitation:

- correlated disturbance modeling introduces additional coupling constraints,
- these constraints interact with the Young-Schur bounds,
- the feasible set shrinks even further compared to the independent case.

The result is a highly conservative controller that is unable to inject sufficient control effort to counteract system dynamics.

9.3.iv Discussion. The numerical evidence clearly shows that the Schur-enhanced Young relaxation, while convex and theoretically sound, is excessively restrictive in practice.

Key observation. Convexity is achieved at the cost of a severe reduction of the feasible controller set.

Mechanism. The restriction arises from the cumulative effect of:

- Young-type bounds on identification errors,
- spectral norm constraints on decision variables,
- Schur complement liftings for inverse bounding,
- distributionally robust constraints on disturbances.

Each step individually preserves convexity, but together they lead to:

$$\mathcal{S}_4 \subset \mathcal{S}_3 \subset \mathcal{S}_2 \subset \mathcal{S}_1 \subset \mathcal{S}_0,$$

with $\mathcal{S}_4$ being extremely small.

Consequence. The optimizer converges to solutions that:

- minimize the worst-case bound,
- but effectively suppress control action,
- resulting in unstable closed-loop behavior.

Conclusion. The Schur-enhanced Young formulation demonstrates that:

convexity alone is not sufficient—overly conservative relaxations can destroy control performance.

This limitation motivates the development of alternative formulations that retain data-driven structure while avoiding excessive shrinkage of the feasible set, as explored in the subsequent chapter.

Part I

Preliminaries and Background

A Additional Experimental Details

A.1 Detailed Data-Driven Experimental Pipeline

Figure A.1 reports the detailed workflow used in the data-driven branch of the experiments, from the generation of multiple open-loop trajectories to the estimation of plant and disturbance quantities and the subsequent execution of the DRO_LMI synthesis pipeline.

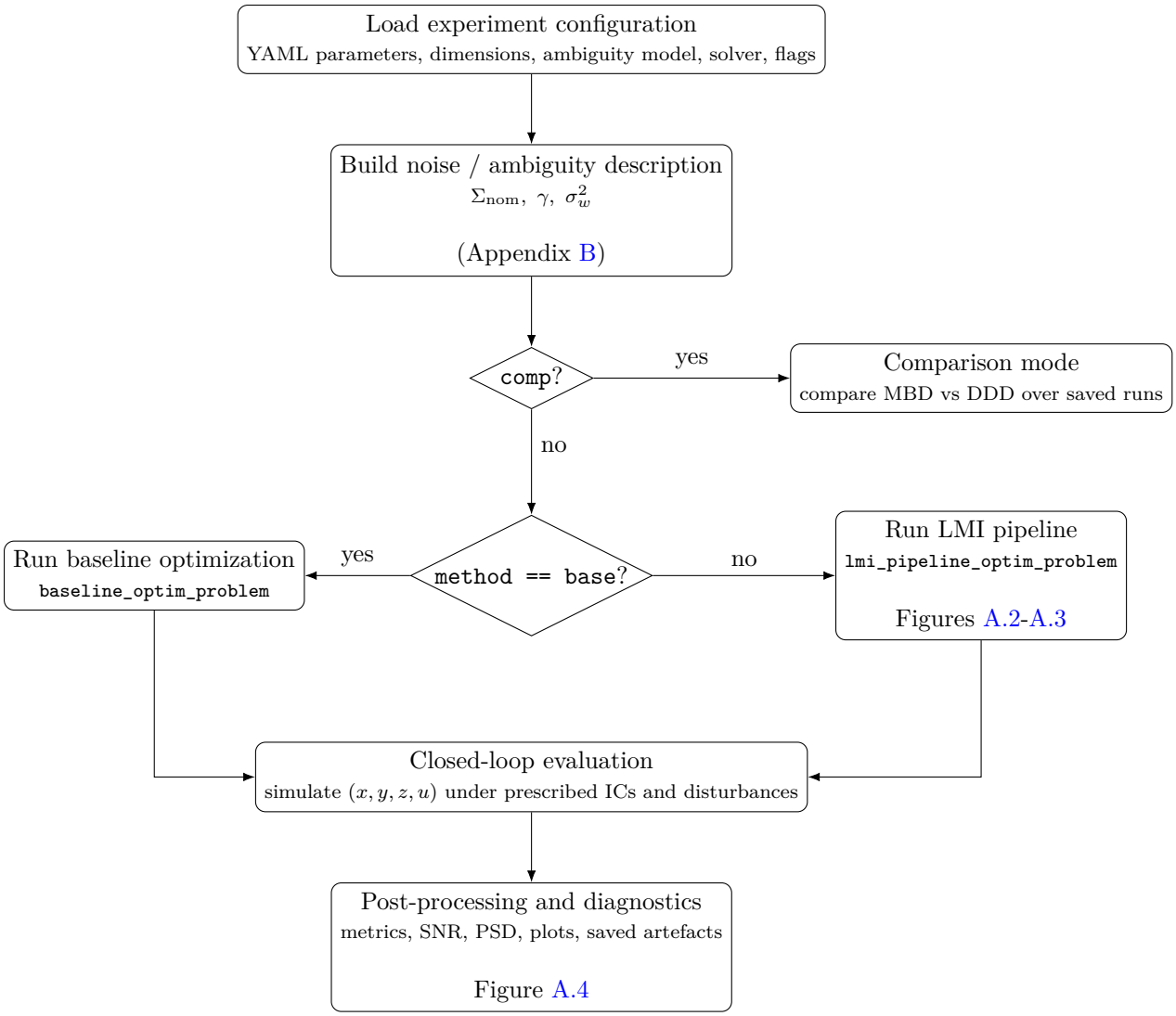


Figure A.1: High-level execution flow of the experimental framework. The main script loads the configuration, defines the ambiguity model, then either runs a comparison routine or executes a single optimisation pipeline. In single-run mode, the framework dispatches either to the baseline optimiser or to the LMI-based synthesis pipeline, followed by closed-loop evaluation and post-processing.

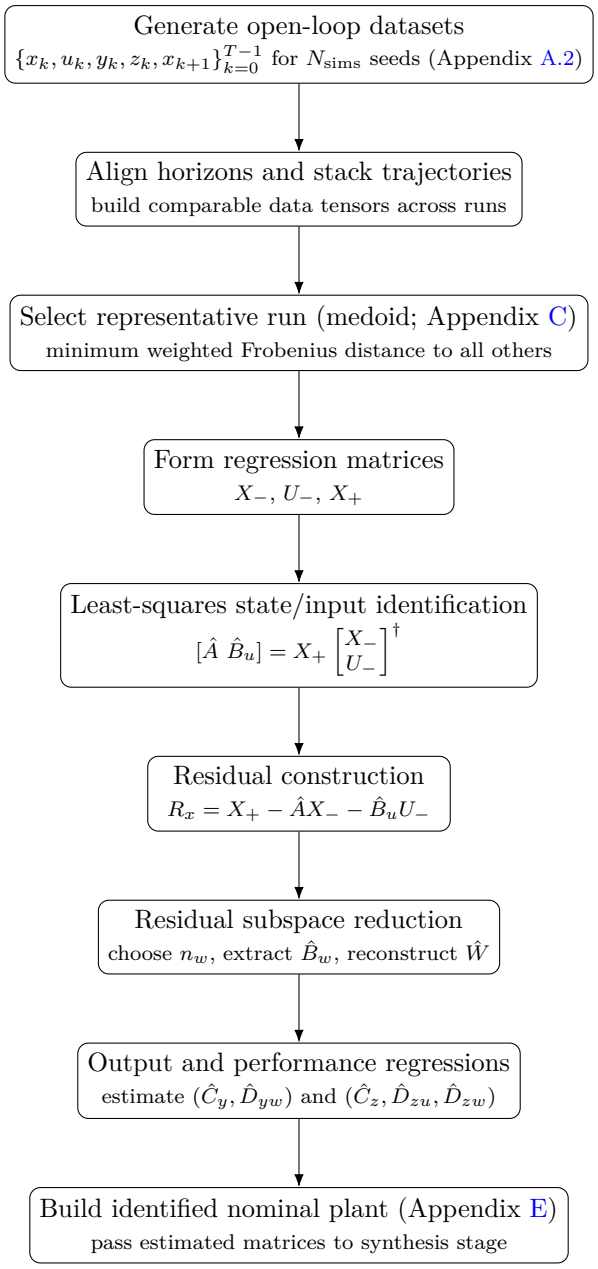


Figure A.2: Detailed data-driven identification workflow. Open-loop datasets are generated for multiple seeds, aligned, and reduced to a representative medoid trajectory. Least-squares regression is then used to estimate the nominal state/input dynamics, while residual analysis provides an effective disturbance channel and reduced-order disturbance realisation used in the subsequent output and performance regressions.

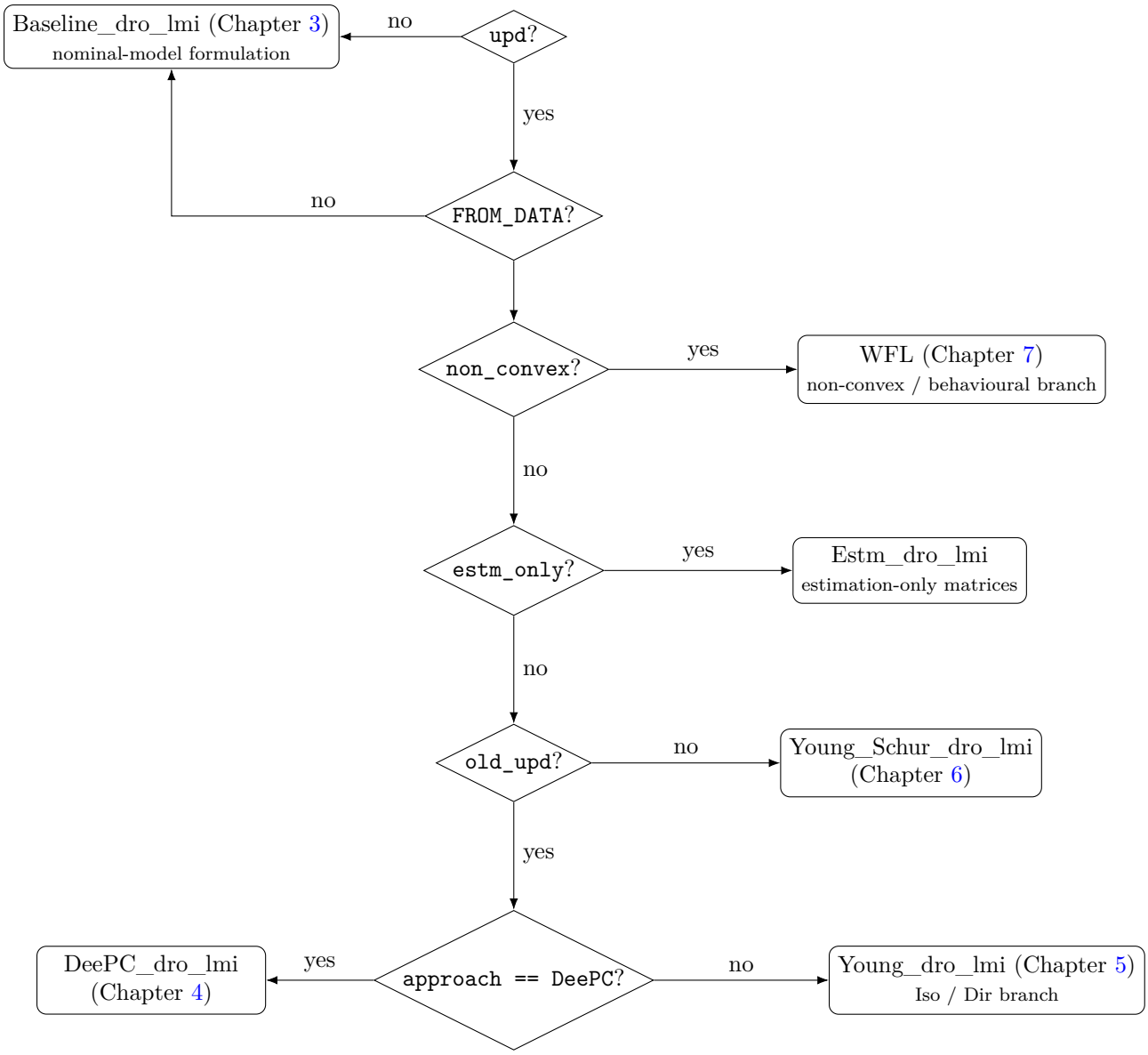


Figure A.3: Method-selection logic inside the LMI synthesis pipeline. The implemented routine dispatches among baseline, DeePC, Young, Young-Schur, estimation-only, and WFL variants according to the configuration flags defined in the parameter file and interpreted by the LMI pipeline.

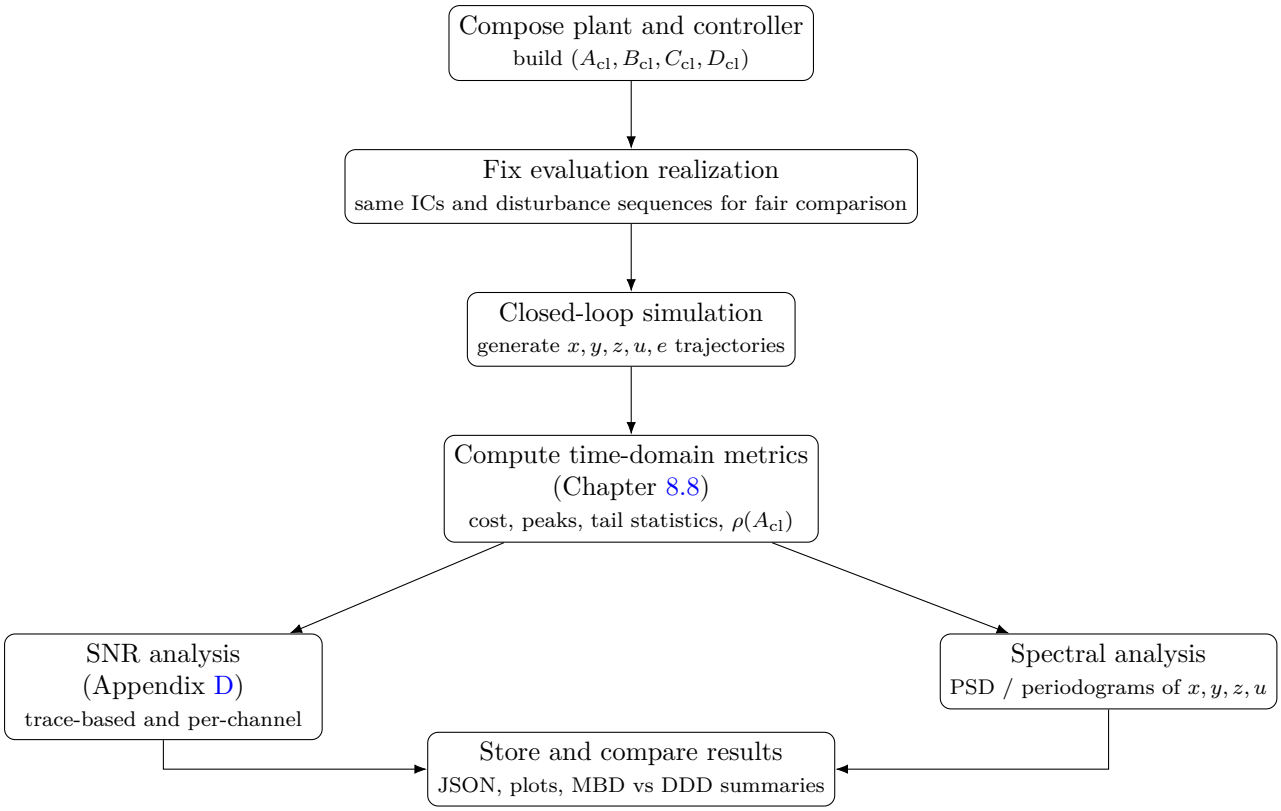


Figure A.4: Evaluation and diagnostics workflow. After synthesis, the plant and controller are interconnected into a closed-loop model, simulated under matched initial conditions and disturbance realizations, and then analyzed through time-domain metrics, SNR indicators, spectral plots, and cross-method summaries.

A.2 Eigenvector-Based Choice of Initial Conditions

The closed-loop simulations are carried out on the augmented state

$$\xi_k = \begin{bmatrix} x_k^\top & (x_k^c)^\top \end{bmatrix}^\top,$$

where x_k is the plant state and x_k^c is the controller state. Instead of drawing the initial condition at random, we construct x_0 and x_0^c so as to excite the most critical dynamical modes of the plant and the controller.

Let $A \in \mathbb{R}^{n_x \times n_x}$ denote the plant state matrix and $A_c \in \mathbb{R}^{n_{x^c} \times n_{x^c}}$ the controller state matrix obtained from the DRO_LMI synthesis. We compute their eigendecompositions

$$AV = V\Lambda, \qquad A_c V_c = V_c \Lambda_c,$$

and select, in each case, the eigenvalue with the largest real part (i.e. the least damped mode),

$$\lambda_* = \arg\max_{\lambda \in \sigma(A)} \Re(\lambda), \qquad \lambda_*^{(c)} = \arg\max_{\lambda \in \sigma(A_c)} \Re(\lambda),$$

with corresponding right eigenvectors $v_* \in \mathbb{C}^{n_x}$, $v_*^{(c)} \in \mathbb{C}^{n_{x^c}}$. Since the model is real, eigenvalues come in complex–conjugate pairs; we build real mode shapes by taking the real part and normalizing,

$$\tilde{v}_* := \frac{\Re(v_*)}{\|\Re(v_*)\|_2}, \qquad \tilde{v}_*^{(c)} := \frac{\Re(v_*^{(c)})}{\|\Re(v_*^{(c)})\|_2}.$$

The initial plant and controller states are then defined as

$$x_0 = r_x\,\tilde{v}_*, \qquad x_0^c = r_c\,\tilde{v}_*^{(c)},$$

with small scalar amplitudes $r_x, r_c > 0$ chosen so that the The resulting deviations remain in the regime where the linear model is valid (e.g. $\|x_0\|_2, \|x_0^c\|_2$ on the order of a few percent of the nominal state magnitudes). The augmented initial condition is $\xi_0 = (x_0, x_0^c)$.

This eigenvector–based choice has two advantages compared to drawing x_0 and x_0^c as random Gaussian vectors:

- it guarantees that the initial condition is aligned with the least damped (i.e. most critical) modes of the plant and controller, which makes the transient a meaningful test of closed–loop damping and robustness;
- it removes the variability induced by arbitrary random initial conditions: all controllers are evaluated starting from the same structured perturbation, so differences in performance cannot be attributed to a "lucky" or "unlucky" random draw.

Random initial conditions are still useful for Monte–Carlo stress testing, but the eigenvector–based initialization provides a reproducible and deliberately challenging operating point for the comparisons presented here.

Given a state matrix $M \in \mathbb{R}^{n\times n}$:

1. Compute eigenvalues and eigenvectors:

$$MV = V\Lambda.$$

2. Select the eigenvalue with largest real part:

$$\lambda_* = \arg\max_{\lambda_i \in \sigma(M)} \Re(\lambda_i).$$

3. Extract the corresponding eigenvector v_*.
4. Build a real mode shape:

$$\tilde{v} = \frac{\Re(v_*)}{\|\Re(v_*)\|_2}.$$

5. Scale it to obtain the initial condition:

$$x_0 = r\,\tilde{v}, \qquad r \text{ small (e.g., } r = 0.05).$$

B Wasserstein Ambiguity Set and Disturbance Evaluation

The disturbance sequences w_t used in both the baseline and the distributionally robust simulations are generated from a *2-Wasserstein ambiguity set* [3], [4] centered at a nominal Gaussian model. This construction captures uncertainty in the second-order statistics of the process noise while preserving a tractable probabilistic structure. In particular, the ambiguity set allows one to perturb the nominal covariance within a prescribed Wasserstein radius and then generate disturbance trajectories that are either temporally independent or temporally correlated, while remaining consistent with the same marginal uncertainty description.

B.1 Gaussian Wasserstein Ambiguity Set

We fix a nominal zero-mean Gaussian distribution

$$P_{\text{nom}} = \mathcal{N}(0, \Sigma_{\text{nom}}), \qquad \Sigma_{\text{nom}} \in \mathbb{S}^{n_w}_{++}.$$

For two zero-mean Gaussian distributions

$$P_1 = \mathcal{N}(0, \Sigma_1), \qquad P_2 = \mathcal{N}(0, \Sigma_2),$$

the squared 2-Wasserstein distance is

$$W_2^2(P_1, P_2) = \operatorname{tr}(\Sigma_1) + \operatorname{tr}(\Sigma_2) - 2\ \operatorname{tr}\Big[\big(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2}\big)^{1/2}\Big].$$

Equivalently,

$$W_2^2(\mathcal{N}(0, S_1), \mathcal{N}(0, S_2)) = \operatorname{tr}(S_1 + S_2) - 2\ \operatorname{tr}\Big[\big(S_2^{1/2}S_1S_2^{1/2}\big)^{1/2}\Big].$$

The corresponding closed Wasserstein ball of radius $\gamma > 0$ around the nominal distribution is

$$\mathcal{B}_\gamma = \Big\{\mathcal{N}(0, \Sigma) \;\Big|\; \Sigma \in \mathbb{S}^{n_w}_{++},\ W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \le \gamma\Big\}.$$

Hence, the Gaussian ambiguity set used in simulation is

$$\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) = \Big\{\mathcal{N}(0, \Sigma) \;\Big|\; W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \le \gamma\Big\},$$

that is, the family of all zero-mean Gaussian distributions whose covariance lies within Wasserstein distance γ from Σ_{nom}.

B.2 Projection of a Candidate Covariance onto the Wasserstein Ball

Given a candidate covariance

$$\Sigma_{\text{raw}} \in \mathbb{S}^{n_w}_{++},$$

we first evaluate its distance to the nominal model:

$$d_{\text{raw}} = W_2(\mathcal{N}(0, \Sigma_{\text{raw}}), \mathcal{N}(0, \Sigma_{\text{nom}}))\,.$$

If $d_{\text{raw}} \leq \gamma$, then the covariance is already admissible and we simply set

$$\Sigma_{\text{use}} = \Sigma_{\text{raw}}.$$

Otherwise, the covariance is projected onto the boundary of the Wasserstein ball along the Bures–Wasserstein geodesic joining Σ_{raw} and Σ_{nom}. Define

$$S = \Sigma_{\text{raw}}^{1/2}, \qquad A = \left(S\Sigma_{\text{nom}}S\right)^{1/2},$$

and consider the path

$$\Sigma(t) = \left((1-t)S + tA\right)^2, \qquad t \in [0,1].$$

Then one finds $t^\star \in (0,1]$ such that

$$W_2(\mathcal{N}(0, \Sigma(t^\star)), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma,$$

and defines

$$\Sigma_{\text{use}} := \Sigma(t^\star).$$

In the implementation, $t^\star$ is obtained by scalar bisection. Therefore, every covariance used for sampling either belongs to the interior of the ball or is the boundary projection of a raw candidate.

B.3 Independent and Correlated Disturbance Models

Once an admissible covariance $\Sigma_{\text{use}} \in \mathcal{B}_\gamma$ has been constructed, two different temporal disturbance models are considered. The key point is that they share the same one-step marginal distribution but differ in temporal dependence.

B.3.i Independent disturbances. In the independent case, the disturbance sequence is generated as

$$w_t \sim \mathcal{N}(0, \Sigma_{\text{use}}), \qquad t = 0, \ldots, T-1,$$

with independence across time. Equivalently, if

$$\Sigma_{\text{use}} = LL^\top$$

is a Cholesky factorization and $z_t \sim \mathcal{N}(0, I_{n_w})$, then

$$w_t = Lz_t, \qquad t = 0, \ldots, T-1.$$

This defines an iid Gaussian sequence and corresponds to the ambiguity class

$$\mathcal{W}_{\text{ind}} = \Big\{\{w_t\}_{t\geq 0} \;\Big|\; w_t \text{ independent across } t,\; P_{w_t} \in \mathcal{W}_2(\Sigma_{\text{nom}}, \gamma)\Big\}.$$

B.3.ii Correlated disturbances. In the correlated case, temporal dependence is introduced through a Gaussian AR(1) process:

$$w_{t+1} = Fw_t + \varepsilon_t, \qquad F = \rho I_{n_w}, \qquad |\rho| < 1,$$

with Gaussian innovations

$$\varepsilon_t \sim \mathcal{N}(0, Q).$$

The matrix Q is chosen so that the stationary covariance equals Σ_{use}. This is enforced by the discrete-time Lyapunov equation

$$\Sigma_{\text{use}} = F\Sigma_{\text{use}}F^\top + Q,$$

which gives

$$Q = \Sigma_{\text{use}} - F\Sigma_{\text{use}}F^\top.$$

Since $F = \rho I_{n_w}$, this simplifies to

$$Q = (1 - \rho^2)\Sigma_{\text{use}}.$$

The process is initialized in stationarity as

$$w_0 \sim \mathcal{N}(0, \Sigma_{\text{use}}),$$

and then propagated according to

$$w_{t+1} = \rho w_t + \varepsilon_t.$$

This defines the correlated ambiguity class

$$\mathcal{W}_{\text{cor}} = \Big\{ \{w_t\}_{t\geq 0} \;\Big|\; P_{w_t} \in \mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) \text{ for all } t \Big\},$$

where the marginal distribution remains admissible at each time, but temporal independence is no longer imposed.

B.3.iii What changes and what does not. Both disturbance models satisfy

$$\mathbb{E}[w_t] = 0, \qquad \text{Cov}(w_t) = \Sigma_{\text{use}},$$

hence they share the same one-step Gaussian marginals. Their difference lies only in temporal covariance:

$$\text{iid:} \qquad \text{Cov}(w_{t+k}, w_t) = 0 \quad (k \neq 0),$$

whereas

$$\text{AR(1):} \qquad \text{Cov}(w_{t+k}, w_t) = \rho^{|k|}\Sigma_{\text{use}}.$$

Thus, the independent and correlated samplers are equivalent from the viewpoint of marginal distribution, but not from the viewpoint of time dependence.

B.4 Sampling Procedure

The overall sampling logic is summarized below.

Algorithm 4 Sampling Gaussian disturbances inside a 2-Wasserstein ball

Require: nominal covariance $\Sigma_{\text{nom}} \in \mathbb{S}^{n_w}_{++}$, radius $\gamma > 0$, time horizon T, mode $\in \{\texttt{independent}, \texttt{correlated}\}$, correlation coefficient $\rho \in (0, 1)$

Choose a candidate covariance $\Sigma_{\text{raw}} \in \mathbb{S}^{n_w}_{++}$

Compute

$$d_{\text{raw}} = W_2(\mathcal{N}(0, \Sigma_{\text{raw}}), \mathcal{N}(0, \Sigma_{\text{nom}}))$$

if $d_{\text{raw}} \leq \gamma$ **then**

  $\Sigma_{\text{use}} \leftarrow \Sigma_{\text{raw}}$

else

  Compute

$$S \leftarrow \Sigma_{\text{raw}}^{1/2}, \qquad A \leftarrow (S\Sigma_{\text{nom}}S)^{1/2}$$

  Define

$$\Sigma(t) = ((1-t)S + tA)^2, \qquad t \in [0, 1]$$

  Find $t^\star$ by bisection such that

$$W_2(\mathcal{N}(0, \Sigma(t^\star)), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma$$

  $\Sigma_{\text{use}} \leftarrow \Sigma(t^\star)$

end if

if mode = independent **then**

  Compute $\Sigma_{\text{use}} = LL^\top$

  for $t = 0, \ldots, T-1$ **do**

    Draw $z_t \sim \mathcal{N}(0, I_{n_w})$

    $w_t \leftarrow Lz_t$

  end for

else

  Set $F \leftarrow \rho I_{n_w}$

  Set $Q \leftarrow \Sigma_{\text{use}} - F\Sigma_{\text{use}}F^\top$

  Draw $w_0 \sim \mathcal{N}(0, \Sigma_{\text{use}})$

  for $t = 0, \ldots, T-2$ **do**

    Draw $\varepsilon_t \sim \mathcal{N}(0, Q)$

    $w_{t+1} \leftarrow Fw_t + \varepsilon_t$

  end for

end if

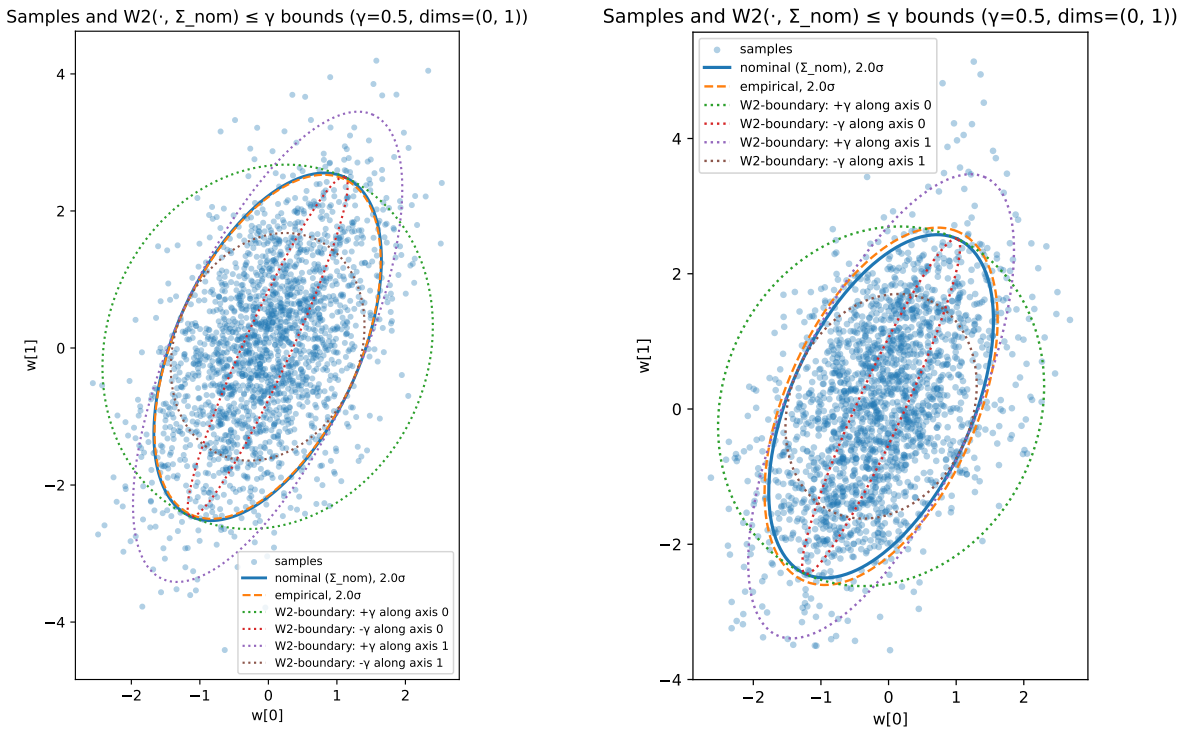
return $\{w_t\}_{t=0}^{T-1}$

B.5 Disturbance Evaluation Through Figures

The following figures illustrate the behavior of the two samplers using the newly generated disturbance-evaluation plots.

B.5.i Geometric visualization of the Wasserstein covariance set. Before examining the sampled time series, it is useful to visualize the geometry of the admissible covariance perturbations in the disturbance plane. Figures B.1a and B.1b show two-dimensional sample clouds together with the nominal covariance ellipse, the empirical covariance ellipse obtained from the generated samples, and several reference ellipses lying on the boundary of the Wasserstein ball.

In both figures, the solid blue ellipse represents the nominal Gaussian covariance Σ_{nom}, while the dashed orange ellipse represents the empirical covariance estimated from the generated disturbance samples. The dotted outer and inner ellipses correspond to covariance perturbations



(a) Independent case: empirical covariance ellipse and Wasserstein boundary directions.

(b) Correlated case: empirical covariance ellipse and Wasserstein boundary directions.

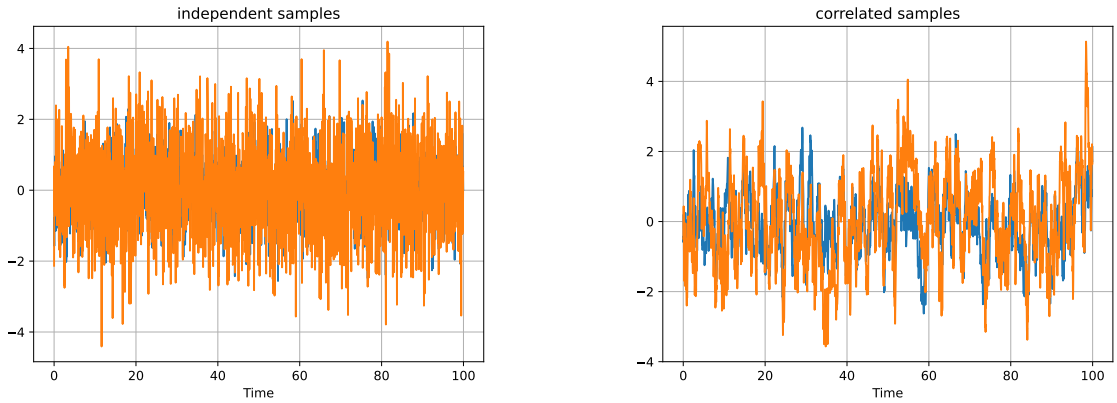
Figure B.1: Geometric interpretation of the Gaussian Wasserstein ambiguity set in the (w_0, w_1) plane. The solid ellipse denotes the nominal covariance, the dashed ellipse the empirical covariance of the generated samples, and the dotted curves illustrate representative covariance perturbations on the Wasserstein boundary for radius $\gamma = 0.5$.

that lie on the boundary of the Wasserstein ball, obtained by moving in positive and negative directions along the selected coordinate axes. In this way, the plots provide a geometric interpretation of the radius γ: rather than defining a pointwise bound on individual samples, the Wasserstein constraint defines an admissible region in covariance space around the nominal Gaussian model.

The main observation is that the empirical covariance remains close to the nominal one and stays inside the family of admissible Wasserstein perturbations. The sample cloud may therefore vary in spread, elongation, and orientation, but only within a controlled covariance neighborhood prescribed by the ambiguity radius. This clarifies the role of the Wasserstein set in the simulation pipeline: the uncertainty acts at the level of second-order distributional structure, not through direct clipping or hard bounds on the disturbance realizations.

The two plots should not be interpreted as confidence ellipses for individual realisations, but rather as projections of admissible covariance models onto the (w_0, w_1) plane. Their purpose is to show how the Wasserstein ball constrains covariance deformation around Σ_{nom}. In particular, the admissible perturbations may enlarge, shrink, or anisotropically deform the nominal ellipse, but only up to the prescribed transportation radius γ. This gives a direct visual meaning to the ambiguity set used later for disturbance generation and robust evaluation.

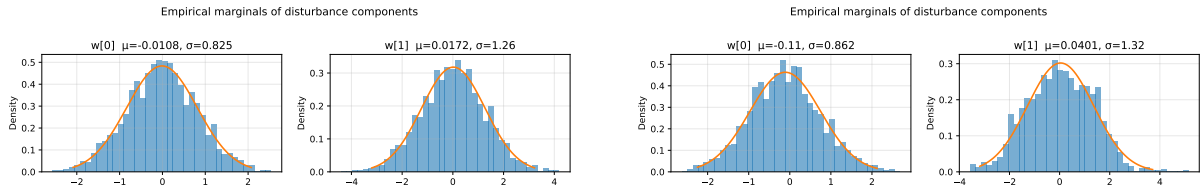
B.5.ii Time-domain samples. Figure B.2 compares representative time traces of the independent and correlated samplers. Although both are centered at zero and share the same stationary covariance, the correlated trajectory exhibits visibly smoother persistence over time, while the iid trajectory fluctuates without serial structure.



(a) Independent samples.

(b) Correlated AR(1) samples.

Figure B.2: Representative disturbance trajectories in time. The iid case shows no temporal persistence, whereas the correlated case exhibits smooth serial dependence.



(a) Independent disturbances: empirical marginal densities.

(b) Correlated disturbances: empirical marginal densities.

Figure B.3: Empirical one-step marginal PDFs. The two samplers share the same Gaussian marginal structure, hence the histograms are nearly indistinguishable.

B.5.iii Marginal PDFs. Figure B.3 shows the empirical marginal distributions of the disturbance components. As expected, the iid and correlated samplers produce nearly identical one-step marginals, since both are constructed to satisfy

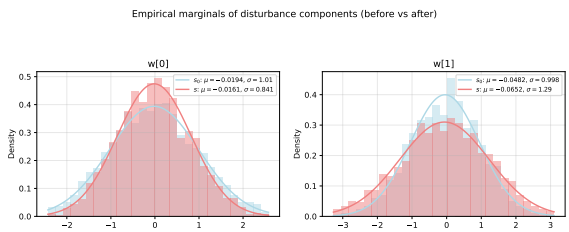
$$w_t \sim \mathcal{N}(0, \Sigma_{\text{use}}).$$

Therefore, the temporal difference between the two models is not visible in marginal histograms alone.

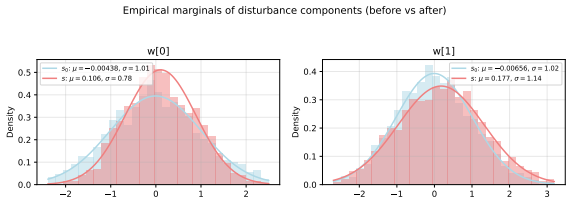
B.5.iv Before/after projection: marginal distributions. To visualise how a raw covariance candidate is modified before sampling, Figures B.4a and B.4b compare the empirical marginals of the original sample cloud and the projected admissible one. Here s_0 denotes the initial raw sample and s the final projected sample used for simulation.

The comparison highlights that the projection acts on covariance geometry rather than on an arbitrary pointwise clipping of the samples. In particular, the distribution remains approximately Gaussian, but the spread and anisotropy are adjusted so that the final covariance becomes Wasserstein-admissible.

B.5.v Before/after projection: scatter structure. A complementary view is provided by the two-dimensional scatter plots in Figure B.5. These plots make visible the geometric effect of the covariance transformation. The projected samples preserve the global Gaussian cloud, but their orientation and elongation change according to the projected covariance.

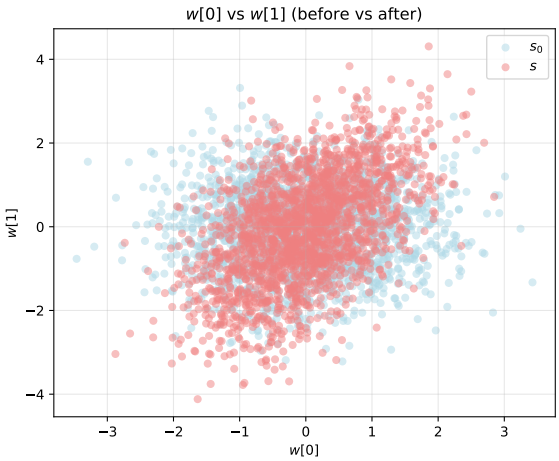


(a) Independent case: before/after marginal distributions.

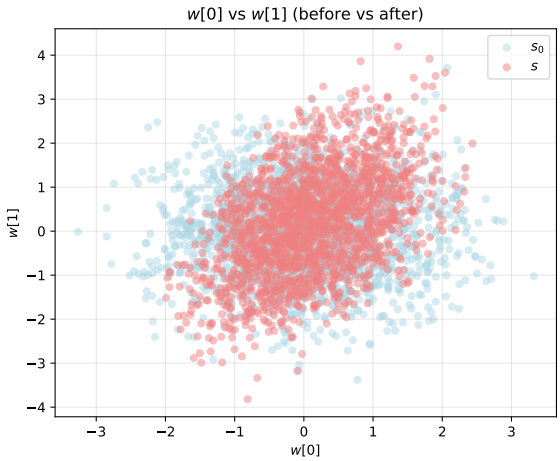


(b) Correlated case: before/after marginal distributions.

Figure B.4: Before/after comparison of empirical marginals. These plots show how the Wasserstein-ball projection modifies the covariance structure while preserving a Gaussian shape.



(a) Independent case: before/after scatter.



(b) Correlated case: before/after scatter.

Figure B.5: Before/after scatter plots in the (w_0, w_1) plane. The projection onto the Wasserstein ball changes the covariance geometry while preserving the overall Gaussian structure.

B.6 Interpretation of the Disturbance Evaluation

The disturbance evaluation figures support the two main structural properties of the sampler.

First, the independent and correlated constructions are consistent with the same marginal ambiguity model. This is seen in the nearly matching histograms of Figure B.3. The difference between the two models does not lie in their one-step statistics, but only in their temporal covariance structure.

Second, the before/after plots illustrate that the code does not simply sample from a fixed nominal covariance. Instead, it may start from a raw covariance candidate, test its Wasserstein distance to the nominal model, and, when necessary, project it onto the admissible set. The resulting disturbance law is therefore not arbitrary Gaussian noise, but a Gaussian law that is explicitly Wasserstein-consistent with the nominal uncertainty description.

These two properties are important for the later robust-control evaluation. They make it possible to separate the effect of *marginal distributional uncertainty* from the effect of *temporal dependence*. In particular:

- the Wasserstein radius γ controls how far the covariance may move away from Σ_{nom};
- the parameter ρ controls temporal persistence in the correlated case;
- the iid and AR(1) samplers can therefore be compared under the same marginal ambiguity constraint.

B.7 Concluding Remarks

In summary, the disturbances used in simulation are sampled from a Gaussian Wasserstein ambiguity set rather than from a fixed covariance model. The procedure first constructs an admissible covariance inside the Wasserstein ball, possibly by projecting a raw candidate onto the boundary, and then generates either:

- an iid Gaussian sequence with covariance Σ_{use}, or
- a correlated AR(1) Gaussian sequence with the same stationary covariance Σ_{use}.

This maintains coherence between the uncertainty model, the simulation pipeline, and the later Monte Carlo evaluation of robustness. Most importantly, it makes explicit the distinction between two different sources of uncertainty:

1. *marginal covariance uncertainty*, encoded by the Wasserstein ball;
2. *temporal dependence*, encoded by the choice between iid and AR(1) sampling.

Such a distinction is essential when interpreting the closed-loop performance of the baseline and distributionally robust controllers.

C Representative Trajectory Selection via Medoid Analysis

C.1 Dataset Construction and Preprocessing

Rather than relying on a single open-loop experiment, we generate multiple trajectories

$$\{\mathcal{D}_i\}_{i=1}^{N}, \qquad \mathcal{D}_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\},$$

where

$$X_i \in \mathbb{R}^{n_x \times T_i}, \quad U_i \in \mathbb{R}^{n_u \times T_i}, \quad Y_i \in \mathbb{R}^{n_y \times T_i}, \quad Z_i \in \mathbb{R}^{n_z \times T_i}, \quad X_{i,+} \in \mathbb{R}^{n_x \times T_i}.$$

All trajectories are truncated to a common horizon

$$T = \min_i T_i.$$

and the initial states are evaluated following the method proposed in Appendix A.2.

Each dataset is initialised following the proposed method in Appendix A.2, to ensure diversity of excitation. Only trajectories satisfying a persistent-excitation condition are retained:

$$\sigma_{\min}\left(\frac{1}{T}\Phi_i\Phi_i^\top\right) > \varepsilon_{\mathrm{PE}}, \qquad \Phi_i = \begin{bmatrix} X_i \\ U_i \end{bmatrix}.$$

C.2 Normalization Across Trajectories

Before selecting a representative dataset, all trajectories are normalized to prevent a single signal block from dominating the distance metric because of scale or units.

For each block $k \in \{X, U, Y, Z, X_+\}$, define

$$s_k = \frac{1}{N}\sum_{i=1}^{N} \|\mathcal{D}_i[k]\|_F + \varepsilon,$$

and rescale each block as

$$\tilde{\mathcal{D}}_i[k] = \frac{\mathcal{D}_i[k]}{s_k}.$$

This normalization ensures that state, input, output, and successor-state data contribute comparably to the medoid criterion.

C.3 Medoid Selection

Among the retained trajectories, the representative dataset is chosen as the medoid, namely the trajectory minimizing the total weighted distance to all others:

$$i^\star = \arg\min_i \sum_k w_k \sum_{j=1}^{N} \left\|\tilde{\mathcal{D}}_j[k] - \tilde{\mathcal{D}}_i[k]\right\|_F^2,$$

where w_k are optional weighting coefficients.

The final matrices used for identification are taken from this single representative trajectory:

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+}).$$

C.4 Motivation

A naive arithmetic average of the trajectories, for instance

$$\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i,$$

is undesirable because opposite-sign components may cancel, thereby shrinking amplitudes and destroying the dynamical consistency between X_- and X_+. The medoid procedure avoids this issue by selecting a real, physically consistent trajectory that is most central with respect to the ensemble.

This strategy preserves the internal structure of the data while reducing sensitivity to random initial conditions, noise, and seed-specific fluctuations.

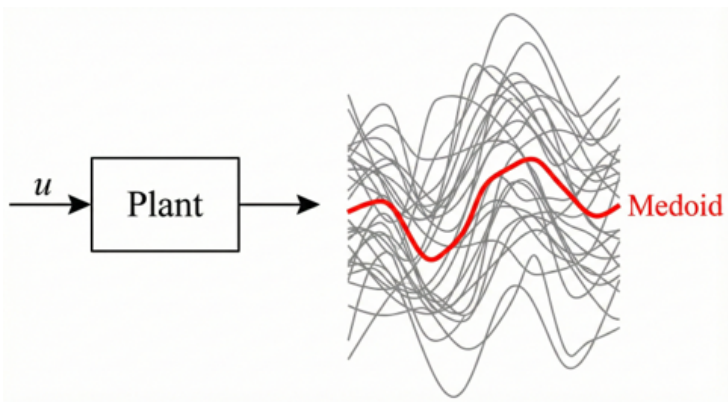


Figure C.1: Illustration of medoid selection. Multiple trajectories generated from the same plant (grey) exhibit variability due to different initial conditions and disturbances. The medoid (red) is the trajectory minimising the aggregate distance to all others, preserving physical consistency while representing the ensemble.

C.5 Algorithmic Summary

Algorithm 5 Dataset Preprocessing and Medoid Selection

Input: N trajectories $\mathcal{D}_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\}$, time lengths T_i
Parameters: persistent-excitation threshold $\varepsilon_{\mathrm{PE}}$, scaling offset ε, weights w_k
(A) Truncation and persistent-excitation filtering
$T \leftarrow \min_i T_i$
for $i = 1$ to N **do**
  Truncate each $\mathcal{D}_i$ to horizon T
  Sample $x_0^{(i)} \sim \mathcal{N}(0, I)$
  Form $\Phi_i = \begin{bmatrix} X_i \\ U_i \end{bmatrix}$
  if $\sigma_{\min}(\frac{1}{T}\Phi_i\Phi_i^\top) \leq \varepsilon_{\mathrm{PE}}$ **then**
    discard $\mathcal{D}_i$
  end if
end for
(B) Normalization
for each block $k \in \{X, U, Y, Z, X_+\}$ **do**
  Compute scale

$$s_k = \frac{1}{N}\sum_{i=1}^{N} \|\mathcal{D}_i[k]\|_F + \varepsilon$$

  for $i = 1$ to N **do**
    $\tilde{\mathcal{D}}_i[k] \leftarrow \mathcal{D}_i[k]/s_k$
  end for
end for
(C) Medoid selection
for $i = 1$ to N **do**
  Compute

$$d_i = \sum_k w_k \sum_{j=1}^{N} \|\tilde{\mathcal{D}}_j[k] - \tilde{\mathcal{D}}_i[k]\|_F^2$$

end for
$i^\star \leftarrow \arg\min_i d_i$
(D) Output
Return

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+})$$

C.6 Influence of $N_{\mathbf{sims}}$ on Closed-Loop Performance

The routine `NsimSweep_FROM_DATA` studies how the number of datasets N_{sims} entering the medoid selection affects the closed-loop behavior of the controller synthesized with `FROM_DATA=True`. For each value of N_{sims} in a prescribed grid, the full synthesis-and-evaluation pipeline is repeated multiple times, while keeping the Wasserstein radius γ and the disturbance model fixed.

For each N_{sims}, the following quantities are collected:

- the closed-loop performance cost J,
- the spectral radius $\rho(A_{\mathrm{cl}})$ of the closed-loop state matrix,
- the optimization objective obj returned by the LMI solver.

For each metric, empirical means and standard deviations are computed over repeated runs. Best values of N_{sims} are then selected through scalar ranking rules favouring small means and small variability. For the spectral radius, only stable designs satisfying $\rho(A_{\text{cl}}) < 1$ are considered admissible.

Figures C.2–C.4 summarize these results. As N_{sims} increases, the variance of both J and obj decreases, and their mean values tend to improve gradually. In contrast, the spectral radius remains close to but below the stability boundary across the entire sweep, indicating that increasing N_{sims} mainly reduces variability rather than fundamentally changing the stability margin.

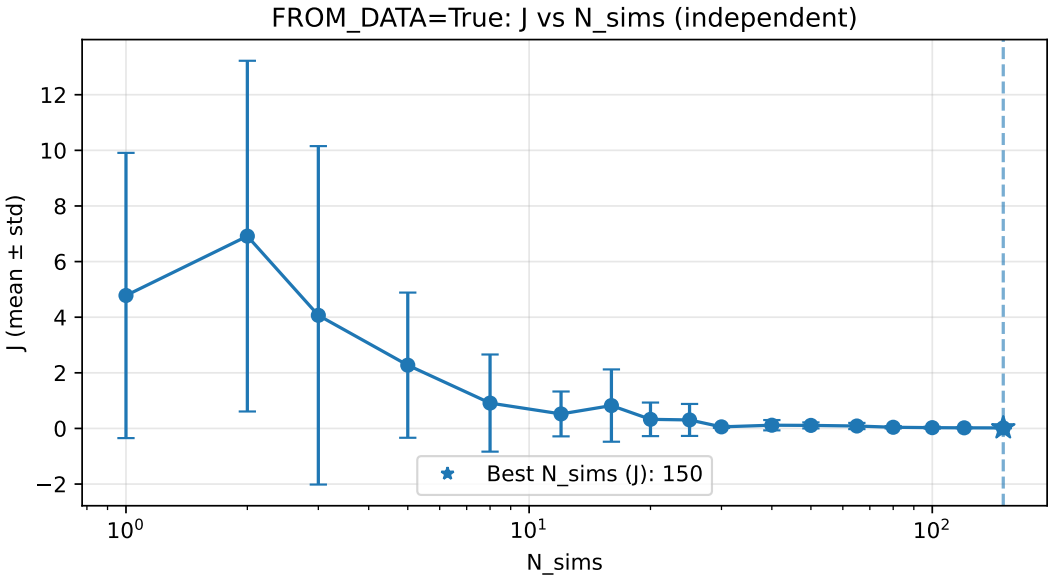


Figure C.2: Closed-loop performance cost J (mean ± standard deviation) versus N_{sims} for `FROM_DATA=True` under the independent disturbance model. The dashed vertical line and star marker indicate the automatically selected best value of N_{sims} according to the ranking based on J.

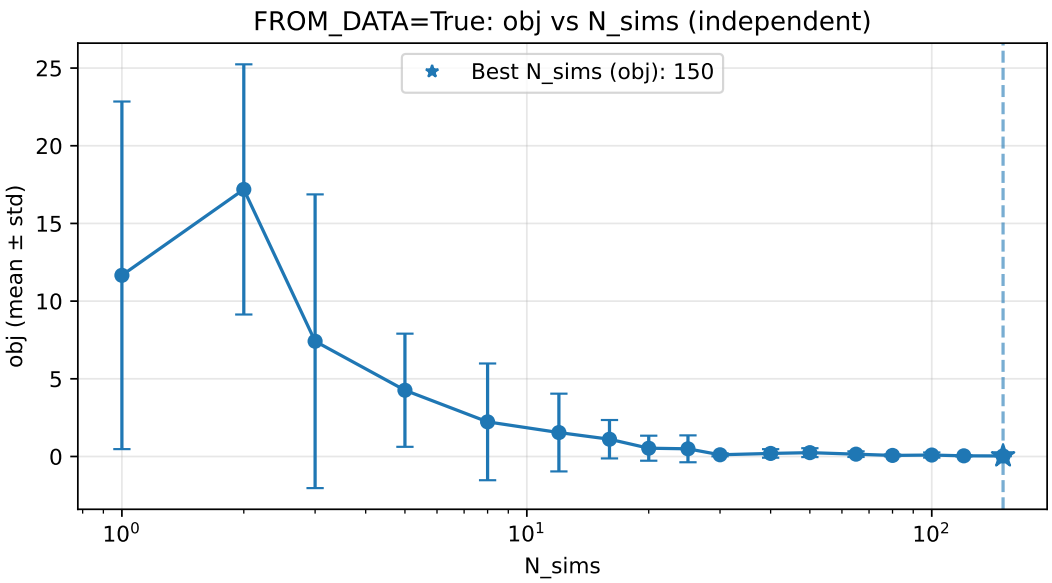


Figure C.3: Optimization objective obj (mean ± standard deviation) versus N_{sims}. Increasing the number of trajectories used in the medoid selection decreases both the variability and the average optimized objective value.

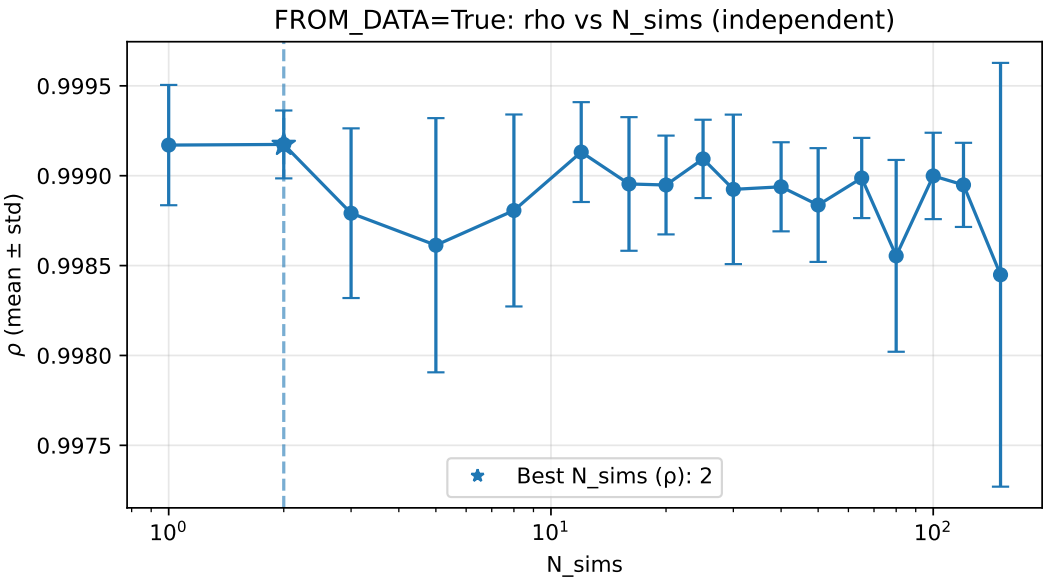


Figure C.4: Spectral radius $\rho(A_{\text{cl}})$ (mean ± standard deviation) versus N_{sims}. All designs remain close to the stability boundary $\rho = 1$, while the spread decreases as N_{sims} increases.

C.7 Matrix-Level Analysis Across N_{sims}

While the preceding sweep focuses on closed-loop scalars, the class `NsimsMatricesAnalyzer` studies how the identified state-space matrices vary with N_{sims}. For each prescribed value of N_{sims}, the analyzer:

1. generates N_{sims} open-loop datasets and selects the medoid,
2. runs the identification pipeline on the representative dataset,

3. stores the estimated matrices

$$A,\ B_u,\ B_w,\ C_y,\ D_{yw},\ C_z,\ D_{zu},\ D_{zw},$$

4. compares them against the corresponding nominal matrices obtained in the model-based case.

The analysis then examines:

- eigenvalue distributions of the identified matrix A,
- the evolution of $\rho(A)$ as a function of N_{sims},
- step and sinusoidal responses of the identified models,
- Frobenius norms, relative errors, and entrywise differences with respect to the reference matrices.

Figure C.5 shows the eigenvalues of the identified matrix A for all values of N_{sims}, together with the unit circle. The eigenvalues remain inside or extremely close to the unit circle across the entire sweep, indicating that the identified dynamics remain near-marginal but consistent. The overlays for small and large N_{sims} almost coincide, which suggests that medoid-based identification is robust to the number of trajectories used.

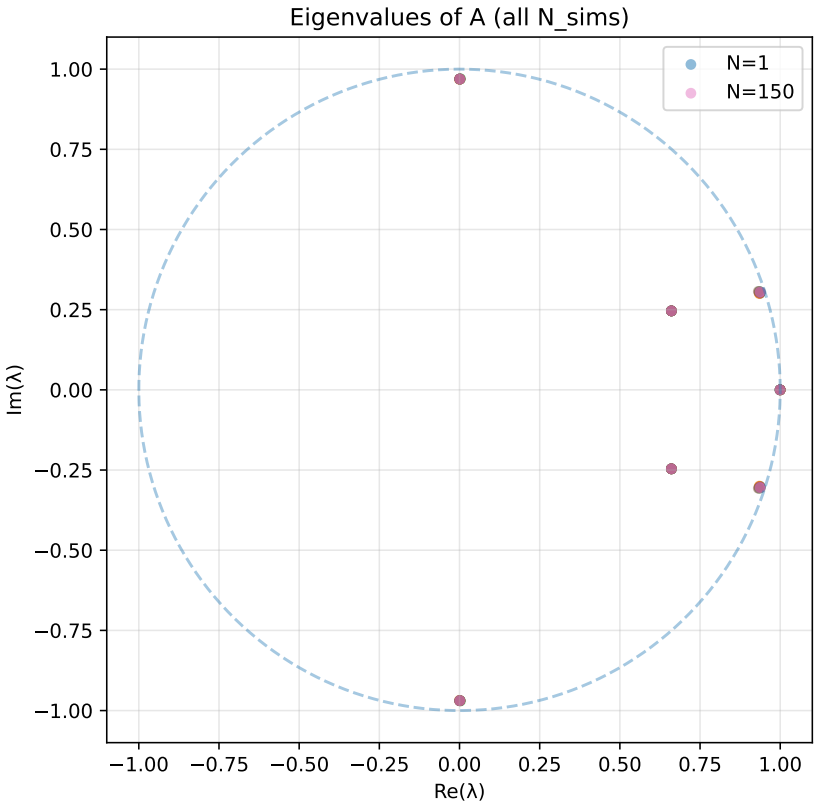


Figure C.5: Eigenvalues of the identified matrix A for all values of N_{sims}, overlaid on the complex plane with the unit circle shown as a dashed line.

Figure C.6 reports the spectral radius of A as a function of N_{sims}. The trend is nearly flat and remains very close to one, confirming that increasing the number of datasets does not drastically alter the identified spectrum, but improves the smoothness and repeatability of the estimates.

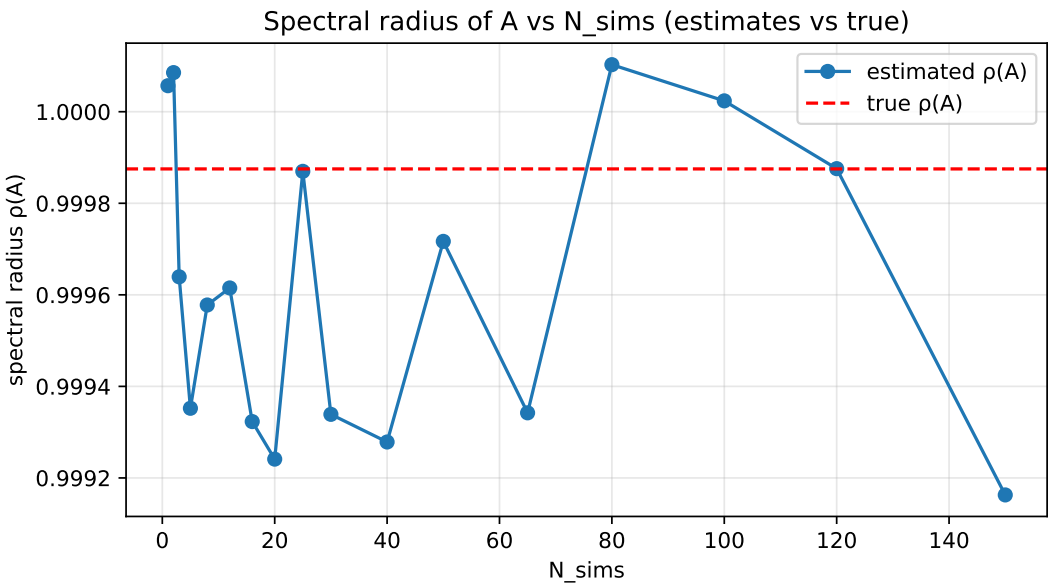
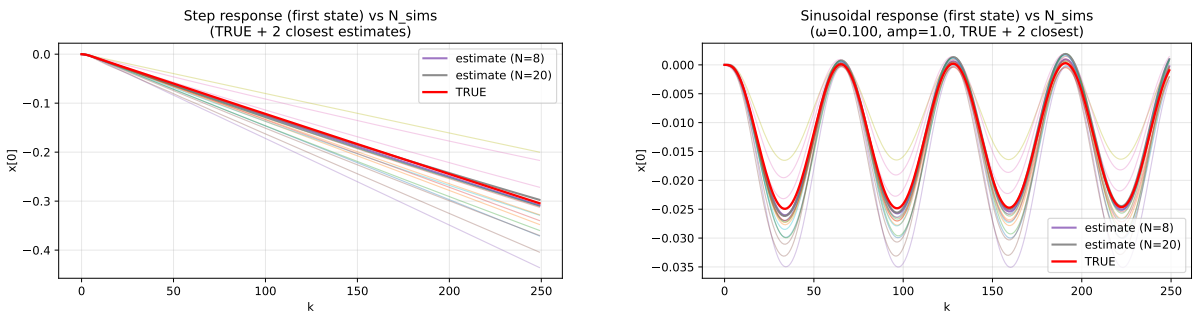


Figure C.6: Spectral radius $\rho(A)$ of the identified state matrix versus N_{sims}. The values remain close to the nominal reference and exhibit only small fluctuations.

Figure C.7 shows the dynamic responses of the first state component for all identified models under both step and sinusoidal inputs. All responses remain coherent across the sweep, while larger N_{sims} yield models that better represent the ensemble.



(a) Step response under unit-step input.

(b) Sinusoidal response.

Figure C.7: Dynamic responses of the first state component for all identified models. Both step and sinusoidal inputs show consistent behaviour across the N_{sims} sweep, with larger datasets yielding more representative models.

C.8 Summary

Taken together, the scalar sweep and the matrix-level analysis show that medoid selection provides a reliable way to extract a representative dataset from multiple open-loop realizations. Increasing N_{sims} mainly reduces variance and improves the centrality of the selected trajectory, rather than fundamentally changing the identified dynamics. This supports the use of medoid-based preprocessing as a robust front end for the subsequent data-driven identification and synthesis pipeline.

D Steady-State Signal-to-Noise Ratio and Spectral Analysis

D.1 Closed-Loop Model

This appendix presents a steady-state signal-to-noise ratio (SNR) analysis of the closed-loop system under additive Gaussian disturbances

$$w_k \sim \mathcal{N}(0, \Sigma),$$

using the controller synthesised in the correlated-noise setting, although the controller is synthesised using a distributionally robust formulation, the present analysis considers Gaussian disturbances in order to obtain interpretable steady-state and spectral metrics.

The plant is described by

$$\begin{aligned} x_{k+1} &= Ax_k + B_u u_k + B_w w_k, \\ y_k &= C_y x_k + D_{yw} w_k, \\ z_k &= C_z x_k + D_{zu} u_k + D_{zw} w_k, \end{aligned}$$

while the controller has the form

$$\begin{aligned} x_{k+1}^c &= A_c x_k^c + B_c y_k, \\ u_k &= C_c x_k^c + D_c y_k. \end{aligned}$$

Introducing the augmented state

$$\chi_k = \begin{bmatrix} x_k \\ x_k^c \end{bmatrix},$$

the closed-loop state equation becomes

$$\chi_{k+1} = A_{\mathrm{cl}} \chi_k + B_{\mathrm{cl}} w_k,$$

with

$$A_{\mathrm{cl}} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \qquad B_{\mathrm{cl}} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix}.$$

The measured output, control input, and performance output can be written in augmented coordinates as

$$\begin{aligned} y_k &= C_{y,\mathrm{cl}} \chi_k + D_{yw} w_k, & C_{y,\mathrm{cl}} &= \begin{bmatrix} C_y & 0 \end{bmatrix}, \\ u_k &= C_{u,\mathrm{cl}} \chi_k + D_{uw} w_k, & C_{u,\mathrm{cl}} &= \begin{bmatrix} D_c C_y & C_c \end{bmatrix}, \quad D_{uw} = D_c D_{yw}, \\ z_k &= C_{z,\mathrm{cl}} \chi_k + D_{zw,\mathrm{cl}} w_k, & C_{z,\mathrm{cl}} &= \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad D_{zw,\mathrm{cl}} = D_{zw} + D_{zu} D_c D_{yw}. \end{aligned}$$

D.2 Steady-State Covariance and SNR Definition

Assuming that the closed-loop matrix is Schur stable,

$$\rho(A_{\mathrm{cl}}) < 1,$$

the stationary covariance P of the augmented state solves the discrete Lyapunov equation

$$P = A_{\mathrm{cl}} P A_{\mathrm{cl}}^{\top} + B_{\mathrm{cl}} \Sigma B_{\mathrm{cl}}^{\top}.$$

For each port $q \in \{y, u, z\}$, the total covariance is split into a signal component propagated through the state and a noise component injected directly by the disturbance:

$$\Sigma_q^{\mathrm{sig}} = C_{q,\mathrm{cl}} P C_{q,\mathrm{cl}}^{\top}, \qquad \Sigma_q^{\mathrm{noise}} = D_{qw} \Sigma D_{qw}^{\top}.$$

Here,

$$D_{yw} = D_{yw}, \qquad D_{uw} = D_c D_{yw}, \qquad D_{zw} = D_{zw,\mathrm{cl}}.$$

The aggregate SNR is defined through traces:

$$\mathrm{SNR}_q = \frac{\mathrm{tr}\big(\Sigma_q^{\mathrm{sig}}\big)}{\mathrm{tr}\big(\Sigma_q^{\mathrm{noise}}\big)}, \qquad \mathrm{SNR}_q^{\mathrm{dB}} = 10 \log_{10}\big(\mathrm{SNR}_q\big).$$

Per-channel SNRs are defined as

$$\mathrm{SNR}_{q,i} = \frac{\big[\Sigma_q^{\mathrm{sig}}\big]_{ii}}{\big[\Sigma_q^{\mathrm{noise}}\big]_{ii}},$$

with numerical guards such as $\max\{\cdot, 0\}$ and a small ε in the denominator when required.

D.3 Aggregate and Per-Channel SNR Plots

Figure D.1 reports the aggregate SNR values for the y, u, and z ports, together with their per-channel decompositions. These bar plots provide a compact summary of how effectively the controller attenuates direct disturbance effects relative to state-borne signal energy.

Higher SNR values indicate stronger attenuation of the direct noise contribution at the corresponding port.

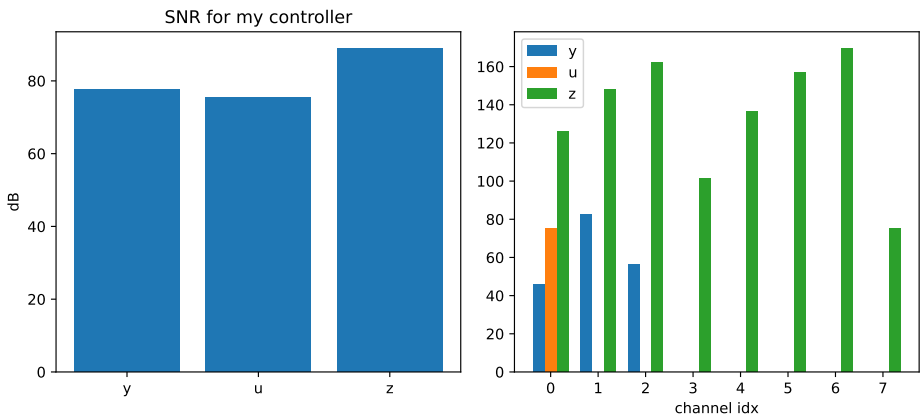


Figure D.1: Trace-based SNR in dB: aggregate values for the y, u, and z ports, together with per-channel decompositions.

D.4 Spectral Analysis via Periodograms

To study how disturbance energy is distributed across frequency, we compute simple periodograms for the state, measured output, performance output, and control input signals. For a discrete-time signal s of length T, the periodogram is defined as

$$\mathrm{PSD}_j(\omega) = \frac{1}{T} \left|\mathcal{F}\{s_j\}(\omega)\right|^2,$$

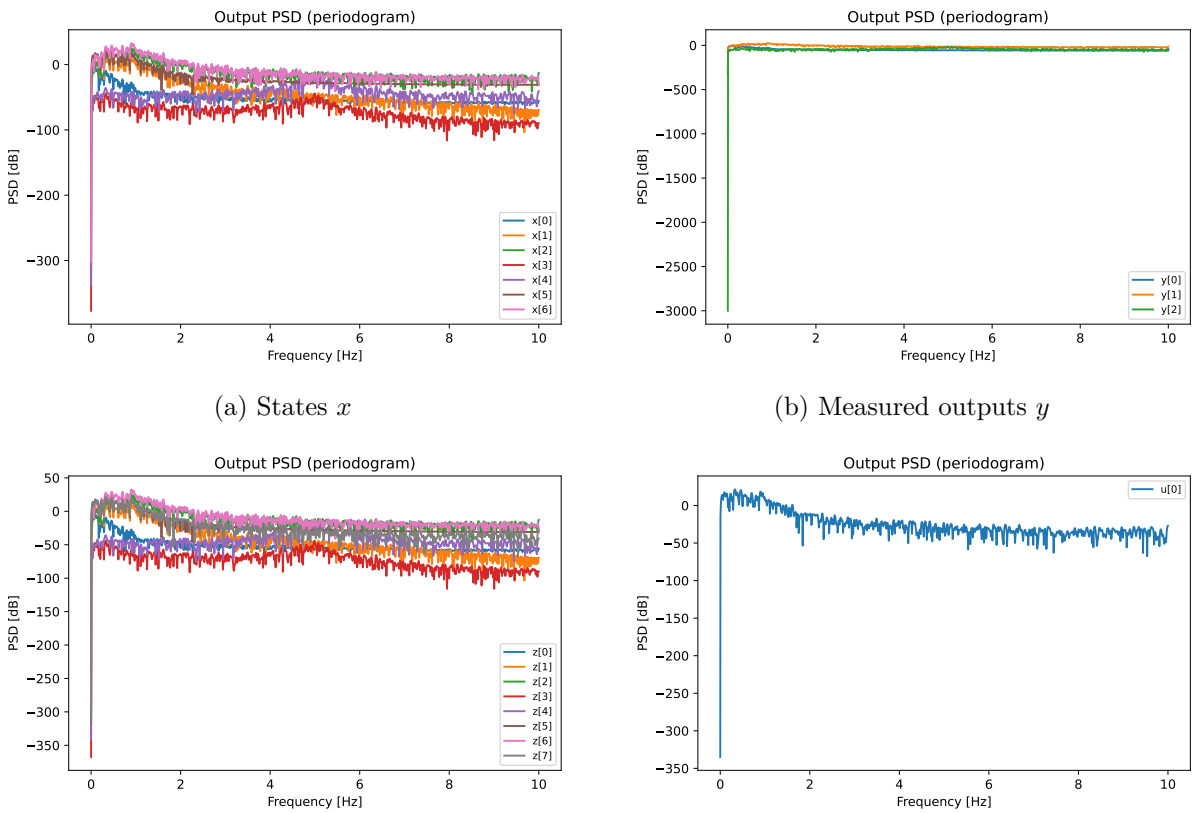
and is plotted in decibels as

$$10 \log_{10}\big(\max\{\mathrm{PSD}_j(\omega), \varepsilon\}\big).$$

These plots reveal how the closed loop shapes disturbance energy across frequency:

- low PSD values indicate strong attenuation,
- higher PSD values indicate frequencies where disturbance energy is transmitted more strongly,
- flat high-frequency tails are typically associated with plant dynamics rather than feedback action.

Figures D.2 report the spectra of states, measured outputs, performance outputs, and control inputs.



(a) States x

(b) Measured outputs y

(c) Performance outputs z

(d) Control input u

Figure D.2: Power spectral densities (in dB) of the system signals: states x, measured outputs y, performance outputs z, and control input u.

D.5 Kernel Representation of SNR

To sweep disturbance orientations without repeatedly solving the Lyapunov equation, we precompute kernels. For each port $q \in \{y, u, z\}$, define X_q as the solution of

$$X_q = A_{\mathrm{cl}}^\top X_q A_{\mathrm{cl}} + C_{q,\mathrm{cl}}^\top C_{q,\mathrm{cl}}.$$

Then define

$$K_q = B_{\mathrm{cl}}^\top X_q B_{\mathrm{cl}}, \qquad R_q = D_{qw}^\top D_{qw}.$$

For any positive semidefinite disturbance covariance $\Sigma \succeq 0$, the SNR can be written as the linear-fractional form

$$\boxed{\mathrm{SNR}_q(\Sigma) = \frac{\mathrm{tr}(K_q \Sigma)}{\mathrm{tr}(R_q \Sigma)}}.$$

This representation is homogeneous in Σ, so the SNR depends only on the orientation of the disturbance covariance, not on its absolute scale.

Figure D.3 illustrates this dependence by rotating Σ in a selected two-dimensional subspace and plotting $\mathrm{SNR}_q(\Sigma)$ in decibels as a function of the rotation angle.

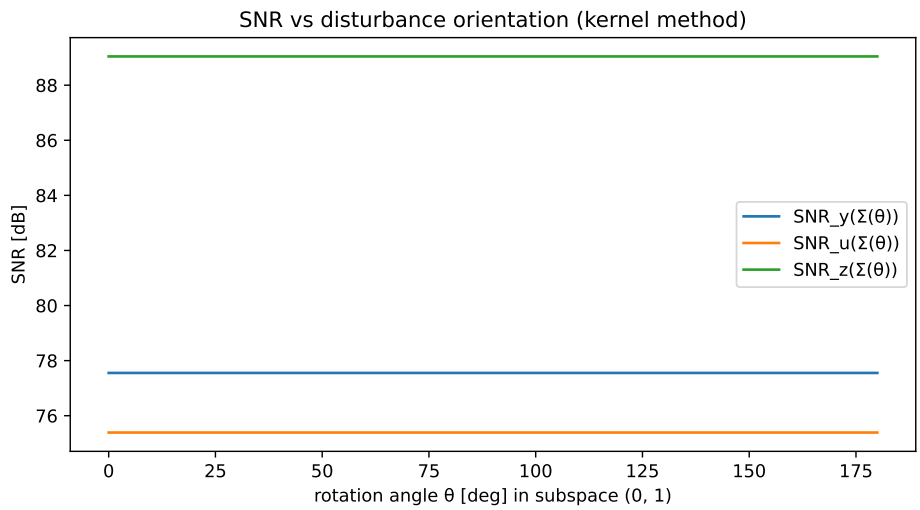


Figure D.3: SNR as a function of disturbance orientation in a selected two-dimensional subspace, computed via the kernel representation.

D.6 Worst- and Best-Case Disturbance Directions

Extremal SNR values under trace-normalized covariances can be characterized through a generalized eigenvalue problem. Let $R_q^{-1/2}$ denote a pseudoinverse square root on the range of R_q, and define

$$S_q = R_q^{-1/2} K_q R_q^{-1/2}.$$

Then the minimum and maximum achievable SNR values over all covariances satisfying $\mathrm{tr}(\Sigma) = 1$ are

$$\mathrm{SNR}_q^{\min} = \lambda_{\min}(S_q), \qquad \mathrm{SNR}_q^{\max} = \lambda_{\max}(S_q).$$

Figure D.4 reports these worst- and best-case values in decibels for the three ports $q \in \{y, u, z\}$.

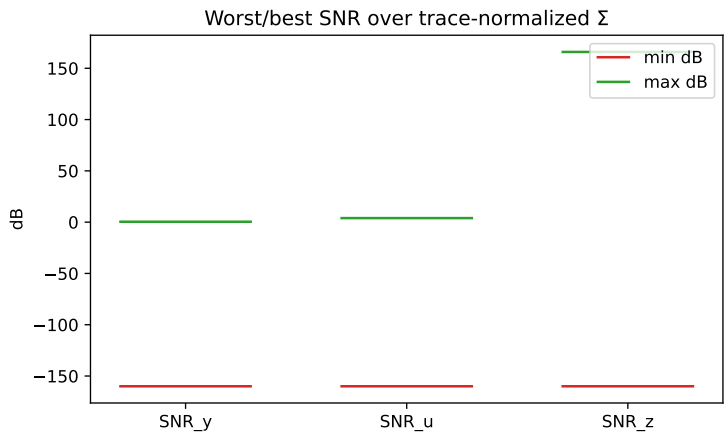


Figure D.4: Worst- and best-case SNR values, in decibels, over trace-normalized disturbance covariances.

D.7 Assumptions and Limitations

The SNR analysis relies on the following assumptions and caveats:

- Steady-state SNR is well defined only if $\rho(A_{\mathrm{cl}}) < 1$.
- Trace-based SNR aggregates across channels; when certain outputs are more important than others, per-channel SNRs should also be examined.
- The periodograms used here are simple, non-Welch estimates; they provide qualitative frequency-shaping information but exhibit higher variance than averaged estimators.
- The kernel representation assumes fixed closed-loop matrices and explores only the orientation of Σ, not changes in the controller itself.
- If R_q is nearly singular, the effective SNR may become very large, and pseudoinverse-based regularization is required in practice.

D.8 Discussion

The spectral analysis confirms that the synthesized controller shapes the closed-loop energy distribution in a meaningful way. Most disturbance energy is concentrated at low frequencies, where feedback actively attenuates it, while higher-frequency content is largely suppressed by the plant dynamics. The performance-output spectra remain consistently below those of the measured outputs, indicating that the controller effectively limits the transmission of disturbance energy to the regulated variables.

Moreover, no significant resonance peaks are observed, which suggests the absence of lightly damped closed-loop modes and supports the interpretation of the design as spectrally robust.

E One-Step System Identification and Residual-Based Disturbance Estimation

E.1 Overview

This appendix presents the identification procedure used throughout the data-driven methodologies developed in this thesis. Its role is to extract, from measured trajectories, a nominal linear model together with an effective disturbance channel that captures the dominant unexplained state variations.

The procedure is intentionally modular and reusable across different chapters. In particular, it is used whenever the synthesis method requires:

- a nominal state-transition model estimated directly from data;
- a disturbance channel constructed from the residual component of the one-step prediction error;
- output and performance maps identified consistently with the same data realisation;
- a practical uncertainty indicator derived from the regression residuals.

The emphasis of this appendix is therefore not on a specific control synthesis method, but on the common identification backbone that supports the subsequent data-driven DRO-LMI formulations.

E.2 Data Matrices and One-Step Regression

Consider a measured trajectory

$$\{x_k, u_k, y_k, z_k\}_{k=0}^{T},$$

where

$$x_k \in \mathbb{R}^{n_x}, \qquad u_k \in \mathbb{R}^{n_u}, \qquad y_k \in \mathbb{R}^{n_y}, \qquad z_k \in \mathbb{R}^{n_z},$$

collected using the Medoid method introduced in Appendix C. From these samples, define the stacked matrices

$$X_- = \begin{bmatrix} x_0 & x_1 & \cdots & x_{T-1} \end{bmatrix}, \qquad U_- = \begin{bmatrix} u_0 & u_1 & \cdots & u_{T-1} \end{bmatrix},$$

$$X_+ = \begin{bmatrix} x_1 & x_2 & \cdots & x_T \end{bmatrix}, \qquad Y_- = \begin{bmatrix} y_0 & y_1 & \cdots & y_{T-1} \end{bmatrix}, \qquad Z_- = \begin{bmatrix} z_0 & z_1 & \cdots & z_{T-1} \end{bmatrix}.$$

The nominal state-update model is identified by a regularised one-step regression. Define the regressor

$$D_x := \begin{bmatrix} X_- \\ U_- \end{bmatrix} \in \mathbb{R}^{(n_x+n_u)\times T},$$

and its regularised right pseudoinverse

$$D_x^{\dagger} := D_x^{\top}\left(D_x D_x^{\top} + \rho I\right)^{-1},$$

where $\rho > 0$ is a small numerical regularisation parameter.

The identified state map is then

$$\hat{O}_x := X_+ D_x^\dagger = \begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix},$$

with

$$\hat{A} \in \mathbb{R}^{n_x \times n_x}, \qquad \hat{B}_u \in \mathbb{R}^{n_x \times n_u}.$$

Thus, the nominal state-update equation is approximated as

$$x_{k+1} \approx \hat{A} x_k + \hat{B}_u u_k.$$

E.3 Residual Construction

The part of the state evolution that is not captured by the nominal one-step regression is collected in the residual matrix

$$R := X_+ - \hat{A} X_- - \hat{B}_u U_- \in \mathbb{R}^{n_x \times T}.$$

Equivalently, if $O_x = [A \;\; B_u]$ denotes the exact stacked state-update operator associated with the data-generating process, then

$$X_+ = O_x D_x + R.$$

Hence, the residual matrix R measures the discrepancy between the measured successor states and the nominal one-step predictor based on the identified matrices $\hat{A}, \hat{B}_u$.

This residual is central to the subsequent data-driven constructions for two reasons:

- it provides a practical indicator of the uncertainty left unexplained by the regression model;
- it contains the information used to build an effective disturbance channel.

E.4 Residual Covariance and Spectral Decomposition

To extract the dominant directions of the unexplained dynamics, consider the empirical residual covariance

$$S_R := \frac{1}{T} R R^\top.$$

In the implementation, this matrix is symmetrized and slightly regularized for numerical stability. Denoting the resulting matrix again by S_R, its eigendecomposition is written as

$$S_R = U_s \operatorname{diag}(s_1, \ldots, s_{n_x}) \, U_s^\top,$$

where the eigenvalues are ordered as

$$s_1 \geq s_2 \geq \cdots \geq s_{n_x} \geq 0.$$

The matrix $U_s \in \mathbb{R}^{n_x \times n_x}$ contains the orthonormal eigenvectors of the residual covariance, while the eigenvalues quantify the residual energy along each direction. This spectral information is later used either:

- to define a reduced disturbance channel;
- or, in more refined relaxations, to weight uncertainty anisotropically according to the dominant residual directions.

E.5 Residual-Based Disturbance Channel Extraction

The effective disturbance dimension n_w is selected by an energy-retention rule. Specifically, one chooses the smallest integer n_w such that

$$\frac{\sum_{i=1}^{n_w} s_i}{\sum_{i=1}^{n_x} s_i} \geq \eta,$$

where $\eta \in (0, 1)$ is a prescribed energy threshold.

Let

$$U_p := U_s(:, 1{:}n_w)$$

denote the matrix containing the first n_w dominant eigenvectors. The code then constructs a disturbance channel $\hat{B}_w$ from this principal residual subspace and uses it to reconstruct an effective disturbance realization. Abstractly, the result is a factorization of the form

$$R \approx \hat{B}_w \hat{W},$$

where

$$\hat{W} := \hat{B}_w^{\dagger} R.$$

Therefore, the nominal state dynamics are enriched as

$$x_{k+1} \approx \hat{A} x_k + \hat{B}_u u_k + \hat{B}_w w_k.$$

The exact algebraic form of $\hat{B}_w$ may depend on the specific factorization convention adopted in the implementation, but its role is always the same: it captures the dominant residual directions and provides a reduced-order disturbance representation consistent with the observed mismatch.

E.6 Output and Performance Regressions

Once the disturbance realization $\hat{W}$ has been obtained, the measured output equation is identified through the regression

$$\begin{bmatrix} \hat{C}_y & \hat{D}_{yw} \end{bmatrix} = Y_- \begin{bmatrix} X_- \\ \hat{W} \end{bmatrix}^{\dagger}.$$

This yields the nominal output relation

$$y_k \approx \hat{C}_y x_k + \hat{D}_{yw} w_k,$$

with

$$\hat{C}_y \in \mathbb{R}^{n_y \times n_x}, \qquad \hat{D}_{yw} \in \mathbb{R}^{n_y \times n_w}.$$

Similarly, the performance or regulated output map is obtained from

$$\begin{bmatrix} \hat{C}_z & \hat{D}_{zu} & \hat{D}_{zw} \end{bmatrix} = Z_- \begin{bmatrix} X_- \\ U_- \\ \hat{W} \end{bmatrix}^{\dagger},$$

which yields

$$z_k \approx \hat{C}_z x_k + \hat{D}_{zu} u_k + \hat{D}_{zw} w_k.$$

Thus, the complete identified nominal model is

$$x_{k+1} \approx \hat{A} x_k + \hat{B}_u u_k + \hat{B}_w w_k, \tag{E.1}$$

$$y_k \approx \hat{C}_y x_k + \hat{D}_{yw} w_k, \tag{E.2}$$

$$z_k \approx \hat{C}_z x_k + \hat{D}_{zu} u_k + \hat{D}_{zw} w_k. \tag{E.3}$$

E.7 Nominal Covariance Used in the DRO Program

Although the disturbance channel is extracted from the residual covariance, the specific optimization routine considered in this thesis does not directly use the empirical covariance of $\hat{W}$ as the nominal covariance in the DRO objective. Instead, once the reduced disturbance dimension n_w has been selected, the nominal covariance is fixed externally as

$$\Sigma_{\text{nom}} = \sigma_w^2 I_{n_w},$$

where σ_w^2 is a scalar variance parameter provided by the noise model or simulation setup.

Likewise, the ambiguity radius γ is not computed inside this identification routine, but is treated as a parameter of the subsequent optimization problem.

Therefore, in the implemented pipeline:

- the residual covariance is used primarily to identify disturbance directions and an effective reduced dimension;
- the nominal covariance matrix Σ_{nom} entering the DRO objective is externally prescribed;
- the ambiguity radius γ is passed directly to the synthesis stage.

E.8 A Practical Radius for Identification Uncertainty

The residual matrix also provides a practical indicator of the uncertainty affecting the nominal state regression. Since

$$X_+ = O_x D_x + R$$

and

$$\hat{O}_x = X_+ D_x^\dagger,$$

one obtains

$$\hat{O}_x = O_x D_x D_x^\dagger + R D_x^\dagger.$$

Motivated by the perturbation term $RD_x^\dagger$, the implementation defines the data-driven uncertainty surrogate

$$\beta := \frac{\|R\|_F}{\max(\sigma_{\min}(D_x), \varepsilon)},$$

where $\sigma_{\min}(D_x)$ is the smallest singular value of the regressor D_x, and $\varepsilon > 0$ is a small safeguard against numerical degeneracy.

This quantity is then split between the state-matrix and input-matrix parts according to the relative block dimensions:

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x + n_u}}, \qquad \beta_B = \beta\sqrt{\frac{n_u}{n_x + n_u}}.$$

These quantities are not intended as exact worst-case identification bounds, but rather as practical, data-consistent surrogates used to parameterize the subsequent convex robustification.

A more explicit motivation can also be drawn from finite-sample identification theory for linear systems under independent rollouts [5]. In particular, for the least-squares estimator of the state and input matrices, non-asymptotic bounds of the form

$$\|\hat{A} - A\| \leq \frac{16\sigma_w}{\sqrt{\lambda_{\min}(\sigma_u^2 G_T G_T^\top + \sigma_w^2 F_T F_T^\top)}} \sqrt{\frac{(n_x + 2n_u)\log(36/\delta)}{N}},$$

and

$$\|\hat{B} - B\| \leq \frac{16\sigma_w}{\sigma_u} \sqrt{\frac{(n_x + 2n_u)\log(36/\delta)}{N}}$$

hold with high probability, provided that the number of rollouts satisfies

$$N \geq 8(n_x + n_u) + 16 \ln\left(\frac{4}{\delta}\right) \;\Big|\; n_x = 7,\ n_u = 1,\ \delta = 0.05$$

$$= 8(7+1) + 16\ln(80) = 64 + 16 \cdot 4.382 \approx 134.11$$

$$\Rightarrow \quad N \geq 135.$$

Here, G_T and F_T denote the finite-horizon controllability operators associated with the input and disturbance channels, respectively, σ_u is the standard deviation of the excitation input, and σ_w is the standard deviation of the process noise. These estimates show that the identification error decays at the standard Monte Carlo rate $N^{-1/2}$, while the bound on $\hat{A}$ is additionally modulated by the conditioning of the state covariance induced over the rollout horizon.

Motivated by this structure, the implementation uses practical uncertainty radii of the form

$$\beta_A := 16\sqrt{\frac{(n_x + 2n_u)\log(36/\delta)}{N}}, \qquad \beta_B := \frac{16\sigma_w}{\sigma_u}\sqrt{\frac{(n_x + 2n_u)\log(36/\delta)}{N}},$$

where N is chosen such that the above sample-size condition is satisfied. Relative to the theoretical bound, the practical radius β_A deliberately omits the factor $\lambda_{\min}\left(\sigma_u^2 G_T G_T^\top + \sigma_w^2 F_T F_T^\top\right)^{-1/2}$, thereby avoiding direct dependence on a model-based conditioning quantity. This produces a simpler and more conservative surrogate, suitable for robust synthesis when only empirical estimates are available.

E.9 Interpretation

The identification pipeline described above should be viewed as a structured decomposition of the observed dynamics into:

1. a nominal state/input model $(\hat{A}, \hat{B}_u)$;
2. an effective residual-driven disturbance channel $\hat{B}_w$;
3. output and performance maps consistent with the same disturbance realization;
4. a computable uncertainty indicator (β_A, β_B) derived from residual magnitude and regressor conditioning.

This decomposition does not claim to recover the true physical disturbance exactly. Rather, it provides a low-order uncertainty representation that is sufficiently rich to support the subsequent convex DRO-LMI design.

E.10 Algorithmic Summary

Algorithm 6 One-Step System Identification and Residual-Based Disturbance Construction

Input: trajectory data $(X_-, U_-, Y_-, Z_-, X_+)$
Parameters: $\rho > 0$, $\eta \in (0,1)$, $\sigma_w^2 > 0$, $\varepsilon > 0$
State regression
Form $D_x = [X_-; U_-]$
Compute $D_x^\dagger = D_x^\top (D_x D_x^\top + \rho I)^{-1}$
Compute $[\hat{A}\ \hat{B}_u] = X_+ D_x^\dagger$
Residual construction
Compute $R = X_+ - \hat{A} X_- - \hat{B}_u U_-$
Compute residual covariance $S_R = \frac{1}{T} R R^\top$
Eigendecompose $S_R = U_s \operatorname{diag}(s_i) U_s^\top$
Disturbance extraction
Select the smallest n_w such that $\sum_{i=1}^{n_w} s_i / \sum_{i=1}^{n_x} s_i \geq \eta$
Build $\hat{B}_w$ from the dominant residual eigenspace
Compute $\hat{W} = \hat{B}_w^\dagger R$
Output regressions
Compute $[\hat{C}_y\ \hat{D}_{yw}] = Y_- [X_-; \hat{W}]^\dagger$
Compute $[\hat{C}_z\ \hat{D}_{zu}\ \hat{D}_{zw}] = Z_- [X_-; U_-; \hat{W}]^\dagger$
Nominal covariance and uncertainty radius
Set $\Sigma_{\text{nom}} = \sigma_w^2 I_{n_w}$
Compute $\beta = \|R\|_F / \max(\sigma_{\min}(D_x), \varepsilon)$
Compute $\beta_A = \beta \sqrt{n_x/(n_x + n_u)}$, $\beta_B = \beta \sqrt{n_u/(n_x + n_u)}$
Output: $(\hat{A}, \hat{B}_u, \hat{B}_w, \hat{C}_y, \hat{D}_{yw}, \hat{C}_z, \hat{D}_{zu}, \hat{D}_{zw}, \Sigma_{\text{nom}}, \beta_A, \beta_B)$

F List of Final Algorithms and Limitations per Methods

The four data-driven formulations developed in Chapters 4–7 were introduced as successive attempts to overcome the main difficulty of robust output-feedback synthesis from data: namely, how to preserve the structure and guarantees of the model-based DRO-LMI framework when the plant matrices are not known exactly and must instead be inferred, approximated, or represented implicitly from measured trajectories.

Although each chapter contributed a meaningful methodological step, none of the resulting formulations fully resolved the problem in a satisfactory way. The reasons are different in each case. Some methods fail at the structural level, because the synthesis problem ceases to be convex once the data-driven parametrization is embedded into the lifted closed-loop matrices. Other methods remain convex, but only at the price of introducing relaxations that are too conservative, too weakly tied to the true identification error, or too dependent on intermediate model reconstruction. The purpose of this section is therefore to summarize, in a unified manner, the main limitation of each chapter and to clarify why none of these formulations can be regarded as a definitive, direct data-driven solution.

F.1 Baseline

$$\begin{aligned} \min_{\theta=\mathrm{vec}(A_c,B_c,C_c,D_c)} \quad & \bar{J}_T(\theta) + \mu_\rho\Big(\max\{0, \rho(\mathcal{A}(\theta)) - \rho_{\max}\}\Big)^2 \\ \text{s.t.} \quad & \underline{\theta}_i \le \theta_i \le \overline{\theta}_i, \qquad i = 1, \dots, n_\theta, \end{aligned}$$

where the Monte Carlo objective is

$$\bar{J}_T(\theta) = \frac{1}{N_s} \sum_{s=1}^{N_s} \frac{1}{T - T_{\text{burn}}} \sum_{t=T_{\text{burn}}}^{T-1} \|z_t^{(s)}\|_2^2,$$

and trajectories are generated according to

$$\chi_{t+1}^{(s)} = \mathcal{A}(\theta)\chi_t^{(s)} + \mathcal{B}(\theta)w_t^{(s)}, \qquad z_t^{(s)} = \mathcal{C}(\theta)\chi_t^{(s)} + \mathcal{D}(\theta)w_t^{(s)},$$

with

$$\chi_0^{(s)} = 0, \qquad w_t^{(s)} \in \mathcal{W}_{Wass}.$$

F.2 Chapter 4: Auxiliary-Inverse Data-Driven DRO-LMI

The formulation of Chapter 4 was motivated by the idea of reconstructing the unknown plant operators directly from trajectory data through auxiliary positive semidefinite variables that approximate inverse regularised data Gramians. At first sight, this appears attractive because it avoids a separate offline inversion step and embeds the estimation mechanism into the optimisation itself.

However, the method does not yield a genuinely convex data-driven synthesis problem. The core obstruction is structural. Even though the regularized inverse penalties are convex in the auxiliary matrices G_x, G_y, G_z, the reconstructed operators enter the lifted closed-loop matrices multiplicatively together with the controller and Lyapunov variables. As a result, the final semidefinite constraints are no longer affine in the decision variables, and the model-based DRO-LMI convexity is lost.

This is the fundamental reason why the formulation did not work as a practical synthesis method. The issue is not merely numerical, but conceptual: the attractive data-driven parametrization cannot be inserted directly into the output-feedback DRO-LMI without generating bilinear couplings. Any attempt to recover tractability, therefore, requires either freezing the inverse variables in advance or replacing the exact parametrisation by outer convex surrogates. In the first case, the method ceases to be purely direct data-driven, because it falls back to a two-stage identify-then-design procedure. In the second case, the resulting feasible set typically becomes too conservative to remain useful. Hence, the failure of this chapter lies in the impossibility of simultaneously achieving directness, exactness, and convexity within this parametrisation.

$$
\begin{aligned}
\min_{\substack{\lambda\geq 0,\ Q\succeq 0,\\ G_x\succ 0,\ G_y\succ 0,\ G_z\succ 0,\\ X=X^\top,\ Y=Y^\top,\ K,\ L,\ M,\ N}} \quad & \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \alpha\, J_{\text{est}} \\
\text{s.t.} \quad & P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \\
& \begin{cases} \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q-\lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, & w \in \mathcal{W}^{\text{corr}}_{\text{Wasserstein}}, \\ \begin{cases} \begin{bmatrix} -P & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -P & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \\ \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q-\lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \end{cases} & w \in \mathcal{W}^{\text{ind}}_{\text{Wasserstein}}. \end{cases}
\end{aligned}
$$

F.3 Chapter 5: Residual-Based Young-Type Relaxation

Chapter 5 addressed the non-convexity of the previous approach by abandoning direct substitution of the data-driven reconstruction into the exact synthesis constraints. Instead, the identification mismatch was interpreted as an uncertainty budget derived from the regression residual, and this uncertainty was injected into the lifted Lyapunov block through Young-type inequalities and auxiliary slack variables.

This modification successfully restores convexity, and for that reason, it constitutes an important methodological advance over the previous chapter. Nevertheless, the formulation remains limited by the nature of the relaxation itself. The uncertain perturbation is no longer represented exactly, but only through upper bounds. Consequently, the synthesis problem no longer corresponds to the true data-driven robust design problem, but to a conservative surrogate of it.

The main reason this method did not fully work is therefore conservatism. In the isotropic version, the uncertainty inflation acts uniformly across all state directions, regardless of the actual geometry of the residual mismatch. This produces a feasible set that can be unnecessarily small, especially when the estimation error is structured or anisotropic. Even in the directional version, only part of the uncertainty geometry is retained, while other components — especially the input-side contribution and the final state-block inflation — remain effectively scalar or isotropic. The method is thus more structured than the auxiliary-inverse formulation, but still too indirect in the way it propagates identification error into the synthesis stage.

A second limitation is that the uncertainty surrogates β_A and β_B are only residual-based proxies. They are computable and data-consistent, but they are not exact worst-case identification bounds. This means that the robustification may be either too pessimistic or insufficiently informative, depending on how well the residual norm reflects the true uncertainty affecting the lifted dynamics.

In short, the Young-based method works mathematically as a convex relaxation, but it does not work as a tight representation of the original problem. Its failure is therefore not loss of solvability, but loss of fidelity: convexity is recovered only by stepping away from the true structure of the identification error.

F.3.i Isotropic program. The isotropic Young-relaxed formulation can be written as

$$\min_{\substack{\lambda\geq 0,\ Q,\ X=X^\top,\ Y=Y^\top,\\ K,\ L,\ M,\ N,\ s_A,s_B,\tau_A,\tau_B,\\ t_K,t_L,t_M,t_N}} \quad \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \mu\left(s_A + \frac{\tau_A}{\beta_A^{\star 2}+\varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2}+\varepsilon} + t_K + t_L + t_M + t_N\right)$$

$$\text{s.t.} \quad P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succeq 0, \qquad Q \succeq 0, \qquad \lambda \geq 0,$$

$$\begin{bmatrix} s_A & \beta_A^\star \\ \beta_A^\star & \tau_A \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0,$$

spectral-norm epigraph (Appendix I.13) constraints for K, L, M, N,

$$S_A = S_B = \begin{bmatrix} I_{n_x} \\ 0 \end{bmatrix}, \qquad \Pi_A = -s_A I_{n_x},$$

$$\text{state_blk} = -P + (\tau_A + \tau_B) I_{2n_x},$$

$$\begin{cases} \mathcal{M}_{\text{corr}} \prec 0, & w \in \mathcal{W}_{\text{Wass}}^{\text{corr}}, \\ \mathcal{M}_1 \prec 0, \quad \mathcal{M}_2 \prec 0, & w \in \mathcal{W}_{\text{Wass}}^{\text{ind}}, \end{cases} \qquad \text{state_blk} \prec 0.$$

F.3.ii Directional program. The directional Young-relaxed formulation can be written as

$$\min_{\substack{\lambda\geq 0,\ Q,\ X=X^\top,\ Y=Y^\top,\\ K,\ L,\ M,\ N,\ \{s_{A,i},\tau_{A,i}\}_{i=1}^{n_x},\ s_B,\tau_B,\\ t_K,t_L,t_M,t_N}} \quad \begin{aligned} &\operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \\ &+ \mu\left(\sum_{i=1}^{n_x} s_{A,i} + \sum_{i=1}^{n_x} \frac{\tau_{A,i}}{\beta_{A,i}^{\star 2}+\varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2}+\varepsilon} + t_K + t_L + t_M + t_N\right)\end{aligned}$$

$$\text{s.t.} \quad P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succeq 0, \qquad Q \succeq 0, \qquad \lambda \geq 0,$$

$$\begin{bmatrix} s_{A,i} & \beta_{A,i}^\star \\ \beta_{A,i}^\star & \tau_{A,i} \end{bmatrix} \succeq 0, \qquad i = 1,\dots,n_x,$$

$$\begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0,$$

spectral-norm epigraph (Appendix I.13) constraints for K, L, M, N,

$$S_A = \begin{bmatrix} I_{n_x} \\ 0 \end{bmatrix} U_A, \qquad S_B = \begin{bmatrix} I_{n_x} \\ 0 \end{bmatrix}, \qquad \Pi_A = -\operatorname{Diag}(s_A),$$

$$\text{state_blk} = -P + \Big(\sum_{i=1}^{n_x} \tau_{A,i} + \tau_B\Big) I_{2n_x},$$

$$\begin{cases} \mathcal{M}_{\text{corr}} \prec 0, & & w \in \mathcal{W}_{\text{Wass}}^{\text{corr}}, \\ \mathcal{M}_1 \prec 0, & \mathcal{M}_2 \prec 0, & w \in \mathcal{W}_{\text{Wass}}^{\text{ind}}, \end{cases} \qquad \text{state_blk} \prec 0.$$

F.4 Chapter 6: Schur-Enhanced Young Relaxation

The Schur-enhanced formulation of Chapter 6 was introduced precisely to reduce the conservatism of the first Young-type relaxation. By augmenting the Lyapunov inequality with additional Schur-complement liftings and auxiliary matrices, the chapter sought to obtain a tighter convex surrogate of the residual-induced uncertainty propagation.

This refinement improves the mathematical sophistication of the relaxation, but it does not eliminate its core limitations. The first reason is that the approach still depends on the same identified nominal state-space model as the previous chapter. In other words, the synthesis problem remains model-mediated rather than fully direct. The uncertainty is not propagated from raw data in a behavioural or nonparametric way, but only through an intermediate realisation and a residual-based summary of its mismatch.

The second reason is that the Schur enhancement increases complexity without fully removing the gap between the relaxed constraints and the true uncertain lifted system. The additional variables S, σ_s, σ_p, H, and P_s improve the convex envelope, but they do not convert the relaxation into an exact uncertainty description. They merely produce a more elaborate conservative bound. Thus, the formulation remains vulnerable to the same conceptual criticism as the earlier Young-based method: the problem being solved is still a surrogate of the true data-driven robust synthesis task.

A third limitation is practical. The tighter lifting introduces more matrix variables, more semidefinite couplings, and more conditioning-sensitive constraints. As a consequence, the formulation may become numerically fragile, solver-dependent, or difficult to tune. In this sense, the method risks exchanging one difficulty for another: it reduces part of the conservatism, but increases the algebraic and numerical burden of the SDP.

Accordingly, this chapter did not fail in the same way as the auxiliary-inverse formulation, since convexity is preserved, nor in the same way as the first Young formulation, since the

robustification is tighter. Its limitation is that the refinement remains fundamentally incremental. It improves the previous relaxation, but does not overcome the deeper dependence on intermediate identification and conservative uncertainty propagation.

$$
\begin{aligned}
\min_{\substack{\lambda\geq 0,\ Q,\ X=X^\top,\ Y=Y^\top,\\ K,\ L,\ M,\ N,\ S,\ H,\ P_s,\\ \sigma_s\geq 0,\ \sigma_p\geq 0,\\ t_K,t_L,t_X,t_Y,t_M,t_N}} \quad & \operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \rho\left(t_K + t_L + t_X + t_Y + t_M + t_N\right) + \varepsilon \\
\text{s.t.} \quad & P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad G = \sigma_p I, \qquad P_{\text{bar}} = P - (\sigma_s + \sigma_p)\beta_E, \\
& Q \succ 0, \quad P \succ 0, \quad S \succ 0, \quad \sigma_s I - S \succ 0, \quad H \succ 0, \quad P_s \succ 0, \quad P_{\text{bar}} \succ 0, \\
& \begin{bmatrix} \sigma_p I & I \\ I & P \end{bmatrix} \succ 0, \qquad \begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0, \qquad \begin{bmatrix} G+H & I \\ I & P_s \end{bmatrix} \succ 0, \\
& \begin{cases} \begin{bmatrix} -P_{\text{bar}} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_{n_w} & \lambda I_{n_w} & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_{n_w} & -Q-\lambda I_{n_w} & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P_s & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_{n_z} \end{bmatrix} \prec 0, & w \in \mathcal{W}^{\text{corr}}_{\text{Wasserstein}}, \\ \begin{cases} \begin{bmatrix} -P_{\text{bar}} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -P_s & 0 \\ \mathbf{C} & 0 & -I_{n_z} \end{bmatrix} \prec 0, \\ \begin{bmatrix} -\lambda I_{n_w} & \lambda I_{n_w} & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_{n_w} & -Q-\lambda I_{n_w} & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I_{n_z} \end{bmatrix} \prec 0, \end{cases} & w \in \mathcal{W}^{\text{ind}}_{\text{Wasserstein}}. \end{cases}
\end{aligned}
$$

F.5 Chapter 7: Willems-Fundamental-Lemma-Based Formulation

The Willems-based chapter pursued the most ambitious departure from the previous methods. Instead of estimating a nominal plant and then robustifying against model error, it aimed to describe directly the set of plant matrices consistent with measured data directly via behavioural constraints inspired by Willems' Fundamental Lemma. Conceptually, this is the most appealing route, because it avoids committing to a single identified realisation and shifts the synthesis problem toward a genuinely data-consistent set-valued formulation.

However, this approach also did not fully work as a complete synthesis method. The main reason is that the behavioral consistency constraints, the lifted output-feedback change of variables, and the distributionally robust closed-loop conditions do not combine naturally into a single tractable exact program. In principle, the Willems-consistent set is rich and appealing, but in practice, one must still introduce relaxations, representative-model selections, frozen factorisations, or post-processing steps in order to obtain something computationally manageable.

This reveals the key limitation of the WFL route: while it reduces dependence on explicit identification, it transfers the difficulty to controller reconstruction and convexification. The exact equivalence between the behavioural representation, the lifted controller variables, and the realizable dynamic controller structure generally relies on restrictive structural assumptions. Once these assumptions are relaxed for numerical tractability, the connection with the original guarantee weakens. In particular, if the factorisation is approximated, if the controller is treated

too freely, or if the reconstruction is performed only after optimisation, then the feasibility of the optimisation problem no longer certifies the feasibility of the intended closed-loop realisation in the original sense.

Thus, the WFL chapter did not fail because the idea is weak; on the contrary, it is arguably the most conceptually promising of the four. It failed because the full behavioural vision is harder to integrate with dynamic output-feedback DRO-LMI synthesis than the nominal-model route. The problem is shifted rather than removed: explicit identification is avoided, but exact reconstruction, tractable convexification, and rigorous certification become much more delicate.

$$
\begin{aligned}
\min_{\substack{\lambda\geq 0,\, Q,\\ X=X^\top,\, Y=Y^\top,\\ P,K,L,M,N,\\ \mathcal{M}_P\in\mathcal{M}_{WFL},\, g}} \quad & \operatorname{trace}(Q\Sigma_{\text{nom}}) + \lambda\,\gamma^2 + \zeta(\mathcal{M}_P) \\
\text{s.t.} \quad & \begin{bmatrix}\mathcal{H}(x^+)\\ \mathcal{H}(y)\end{bmatrix} g = \mathcal{M}_P \begin{bmatrix}\mathcal{H}(x)\\ \mathcal{H}(u)\end{bmatrix} g + w, \\
& \mathbf{P} = \begin{bmatrix} P & \mathbf{0}_{2n_x\times n_z} \\ \mathbf{0}_{n_z\times 2n_x} & \mathbf{I}_{n_z}\end{bmatrix}, \quad P \succ 0 \\
& \begin{cases}
\begin{bmatrix} -\mathbf{P}\Big(\mathbf{I}_{2n_x+n_w} + (\lambda-1)\,\Psi_w\Psi_w^\top\Big) & \lambda\,\Psi_w & (\mathbf{P}\mathcal{M}_{CL})^\top \\ \lambda\,\Psi_w^\top & -Q-\lambda\,\mathbf{I}_{n_w} & \mathbf{0}_{n_w\times(2n_x+n_z)} \\ \mathbf{P}\mathcal{M}_{CL} & \mathbf{0}_{n_z\times n_w} & -\mathbf{P}\end{bmatrix} \prec 0, & w\in\mathcal{W}^{corr}_{\text{Wasserstein}} \\
\begin{cases}
\begin{bmatrix} -\Psi_z^\top\,\mathbf{P}\,\Psi_z & \begin{bmatrix}\mathbf{P}\;\mathcal{M}_{CL}\;\Psi_{2n_x}\end{bmatrix}^\top \\ \mathbf{P}\;\mathcal{M}_{CL}\;\Psi_{2n_x} & -\mathbf{P}\end{bmatrix} \prec 0 \\
\begin{bmatrix} -\lambda\,\mathbf{I}_{n_w} & \lambda\,\mathbf{I}_{n_w} & \begin{bmatrix}\mathbf{P}\;\mathcal{M}_{CL}\;\Psi_{n_w}\end{bmatrix}^\top \\ \lambda\,\mathbf{I}_{n_w} & -Q\;-\lambda\,\mathbf{I}_{n_w} & \mathbf{0}_{n_w\times(2n_x+n_z)} \\ \mathbf{P}\;\mathcal{M}_{CL}\;\Psi_{n_w} & \mathbf{0}_{(2n_x+n_z)\times n_w} & -\mathbf{P}\end{bmatrix} \prec 0
\end{cases} & w\in\mathcal{W}^{ind}_{\text{Wasserstein}}
\end{cases} \\
& \mathcal{M}_{CL} = \mathcal{M}_{CL}(\mathcal{M}_P, A_c, B_c, C_c, D_c),
\end{aligned}
$$

F.6 Overall Interpretation

Taken together, the four chapters reveal a recurring trade-off that appears intrinsic to this research direction.

If one seeks a direct and expressive data-driven parametrisation, as in Chapter 4, then convexity is generally lost. If one restores convexity through uncertainty relaxations, as in Chapters 5 and 6, then the formulation becomes conservative and remains dependent on an intermediate nominal model. If one attempts to avoid identification entirely through a behavioural description, as in Chapter 7, then the main difficulty reappears in the form of reconstruction, tractability, and loss of exact certification under practical relaxations.

In this sense, none of the four methods fully solves the direct data-driven distributionally robust output-feedback problem. Instead, each chapter identifies a different obstruction:

- Chapter 4: failure due to structural non-convexity;
- Chapter 5: failure due to coarse conservative relaxation;
- Chapter 6: failure due to tighter but still model-mediated and numerically heavier relaxation;

- Chapter 7: failure due to the difficulty of combining behavioural consistency, realisable controller reconstruction, and tractable robust synthesis in one exact framework.

The value of these chapters, therefore, lies not in having produced a final definitive method, but in mapping the methodological boundary of the problem. Each formulation clarifies which ingredients can be preserved simultaneously and which cannot. This negative knowledge is itself important: it shows that the tension between data-driven directness, convex tractability, and robust certification is not accidental, but central to the problem under study.

Method	**Convex**	**Direct**	**Fidelity**	**Numerical Robustness**	**Overall**
Baseline	×	△	✓	△	△
DeePC	×	✓	✓	×	×
Young	✓	△	×	✓	△
Young-Schur	✓	△	△	△	△
WFL	△	✓	△	×	△

Table F.1: Qualitative comparison of the considered methods in terms of convexity, direct data-driven formulation, fidelity to the true problem, and numerical robustness. Symbols: ✓ (satisfactory), △ (partial/limited), × (not satisfied).

Part II

Method-Specific Derivations

G Implementation of the Simulation-Based Baseline

G.1 Overview of the Computational Pipeline

The baseline approach seeks a dynamic output-feedback controller

$$\begin{bmatrix} x_{c,t+1} \\ u_t \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} x_{c,t} \\ y_t \end{bmatrix}$$

that minimizes the asymptotic closed-loop performance under stochastic excitation. Recalling the lifted closed-loop representation,

$$\chi_{t+1} = \mathcal{A}(\theta)\chi_t + \mathcal{B}(\theta)w_t, \tag{G.1}$$

$$z_t = \mathcal{C}(\theta)\chi_t + \mathcal{D}(\theta)w_t, \tag{G.2}$$

where

$$\chi_t = \begin{bmatrix} x_t^\top & x_{c,t}^\top \end{bmatrix}^\top, \qquad \theta \leftrightarrow (A_c, B_c, C_c, D_c),$$

the ideal optimization problem can be written as

$$\min_\theta J(\theta) \qquad \text{subject to} \qquad \rho(\mathcal{A}(\theta)) < 1, \tag{G.3}$$

where $\rho(\cdot)$ denotes the spectral radius and $J(\theta)$ is the steady-state average quadratic cost.

In general, (G.3) is not directly tractable because:

1. the cost is defined over an infinite horizon;
2. the expectation is taken with respect to a stochastic disturbance process;
3. the dependence of the cost on θ is nonlinear and non-convex;
4. the stability constraint $\rho(\mathcal{A}(\theta)) < 1$ is itself non-convex.

The implemented baseline therefore replaces the ideal problem by a computational pipeline based on:

1. finite-horizon closed-loop simulation;
2. Monte Carlo approximation of the stochastic objective;
3. spectral-radius penalization for instability;
4. bounded gradient-based numerical optimization.

At a high level, the procedure is:

Controller parameters θ $\longrightarrow$ closed-loop matrices $(\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D})$ $\longrightarrow$ Monte Carlo simulation $\longrightarrow$ empirical cost estimate $\longrightarrow$ stability penalty $\longrightarrow$ numerical optimiser

G.2 Controller Parameterization

The optimization variables are the entries of the dynamic output-feedback controller matrices

$$A_c \in \mathbb{R}^{n_c \times n_c}, \qquad B_c \in \mathbb{R}^{n_c \times n_y}, \qquad C_c \in \mathbb{R}^{n_u \times n_c}, \qquad D_c \in \mathbb{R}^{n_u \times n_y},$$

where n_c denotes the controller state dimension.

For numerical optimization, all controller matrices are stacked into a single parameter vector

$$\theta = \mathrm{vec}(A_c, B_c, C_c, D_c) \in \mathbb{R}^{n_\theta}. \tag{G.4}$$

Equivalently,

$$n_\theta = n_c^2 + n_c n_y + n_u n_c + n_u n_y.$$

This vectorized representation allows the use of a standard nonlinear optimization routine. At each iteration, the current vector θ is unpacked back into the matrices (A_c, B_c, C_c, D_c), from which the closed-loop matrices are assembled according to the formulas introduced in Section 2.1.

This parameterization has the advantage of being direct and implementation-friendly. However, it does not impose any structural guarantee of closed-loop stability or convexity. The resulting optimization problem is therefore solved as a general constrained nonlinear search in parameter space.

G.3 Finite-Horizon Approximation of the Cost

The ideal performance index is the long-run average output energy

$$J(\theta) = \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\Big[\|z_t\|_2^2\Big]. \tag{G.5}$$

In the implementation, this quantity is approximated by a finite-horizon empirical average. For a fixed disturbance realization and a finite simulation horizon T, the cost estimate is

$$\hat{J}_T(\theta) = \frac{1}{T - T_{\text{burn}}} \sum_{t=T_{\text{burn}}}^{T-1} \|z_t\|_2^2, \qquad T_{\text{burn}} < T, \tag{G.6}$$

where T_{burn} is a burn-in interval used to suppress transient effects.

The burn-in phase is important because the closed-loop state is initialized at zero in the simulations. Since the asymptotic cost of interest is a steady-state quantity, including the first samples in the average would bias the estimate toward transient behavior. By discarding the first T_{burn} samples, the empirical objective more closely approximates the stationary regime.

G.4 Closed-Loop Simulation

Given a candidate controller θ, the corresponding closed-loop matrices

$$\mathcal{A}(\theta), \quad \mathcal{B}(\theta), \quad \mathcal{C}(\theta), \quad \mathcal{D}(\theta)$$

are first assembled. The augmented state is initialized at

$$\chi_0 = 0. \tag{G.7}$$

Then, for each time step $t = 0, \ldots, T-1$, the simulation proceeds according to

$$z_t = \mathcal{C}(\theta)\chi_t + \mathcal{D}(\theta) w_t, \tag{G.8}$$

$$\chi_{t+1} = \mathcal{A}(\theta)\chi_t + \mathcal{B}(\theta) w_t. \tag{G.9}$$

At each step, the instantaneous quadratic contribution $\|z_t\|_2^2$ is accumulated. After discarding the burn-in phase, the empirical average (G.6) is returned as the estimated cost for that simulation.

The simulation procedure is summarized in Algorithm 7.

Algorithm 7 Finite-Horizon Cost Estimation for a Candidate Controller

Require: Controller parameters θ, nominal covariance Σ_w, horizon T, burn-in T_{burn}

```
Construct A(θ), B(θ), C(θ), D(θ)
Compute a factor L such that Σ_w ≈ LL^⊤
Initialize χ_0 = 0
Set cost ← 0, count ← 0
for t = 0, ..., T − 1 do
    Sample ξ_t ~ N(0, I) and set w_t = Lξ_t
    Compute z_t = C(θ)χ_t + D(θ)w_t
    Update χ_{t+1} = A(θ)χ_t + B(θ)w_t
    if t ≥ T_burn then
        cost ← cost + ||z_t||_2^2
        count ← count + 1
    end if
end for
return Ĵ_T(θ) = cost/count
```

G.5 Monte Carlo Averaging

Because a single finite-horizon simulation may exhibit non-negligible variance, the implementation uses multiple random seeds and averages the resulting empirical costs. Let N_s denote the number of Monte Carlo runs. For seed $s \in \{1, \dots, N_s\}$, let $\hat{J}_T^{(s)}(\theta)$ denote the empirical cost produced by Algorithm 7. The Monte Carlo objective is then

$$\bar{J}_T(\theta) = \frac{1}{N_s} \sum_{s=1}^{N_s} \hat{J}_T^{(s)}(\theta). \tag{G.10}$$

Averaging over multiple seeds reduces the sensitivity of the objective to a particular disturbance realization and yields a more reliable estimate of the underlying stochastic performance. This is especially important when the optimiser explores parameter regions near the stability boundary, where sample-path variability may become significant.

In the numerical implementation, two distinct uses of simulation are considered:

1. an *initial evaluation* of the starting controller, typically with a given horizon and a single seed;
2. an *optimization objective*, evaluated repeatedly using multiple seeds and a possibly different horizon.

This distinction is practical: the initial evaluation is primarily diagnostic, whereas the optimization stage benefits from variance reduction through averaging.

G.6 Stability Assessment and Penalization

The exact baseline problem requires the constraint

$$\rho(\mathcal{A}(\theta)) < 1,$$

which guarantees asymptotic stability of the discrete-time closed loop. Since this condition is non-convex and not directly imposed through LMIs in the baseline implementation, stability is handled heuristically.

G.6.i Spectral-radius check. For each candidate controller, the closed-loop matrix $\mathcal{A}(\theta)$ is formed and its spectral radius is computed:

$$\rho(\mathcal{A}(\theta)) = \max_i |\lambda_i(\mathcal{A}(\theta))|. \tag{G.11}$$

A controller is declared stable if

$$\rho(\mathcal{A}(\theta)) < \rho_{\max}, \tag{G.12}$$

where $\rho_{\max}$ is chosen slightly below 1 in order to include a numerical safety margin.

G.6.ii Penalty formulation. Instead of enforcing (G.12) as a hard constraint, the implementation augments the objective with a quadratic penalty whenever the spectral radius exceeds the prescribed threshold. The penalized objective can be written as

$$J_{\text{pen}}(\theta) = \bar{J}_T(\theta) + \mu_\rho \Big(\max\{0, \rho(\mathcal{A}(\theta)) - \rho_{\max}\} \Big)^2, \tag{G.13}$$

where $\mu_\rho > 0$ is a user-defined penalty coefficient.

This construction preserves the original empirical objective in the stable region and discourages the optimiser from exploring strongly unstable controllers. Although this does not provide a formal stability guarantee, it is simple to implement and works effectively as a soft barrier in practice.

G.6.iii Post-optimization projection. After the numerical optimization terminates, the final controller is checked once more for stability. If the resulting closed loop is still unstable, a simple projection heuristic is applied by shrinking selected controller matrices:

$$A_c \leftarrow \alpha A_c, \qquad D_c \leftarrow \alpha D_c, \qquad 0 < \alpha < 1. \tag{G.14}$$

The purpose of this step is not to solve a secondary optimization problem, but rather to provide a last-resort stabilization heuristic. By damping the internal controller dynamics and direct feedthrough, the method attempts to reduce the closed-loop spectral radius without redesigning the controller from scratch.

This mechanism highlights one of the main weaknesses of the baseline formulation: stability is not encoded structurally, but only checked and corrected a posteriori.

G.7 Numerical Optimization Routine

The baseline problem is ultimately solved as a bounded nonlinear optimization over the vector θ. Denoting by

$$\theta_0$$

the initial controller parameters, the implementation uses a quasi-Newton optimization method of L-BFGS-B type.

The numerical problem can be written as

$$\min_{\theta \in \Theta} J_{\text{pen}}(\theta), \tag{G.15}$$

where Θ is a box-constrained parameter set of the form

$$\Theta = \Big\{ \theta \in \mathbb{R}^{n_\theta} : \underline{\theta}_i \le \theta_i \le \overline{\theta}_i, \;\; i = 1, \dots, n_\theta \Big\}. \tag{G.16}$$

These bounds serve a practical regularization role. Since the optimization is non-convex and simulation-based, constraining the search region prevents the controller parameters from taking

excessively large values that would be difficult to interpret, numerically fragile, or physically implausible.

Algorithm 8 summarizes the complete optimization routine.

Algorithm 8 Simulation-Based Baseline Optimization

Require: Initial controller θ_0, covariance Σ_w, simulation parameters, optimization bounds

```
Set θ ← θ_0
repeat
    Unpack θ into (A_c, B_c, C_c, D_c)
    Assemble closed-loop matrices (𝒜, ℬ, 𝒞, 𝒟)
    Compute ρ(𝒜)
    Estimate J̄_T(θ) via Monte Carlo simulation
    Form penalized objective J_pen(θ)
    Update θ using L-BFGS-B
until convergence or maximum iterations reached
Recover the final controller matrices
if the resulting controller is unstable then
    Apply the projection heuristic (G.14)
end if
return optimized controller
```

Since the objective is simulation-based and potentially noisy, the optimiser should not be interpreted as solving the exact stochastic control problem. Rather, it computes a locally improved controller with respect to the chosen empirical surrogate objective.

G.8 Baseline Evaluation Workflow

Once the optimized controller has been obtained, the implementation performs a sequence of validation and bookkeeping steps:

1. evaluation of the initial controller cost;
2. optimization of the controller parameters;
3. evaluation of the optimized controller on a longer simulation;
4. computation of the final closed-loop spectral radius;
5. storage of the controller and plant data for reproducibility;
6. re-simulation of the closed loop using the saved configuration.

This workflow is useful for separating the *design phase* from the *validation phase*. In particular, it ensures that the reported performance is not only associated with transient optimization output, but also with a reproducible controller-plant pair that can later be reused for comparison against the LMI-based and data-driven approaches.

The stored data typically include:

- the nominal covariance Σ_w;
- the baseline cost of the initial controller;
- the optimiser status;
- the optimized cost;

- the final spectral radius;
- the optimized controller matrices;
- the plant matrices and the resulting composite closed-loop model.

This organisation supports a transparent experimental pipeline and simplifies subsequent post-processing, plotting, and cross-method comparisons.

H DeePC: Auxiliary Inverse Variables and Loss of Convexity

H.1 Variational Interpretation of the Auxiliary Matrices

This appendix derives the auxiliary inverse parametrization used in Chapter 4. Consider a positive definite matrix

$$S \in \mathbb{S}^n_{++},$$

and the function

$$\phi(G; S) := \operatorname{tr}(SG) - \log\det(G), \qquad G \in \mathbb{S}^n_{++}.$$

Its differential is

$$d\phi = \operatorname{tr}(S\, dG) - \operatorname{tr}(G^{-1} dG) = \operatorname{tr}\big((S - G^{-1})\, dG\big),$$

hence

$$\nabla_G \phi(G; S) = S - G^{-1}.$$

The stationary condition $\nabla_G \phi = 0$ yields

$$S - G^{-1} = 0 \qquad \Longleftrightarrow \qquad G = S^{-1}.$$

Since $\operatorname{tr}(SG)$ is linear in G and $-\log\det(G)$ is strictly convex on $\mathbb{S}^n_{++}$, the minimizer is unique:

$$\arg\min_{G \succ 0} \ \operatorname{tr}(SG) - \log\det(G) = S^{-1}.$$

Applying this construction to the regularized data Gramians

$$S_x := D_x D_x^\top + r_x I, \qquad S_y := D_y D_y^\top + r_y I, \qquad S_z := D_z D_z^\top + r_z I,$$

gives the intended meanings

$$G_x \approx S_x^{-1}, \qquad G_y \approx S_y^{-1}, \qquad G_z \approx S_z^{-1}.$$

Therefore the estimation objective

$$J_{\text{est}} = \mu_x\big(\operatorname{tr}(S_x G_x) - \log\det G_x\big) + \mu_y\big(\operatorname{tr}(S_y G_y) - \log\det G_y\big) + \mu_z\big(\operatorname{tr}(S_z G_z) - \log\det G_z\big)$$

is precisely a convex mechanism for promoting inverse-Gram behaviour.

Using these matrices, the data-driven operators are defined as

$$O_x = X_+ D_x^\top G_x, \qquad O_y = Y D_y^\top G_y, \qquad O_z = Z D_z^\top G_z,$$

with block extraction

$$\begin{bmatrix} A_x & B_u & B_w \end{bmatrix} = O_x, \qquad \begin{bmatrix} C_y & D_{yw} \end{bmatrix} = O_y, \qquad \begin{bmatrix} C_z & D_{zu} & D_{zw} \end{bmatrix} = O_z.$$

H.2 Origin of the Nonconvexity in the Lifted LMIs

Although the objective is convex in the auxiliary matrices, the full synthesis problem is not convex because the semidefinite constraints involve products between the reconstructed matrices and the controller variables.

From the definitions,

$$A_x = X_+ D_x^\top G_x^{(A)}, \qquad B_u = X_+ D_x^\top G_x^{(B_u)}, \qquad B_w = X_+ D_x^\top G_x^{(B_w)},$$

where $G_x^{(A)}, G_x^{(B_u)}, G_x^{(B_w)}$ denote the relevant column blocks of G_x. Similarly,

$$C_y = Y D_y^\top G_y^{(C_y)}, \qquad D_{yw} = Y D_y^\top G_y^{(D_{yw})},$$

and

$$C_z = Z D_z^\top G_z^{(C_z)}, \quad D_{zu} = Z D_z^\top G_z^{(D_{zu})}, \quad D_{zw} = Z D_z^\top G_z^{(D_{zw})}.$$

Substituting these expressions into the lifted matrices gives

$$\mathcal{A} = \begin{bmatrix} A_x Y + B_u M & A_x + B_u N C_y \\ K & X A_x + L C_y \end{bmatrix},$$

$$\mathcal{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ X B_w + L D_{yw} \end{bmatrix},$$

$$\mathcal{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \qquad \mathcal{D} = D_{zw} + D_{zu} N D_{yw}.$$

Hence, terms of the following types appear:

$$A_x Y = (X_+ D_x^\top G_x^{(A)}) Y, \qquad B_u M = (X_+ D_x^\top G_x^{(B_u)}) M,$$

$$B_u N C_y = (X_+ D_x^\top G_x^{(B_u)})\, N\, (Y D_y^\top G_y^{(C_y)}),$$

$$X A_x = X (X_+ D_x^\top G_x^{(A)}), \qquad L C_y = L (Y D_y^\top G_y^{(C_y)}),$$

$$D_{zu} M = (Z D_z^\top G_z^{(D_{zu})}) M,$$

$$D_{zu} N C_y = (Z D_z^\top G_z^{(D_{zu})})\, N\, (Y D_y^\top G_y^{(C_y)}).$$

These expressions are bilinear or trilinear in the decision variables. Therefore the coefficients of the semidefinite constraints are not affine in the unknowns. As a result, the matrix inequalities are not LMIs in the usual convex SDP sense.

The source of the nonconvexity is thus not merely the inverse representation itself, but the multiplicative interaction between:

- the auxiliary inverse variables G_x, G_y, G_z,
- the reconstructed plant matrices,
- and the controller synthesis variables X, Y, K, L, M, N.

H.3 Why Convex Relaxations Become Restrictive and Indirect

A natural idea is to recover convexity by replacing the exact inverse representation with convex constraints. For instance, one may impose Schur-complement bounds such as

$$\begin{bmatrix} S_x & I \\ I & G_x \end{bmatrix} \succeq 0 \qquad \Longrightarrow \qquad G_x \succeq S_x^{-1},$$

and similarly for G_y, G_z.

Such relaxations may help preserve convexity in some subproblems, but they introduce substantial conservatism. Indeed, the matrix G_x is no longer forced to equal the desired inverse; it is only constrained to dominate it in Loewner order. Once inserted into the reconstructed model matrices, this over-approximation propagates through the lifted closed-loop realization and generally strengthens the semidefinite constraints beyond what is intrinsically required.

Likewise, another possibility is to solve first

$$G_x^\star = S_x^{-1}, \qquad G_y^\star = S_y^{-1}, \qquad G_z^\star = S_z^{-1},$$

or equivalently identify the nominal matrices offline, and then solve the DRO-LMI with these quantities frozen. This restores convexity of the synthesis stage, but at the cost of separating estimation and control. In that case, the overall method is no longer a direct data-driven synthesis formulation; rather, it becomes a model-based LMI design applied to a data-identified surrogate model.

Therefore, the convex alternatives suffer from two limitations:

1. they are more conservative and may become too restrictive in practice;
2. they are not purely direct data-driven, because they rely on an intermediate parametric model reconstruction.

This explains why the auxiliary inverse formulation, although mathematically appealing, does not fully resolve the structural incompatibility between inverse-Gram identification and convex direct LMI synthesis.

I Supplementary Derivations for the Residual-Based Young Methodology

I.1 Overview

This appendix derives the convex Young-based relaxations used in the data-driven DRO-LMI synthesis procedure. It builds upon the identification framework of Appendix E, where the nominal matrices $(\hat{A}, \hat{B}_u, \hat{B}_w, \hat{C}_y, \hat{D}_{yw}, \hat{C}_z, \hat{D}_{zu}, \hat{D}_{zw})$ and the residual-based uncertainty surrogates (β_A, β_B) are obtained.

The goal of the present appendix is to show how the state-regression uncertainty is injected into the lifted output-feedback DRO-LMI without destroying convexity. Two variants are considered:

- an isotropic Young-type relaxation, based on scalar slack variables;
- a residual-directional relaxation, where the uncertainty associated with the state matrix is distributed along the principal directions of the residual covariance.

The exposition is divided into:

1. construction of the nominal lifted closed-loop matrices;
2. explicit decomposition of the uncertain lifted state block;
3. Young-type convexification of the mixed terms;
4. LMI embedding through selector augmentation and Schur complements;
5. final correlated and independent formulations.

I.2 Nominal Lifted Closed-Loop Matrices

We start from the standard output-feedback change of variables

$$P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad X = X^\top, \quad Y = Y^\top.$$

The controller decision variables are

$$K \in \mathbb{R}^{n_x \times n_x}, \qquad L \in \mathbb{R}^{n_x \times n_y}, \qquad M \in \mathbb{R}^{n_u \times n_x}, \qquad N \in \mathbb{R}^{n_u \times n_y}.$$

Using the identified nominal plant and output matrices, define the lifted closed-loop blocks

$$\mathbf{A} = \begin{bmatrix} \hat{A}Y + \hat{B}_u M & \hat{A} + \hat{B}_u N \hat{C}_y \\ K & X\hat{A} + L\hat{C}_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} \hat{B}_w + \hat{B}_u N \hat{D}_{yw} \\ X\hat{B}_w + L\hat{D}_{yw} \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} \hat{C}_z Y + \hat{D}_{zu} M & \hat{C}_z + \hat{D}_{zu} N \hat{C}_y \end{bmatrix},$$

$$\mathbf{D} = \hat{D}_{zw} + \hat{D}_{zu} N \hat{D}_{yw}.$$

The DRO variables are

$$\lambda \geq 0, \qquad Q \succeq 0.$$

The nominal objective is

$$J_{\mathrm{DRO}} = \mathrm{tr}(Q\Sigma_{\mathrm{nom}}) + \lambda\gamma^2.$$

I.3 Nominal Correlated and Independent Kernels

In the correlated case, the nominal DRO-LMI kernel is

$$\Xi_{\mathrm{corr}}(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) := \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0. \tag{I.1}$$

In the independent case, the nominal structure separates into two LMIs:

$$\Xi_{\mathrm{ind},1}(\mathbf{A}, \mathbf{C}) := \begin{bmatrix} -P & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -P & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \tag{I.2}$$

and

$$\Xi_{\mathrm{ind},2}(\mathbf{B}, \mathbf{D}) := \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0. \tag{I.3}$$

The nominal state block $-P$ appearing in these matrices is the location where the Young-type regularization will be injected.

I.4 Isolation of the Lyapunov Generator

The point at which the identification uncertainty enters the synthesis problem can be understood by isolating the Lyapunov-generating part of the nominal DRO-LMI kernel. In the correlated case, this corresponds to the $(1, 4)$–$(4, 1)$ corner

$$M(\mathbf{A}, P) := \begin{bmatrix} -P & \mathbf{A}^\top \\ \mathbf{A} & -P \end{bmatrix} \prec 0.$$

In the independent formulation, the same structure appears in the first LMI, namely in the block coupling $-P$ and $\mathbf{A}$. Therefore, in both formulations, the state block $\mathbf{A}$ interacts with the Lyapunov variable P through the same fundamental matrix inequality.

Taking the Schur complement of the lower-right block $-P$, one obtains

$$-P - \mathbf{A}^\top(-P)^{-1}\mathbf{A} = -P + \mathbf{A}^\top P^{-1}\mathbf{A} \prec 0.$$

This is a discrete-time stability condition written in inverse-P coordinates.

To recover the more familiar Lyapunov form, apply a congruence transformation with $P^{1/2}$:

$$P^{1/2}\big(-P + \mathbf{A}^\top P^{-1}\mathbf{A}\big)P^{1/2} \prec 0.$$

Introduce the scaled state matrix

$$\widetilde{\mathbf{A}} := P^{1/2}\mathbf{A}P^{-1/2}.$$

Then

$$\widetilde{\mathbf{A}}^\top\widetilde{\mathbf{A}} = (P^{-1/2}\mathbf{A}^\top P^{1/2})(P^{1/2}\mathbf{A}P^{-1/2}) = P^{-1/2}\mathbf{A}^\top P\mathbf{A}P^{-1/2},$$

which implies

$$\mathbf{A}^\top P\mathbf{A} = P^{1/2}\widetilde{\mathbf{A}}^\top\widetilde{\mathbf{A}}P^{1/2}.$$

Substituting this into the previous expression gives

$$\widetilde{\mathbf{A}}^\top\widetilde{\mathbf{A}} - I \prec 0,$$

which is the normalised discrete-time contraction condition. Undoing the normalisation yields the standard Lyapunov inequality

$$\mathbf{A}^\top P\mathbf{A} - P \prec 0.$$

This derivation clarifies why the uncertainty treatment developed in this appendix is injected precisely into the block $-P$ and the coupling terms involving $\mathbf{A}$: these are exactly the terms that generate the Lyapunov stability condition. Consequently, once $\mathbf{A}$ is perturbed by the identification errors associated with A and B_u, the resulting non-convex terms must be controlled at this location of the kernel.

I.5 Uncertain Lifted State Block

Assume that the true state and input matrices are

$$A = \hat{A} + \Delta_A, \qquad B_u = \hat{B}_u + \Delta_B.$$

Since the lifted block $\mathbf{A}$ depends affinely on A and B_u, the true lifted state block can be written as

$$\mathbf{A}_{\text{true}} = \mathbf{A} + E_A + E_B,$$

where E_A collects the contribution of Δ_A, and E_B the contribution of Δ_B.

By direct substitution one obtains

$$E_A = \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix}, \qquad E_B = \begin{bmatrix} \Delta_B M & \Delta_B N\hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

Indeed, substituting $A = \hat{A} + \Delta_A$ in $\mathbf{A}$ gives

$$\begin{bmatrix} (\hat{A}+\Delta_A)Y + \hat{B}_u M & (\hat{A}+\Delta_A) + \hat{B}_u N\hat{C}_y \\ K & X(\hat{A}+\Delta_A) + L\hat{C}_y \end{bmatrix} = \mathbf{A} + \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix},$$

while substituting $B_u = \hat{B}_u + \Delta_B$ gives

$$\begin{bmatrix} \hat{A}Y + (\hat{B}_u+\Delta_B)M & \hat{A} + (\hat{B}_u+\Delta_B)N\hat{C}_y \\ K & X\hat{A} + L\hat{C}_y \end{bmatrix} = \mathbf{A} + \begin{bmatrix} \Delta_B M & \Delta_B N\hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

I.6 Expansion of the Lyapunov Generator

The non-convex difficulty appears when replacing $\mathbf{A}$ by $\mathbf{A}_{\text{true}}$ inside the Lyapunov part. Formally,

$$\mathbf{A}_{\text{true}}^\top P\mathbf{A}_{\text{true}} - P = (\mathbf{A} + E_A + E_B)^\top P(\mathbf{A} + E_A + E_B) - P.$$

Expanding gives

$$\begin{aligned}\mathbf{A}_{\text{true}}^\top P\mathbf{A}_{\text{true}} - P = \mathbf{A}^\top P\mathbf{A} - P \\ &+ \mathbf{A}^\top PE_A + E_A^\top P\mathbf{A} \\ &+ \mathbf{A}^\top PE_B + E_B^\top P\mathbf{A} \\ &+ E_A^\top PE_A + E_B^\top PE_B \\ &+ E_A^\top PE_B + E_B^\top PE_A. \end{aligned} \tag{I.4}$$

The troublesome terms are the mixed bilinear contributions

$$\mathbf{A}^\top PE_A + E_A^\top P\mathbf{A}, \qquad \mathbf{A}^\top PE_B + E_B^\top P\mathbf{A},$$

because they couple uncertainty and decision variables.

Introducing

$$Z := P\mathbf{A},$$

they can be written as Frobenius inner products:

$$\mathbf{A}^\top PE_A + E_A^\top P\mathbf{A} = 2\langle E_A, Z\rangle,$$

$$\mathbf{A}^\top PE_B + E_B^\top P\mathbf{A} = 2\langle E_B, Z\rangle.$$

The higher-order terms in (I.4) are conservatively absorbed into the relaxation, which is why the final formulation is convex but conservative.

I.7 Young-Type Relaxation of the Mixed Terms

For any matrices U, V and any scalar $s > 0$,

$$2\langle U, V\rangle \le s\|U\|_F^2 + \frac{1}{s}\|V\|_F^2.$$

Applying this inequality to the two mixed terms yields

$$2\langle E_A, Z\rangle \le s_A\|E_A\|_F^2 + \frac{1}{s_A}\|Z\|_F^2, \qquad s_A > 0,$$

and

$$2\langle E_B, Z\rangle \le s_B\|E_B\|_F^2 + \frac{1}{s_B}\|Z\|_F^2, \qquad s_B > 0.$$

Hence, once computable bounds on $\|E_A\|_F$ and $\|E_B\|_F$ are available, the mixed bilinear terms can be replaced by convex upper bounds.

I.8 Bounds on E_A and E_B

Using the submultiplicativity of the Frobenius norm,

$$\|UVW\|_F \le \|U\|_2\,\|V\|_F\,\|W\|_2,$$

we obtain

$$\begin{aligned}\|E_A\|_F^2 &= \|\Delta_A Y\|_F^2 + \|\Delta_A\|_F^2 + \|X\Delta_A\|_F^2 \\ &\le (\|Y\|_2^2 + 1 + \|X\|_2^2)\|\Delta_A\|_F^2, \end{aligned} \tag{I.5}$$

and

$$\begin{aligned}\|E_B\|_F^2 &= \|\Delta_B M\|_F^2 + \|\Delta_B N\hat{C}_y\|_F^2 \\ &\le (\|M\|_2^2 + \|N\hat{C}_y\|_2^2)\|\Delta_B\|_F^2.\end{aligned} \tag{I.6}$$

Therefore,

$$\|E_A\|_F \le \sqrt{1 + \|X\|_2^2 + \|Y\|_2^2}\,\|\Delta_A\|_F,$$

$$\|E_B\|_F \le \sqrt{\|M\|_2^2 + \|N\hat{C}_y\|_2^2}\,\|\Delta_B\|_F.$$

In practice, the code does not optimize these decision-dependent norm multipliers exactly. Instead, it replaces them with computable surrogates derived from the residual-based uncertainty radii:

$$\beta_A, \qquad \beta_B.$$

Depending on the selected relaxation, these surrogates are used either isotropically or directionally.

I.9 Isotropic Young-Type Relaxation

The isotropic version uses scalar slacks

$$s_A \ge 0, \qquad \tau_A \ge 0, \qquad s_B \ge 0, \qquad \tau_B \ge 0.$$

The relation between s_A, τ_A and the uncertainty radius $\beta_A^\star$ is enforced through the 2×2 PSD condition

$$\begin{bmatrix} s_A & \beta_A^\star \\ \beta_A^\star & \tau_A \end{bmatrix} \succeq 0,$$

and similarly

$$\begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0.$$

By the Schur complement, these imply

$$\tau_A \ge \frac{\beta_A^{\star 2}}{s_A}, \qquad \tau_B \ge \frac{\beta_B^{\star 2}}{s_B}.$$

The state-corner regularization is then defined as

$$\text{state_blk} = -P + (\tau_A + \tau_B) I_{2n_x}.$$

The selector matrices are

$$S_A = S_B = \begin{bmatrix} I_{n_x} \\ 0_{n_x \times n_x} \end{bmatrix},$$

and the Young slack block associated with the A-uncertainty is

$$\Pi_A = -s_A I_{n_x}.$$

I.10 Residual-Directional Relaxation

The isotropic relaxation can be refined by using the anisotropy of the residual covariance. Let

$$S_R = U_A \operatorname{diag}(\sigma_1, \dots, \sigma_{n_x})\, U_A^\top$$

be the eigendecomposition of the residual covariance, with $\sigma_1 \geq \cdots \geq \sigma_{n_x} \geq 0$.

Define normalized directional weights

$$w_A^{(i)} = \frac{\sqrt{\sigma_i}}{\max_j \sqrt{\sigma_j}}, \qquad i = 1, \dots, n_x.$$

Then the uncertainty budget associated with Δ_A is distributed as

$$\beta_{A,i}^\star = \beta_A w_A^{(i)}, \qquad i = 1, \dots, n_x.$$

The A-part now uses vector slacks

$$s_A \in \mathbb{R}_{\geq 0}^{n_x}, \qquad \tau_A \in \mathbb{R}_{\geq 0}^{n_x},$$

and imposes, for each $i = 1, \dots, n_x$,

$$\begin{bmatrix} s_{A,i} & \beta_{A,i}^\star \\ \beta_{A,i}^\star & \tau_{A,i} \end{bmatrix} \succeq 0.$$

The B-part remains scalar:

$$\begin{bmatrix} s_B & \beta_B^\star \\ \beta_B^\star & \tau_B \end{bmatrix} \succeq 0.$$

The selector associated with the A-uncertainty is rotated along the residual eigendirections:

$$S_A = \begin{bmatrix} I_{n_x} \\ 0_{n_x \times n_x} \end{bmatrix} U_A,$$

while

$$S_B = \begin{bmatrix} I_{n_x} \\ 0_{n_x \times n_x} \end{bmatrix}.$$

The block associated with the A-part is now

$$\Pi_A = -\operatorname{Diag}(s_A),$$

and the regularized state corner becomes

$$\text{state_blk} = -P + \Big(\sum_{i=1}^{n_x} \tau_{A,i} + \tau_B\Big) I_{2n_x}.$$

Thus, the directional variant preserves convexity while replacing isotropic slack allocation with an anisotropic one informed by the empirical geometry of the residuals.

I.11 Selector Augmentation and Schur-Complement Embedding

The terms $\frac{1}{s_A}\|Z\|_F^2$ and $\frac{1}{s_B}\|Z\|_F^2$ are embedded into the final LMI through an augmented Schur-complement construction.

In the correlated case, this produces

$$\mathcal{M}_{\text{corr}} = \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top & 0 & 0 \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top & 0 & 0 \\ 0 & \lambda I_w & -Q-\lambda I_w & 0 & 0 & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & \text{state_blk} & 0 & S_A & S_B \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z & 0 & 0 \\ 0 & 0 & 0 & S_A^\top & 0 & \Pi_A & 0 \\ 0 & 0 & 0 & S_B^\top & 0 & 0 & -s_B I_{n_x} \end{bmatrix} \prec 0.$$

By Schur complement, the augmented rows and columns encode the negative quadratic penalties associated with the Young terms. Intuitively, the selectors S_A, S_B determine where in the lifted Lyapunov structure the uncertainty acts; in this implementation, they affect the state coordinates only.

I.12 Independent Formulation

In the independent case, the same Young-type construction is applied, but the robust condition is split into two LMIs.

The first one contains the Lyapunov and Young-relaxation terms:

$$\mathcal{M}_1 = \begin{bmatrix} -P & \mathbf{A}^\top & \mathbf{C}^\top & 0 & 0 \\ \mathbf{A} & \text{state_blk} & 0 & S_A & S_B \\ \mathbf{C} & 0 & -I_z & 0 & 0 \\ 0 & S_A^\top & 0 & \Pi_A & 0 \\ 0 & S_B^\top & 0 & 0 & -s_B I_{n_x} \end{bmatrix} \prec 0.$$

The second one contains the disturbance-DRO coupling:

$$\mathcal{M}_2 = \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q-\lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0.$$

Hence, the difference between the correlated and independent models lies in the way the disturbance uncertainty couples with the Lyapunov block:

- in the correlated case, all terms appear in a single augmented LMI;
- in the independent case, the disturbance part is separated into a second condition.

I.13 Spectral-Norm Epigraph Regularization

To prevent the controller-related matrices from becoming excessively large, the implementation introduces spectral norm epigraph variables

$$t_K, \quad t_L, \quad t_M, \quad t_N \geq 0,$$

together with the LMIs

$$\begin{bmatrix} t_K I_{n_x} & K \\ K^\top & t_K I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_L I_{n_x} & L \\ L^\top & t_L I_{n_y} \end{bmatrix} \succeq 0,$$

$$\begin{bmatrix} t_M I_{n_u} & M \\ M^\top & t_M I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_N I_{n_u} & N \\ N^\top & t_N I_{n_y} \end{bmatrix} \succeq 0.$$

These imply

$$\|K\|_2 \le t_K, \qquad \|L\|_2 \le t_L, \qquad \|M\|_2 \le t_M, \qquad \|N\|_2 \le t_N.$$

I.14 Convex Regularization in the Objective

The total objective solved by the code is

$$\min \ \mathrm{tr}(Q\Sigma_{\mathrm{nom}}) + \lambda\gamma^2 + J_{\mathrm{reg}},$$

where the regularization term has two components:

$$J_{\mathrm{reg}} = J_{\mathrm{slack}} + J_{\mathrm{spec}}.$$

In the isotropic case,

$$J_{\mathrm{slack}} = \mu \left(s_A + \frac{\tau_A}{\beta_A^{\star 2} + \varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2} + \varepsilon} \right),$$

whereas in the directional case,

$$J_{\mathrm{slack}} = \mu \left(\sum_{i=1}^{n_x} s_{A,i} + \sum_{i=1}^{n_x} \frac{\tau_{A,i}}{\beta_{A,i}^{\star 2} + \varepsilon} + s_B + \frac{\tau_B}{\beta_B^{\star 2} + \varepsilon} \right).$$

The spectral regularization is

$$J_{\mathrm{spec}} = \mu(t_K + t_L + t_M + t_N).$$

This regularization has a practical rather than theoretical role: it discourages excessively large controller matrices and keeps the slack variables bounded, thereby improving numerical behavior and solution conditioning.

I.15 Final Summary of the Relaxation

The complete residual-based Young methodology can now be summarized as follows:

1. identify the nominal matrices and residual-based uncertainty quantities from data;
2. construct the lifted nominal closed-loop matrices $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$;
3. decompose the uncertainty in the lifted state block into an A-part and a B-part;
4. upper-bound the mixed uncertainty terms by Young inequalities;
5. encode those bounds through PSD slack constraints and selector augmentation;
6. solve either the correlated or independent augmented DRO-LMI;
7. recovery of the matrices-error post-optimisation through the method proposed in the Appendix J.4.

This produces a convex, data-consistent synthesis method that incorporates identification uncertainty directly into the controller design stage, at the price of a conservative relaxation.

J Supplementary Derivations for the Schur-Enhanced Young-Type DRO-LMI

J.1 Detailed derivation of the Schur-enhanced relaxation

This appendix provides the detailed mathematical derivation underlying the Schur-enhanced Young-type relaxation presented in Chapter 6. The goal is to explain how the data-driven identification error is embedded into the lifted Lyapunov inequality, and how the final convex semidefinite constraints arise from successive upper bounds and Schur-complement arguments.

J.1.i Nominal lifted matrices. Recall the lifted variables

$$P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0,$$

and the nominal lifted closed-loop matrices

$$\mathbf{A} = \begin{bmatrix} \hat{A}_x Y + \hat{B}_u M & \hat{A}_x + \hat{B}_u N \hat{C}_y \\ K & X\hat{A}_x + L\hat{C}_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} \hat{B}_w + \hat{B}_u N \hat{D}_{yw} \\ X\hat{B}_w + L\hat{D}_{yw} \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} \hat{C}_z Y + \hat{D}_{zu} M & \hat{C}_z + \hat{D}_{zu} N \hat{C}_y \end{bmatrix}, \qquad \mathbf{D} = \hat{D}_{zw} + \hat{D}_{zu} N \hat{D}_{yw}.$$

The nominal correlated-kernel DRO-LMI is

$$\Xi(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) := \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I & -Q - \lambda I & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0. \tag{J.1}$$

The independent-noise case is obtained by splitting the state/performance part and the disturbance part into two separate LMIs, but the derivation of the uncertainty treatment begins from the same lifted state block.

J.1.ii Lyapunov core and perturbation injection. The core Lyapunov block is

$$M(\mathbf{A}_{\text{true}}, P) := \begin{bmatrix} -P & \mathbf{A}_{\text{true}}^\top \\ \mathbf{A}_{\text{true}} & -P \end{bmatrix} \prec 0.$$

By the Schur complement, this is equivalent to

$$-P + \mathbf{A}_{\text{true}}^\top P^{-1} \mathbf{A}_{\text{true}} \prec 0.$$

Now let

$$A_x = \hat{A}_x + \Delta_A, \qquad B_u = \hat{B}_u + \Delta_B.$$

Since the lifted matrix $\mathbf{A}$ depends affinely on A_x and B_u, one can write

$$\mathbf{A}_{\text{true}} = \mathbf{A} + E,$$

where $\mathbf{A}$ is the nominal lifted matrix built from $(\hat{A}_x, \hat{B}_u)$, and E collects the perturbation terms generated by Δ_A, Δ_B. Expanding the dependence gives

$$E = \begin{bmatrix} \Delta_A Y + \Delta_B M & \Delta_A + \Delta_B N \hat{C}_y \\ 0 & X\Delta_A \end{bmatrix}.$$

Substituting into the inverse-metric inequality yields

$$-P + (\mathbf{A} + E)^\top P^{-1} (\mathbf{A} + E) \prec 0,$$

that is,

$$-P + \mathbf{A}^\top P^{-1}\mathbf{A} + \mathbf{A}^\top P^{-1} E + E^\top P^{-1}\mathbf{A} + E^\top P^{-1} E \prec 0.$$

Therefore the nominal term is accompanied by:

- a quadratic perturbation term $E^\top P^{-1} E$,
- a mixed perturbation term $\mathbf{A}^\top P^{-1} E + E^\top P^{-1}\mathbf{A}$.

J.1.iii Residual-based bound on the perturbation block. At the implementation level, the code constructs residual-based surrogates Δ_a, Δ_b and defines

$$\kappa_A := \|\Delta_a\|_2, \qquad \kappa_B := \|\Delta_b\|_2, \qquad c_y := \|\hat{C}_y\|_2.$$

Assuming bounds

$$\|Y\|_2 \le t_Y, \qquad \|M\|_2 \le t_M, \qquad \|N\|_2 \le t_N, \qquad \|X\|_2 \le t_X,$$

one obtains

$$\begin{aligned} \|\Delta_A Y + \Delta_B M\|_2 &\le \|\Delta_A\|_2 \|Y\|_2 + \|\Delta_B\|_2 \|M\|_2 \le \kappa_A t_Y + \kappa_B t_M =: e_{11}, \\ \|\Delta_A + \Delta_B N \hat{C}_y\|_2 &\le \|\Delta_A\|_2 + \|\Delta_B\|_2 \|N\|_2 \|\hat{C}_y\|_2 \le \kappa_A + \kappa_B t_N c_y =: e_{12}, \\ \|X\Delta_A\|_2 &\le \|X\|_2 \|\Delta_A\|_2 \le t_X \kappa_A =: e_{22}. \end{aligned}$$

Consequently,

$$\|E\|_F^2 \le e_{11}^2 + e_{12}^2 + e_{22}^2 =: \gamma_E,$$

and therefore

$$E^\top E \preceq \gamma_E I_{2n_x}.$$

Define

$$\beta_E := \gamma_E I_{2n_x}.$$

This is exactly the isotropic perturbation-energy bound used in the implemented `new=True` branch.

J.1.iv Upper bound on the inverse Lyapunov metric. To control the quadratic term $E^\top P^{-1}E$, introduce $\sigma_p > 0$ such that

$$P^{-1} \prec \sigma_p I.$$

By the Schur complement, this is equivalent to

$$\begin{bmatrix} \sigma_p I & I \\ I & P \end{bmatrix} \succ 0.$$

Hence

$$E^\top P^{-1}E \preceq \sigma_p E^\top E \preceq \sigma_p \beta_E.$$

J.1.v Bounding the mixed term. For any $S \succ 0$, Young's inequality yields

$$\mathbf{A}^\top P^{-1}E + E^\top P^{-1}\mathbf{A} \preceq \mathbf{A}^\top P^{-1}S^{-1}P^{-1}\mathbf{A} + E^\top SE.$$

Now introduce a diagonal slack matrix

$$S = \operatorname{diag}(s_A, s_B) \succ 0,$$

and impose

$$\sigma_s I - S \succ 0$$

for some $\sigma_s > 0$. Then

$$S \prec \sigma_s I,$$

and consequently

$$E^\top SE \preceq \sigma_s E^\top E \preceq \sigma_s \beta_E.$$

Combining the previous bounds gives

$$-P + \mathbf{A}^\top P^{-1}\mathbf{A} + \mathbf{A}^\top P^{-1}S^{-1}P^{-1}\mathbf{A} + (\sigma_s + \sigma_p)\beta_E \prec 0.$$

J.1.vi Definition of the shrunken Lyapunov block. Define

$$\bar{P} := P - (\sigma_s + \sigma_p)\beta_E.$$

Then the previous inequality becomes

$$-\bar{P} + \mathbf{A}^\top \Big(P^{-1} + P^{-1}S^{-1}P^{-1}\Big)\mathbf{A} \prec 0.$$

The remaining difficulty is that the term

$$P^{-1} + P^{-1}S^{-1}P^{-1}$$

is not directly affine in the optimization variables. The Schur-enhanced formulation replaces this by a convex upper-bound construction.

J.1.vii Schur lifting with G, H, and P_s. Define

$$G := \sigma_p I.$$

Since $P^{-1} \prec \sigma_p I$, one has

$$P^{-1} \prec G.$$

Next introduce a matrix $H \succ 0$ such that

$$\begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0.$$

By the Schur complement,

$$H - GS^{-1}G \succ 0,$$

hence

$$GS^{-1}G \prec H.$$

Since $P^{-1} \prec G$, this provides the convex upper-bound chain

$$P^{-1}S^{-1}P^{-1} \;\prec\; GS^{-1}G \;\prec\; H.$$

Therefore

$$P^{-1} + P^{-1}S^{-1}P^{-1} \prec G + H.$$

To obtain an LMI representation of this upper bound, introduce $P_s \succ 0$ satisfying

$$\begin{bmatrix} G + H & I \\ I & P_s \end{bmatrix} \succ 0.$$

Again by the Schur complement,

$$P_s^{-1} \prec G + H.$$

Thus P_s^{-1} acts as a convex surrogate upper bound on

$$P^{-1} + P^{-1}S^{-1}P^{-1}.$$

Substituting this bound into the robustified Lyapunov inequality gives

$$-\bar{P} + \mathbf{A}^\top P_s^{-1}\mathbf{A} \prec 0,$$

which is equivalent to

$$\begin{bmatrix} -\bar{P} & \mathbf{A}^\top \\ \mathbf{A} & -P_s \end{bmatrix} \prec 0.$$

J.1.viii Insertion into the full DRO-LMI. Replacing the nominal Lyapunov blocks in the correlated kernel yields

$$\begin{bmatrix} -\bar{P} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I & -Q - \lambda I & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P_s & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0.$$

For the independent model, the same replacement leads to

$$\begin{bmatrix} -\bar{P} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -P_s & 0 \\ \mathbf{C} & 0 & -I \end{bmatrix} \prec 0,$$

while the disturbance block is handled separately through

$$\begin{bmatrix} -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I & -Q - \lambda I & 0 & 0 \\ \mathbf{B} & 0 & -P & 0 \\ \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0.$$

J.2 Spectral-norm regularization details

When regularization is enabled, the code introduces epigraph variables

$$t_K, t_L, t_X, t_Y, t_M, t_N \geq 0$$

and enforces

$$\begin{bmatrix} t_J I_{n^J_{\text{row}}} & J \\ J^\top & t_J I_{n^J_{\text{col}}} \end{bmatrix} \succeq 0, \qquad J \in \{K, L, X, Y, M, N\}.$$

By the Schur complement, each of these implies

$$\|J\|_2 \leq t_J.$$

The regularization term is therefore

$$\mathcal{R}_{\text{spec}} = \rho\,(t_K + t_L + t_X + t_Y + t_M + t_N),$$

and the full objective is

$$\operatorname{tr}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2 + \mathcal{R}_{\text{spec}} + \varepsilon.$$

J.3 Post-optimization recovery of Δ_A and Δ_B

After solving the semidefinite program, one may reconstruct representative perturbations Δ_A and Δ_B that are consistent with the optimized uncertainty directions.

Let the optimized matrices be

$$P^\star, \quad X^\star, \quad Y^\star, \quad M^\star, \quad N^\star,$$

and let the nominal model be $(\hat{A}_x, \hat{B}_u, \hat{C}_y)$. The perturbation structure is

$$E_A = \begin{bmatrix} \Delta_A Y^\star & \Delta_A \\ 0 & X^\star \Delta_A \end{bmatrix}, \qquad E_B = \begin{bmatrix} \Delta_B M^\star & \Delta_B N^\star \hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

A representative worst-case perturbation may be extracted from the optimized uncertainty budget and projected back onto these two structures. This leads to:

- a Sylvester equation for Δ_A,
- a right-side Gram solve for Δ_B.

In particular, the recovery of Δ_A is obtained from

$$(X^{\star\top} X^\star)\Delta_A + \Delta_A(Y^\star Y^{\star\top} + I) = A_2 + A_1 Y^{\star\top} + X^{\star\top} A_3,$$

while Δ_B is computed from

$$\Delta_B G_B = B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top},$$

where

$$G_B := M^\star M^{\star\top} + N^\star \hat{C}_y \hat{C}_y^\top N^{\star\top}.$$

Finally, the nominal matrices are updated as

$$A_{x,\text{aug}} = \hat{A}_x + \Delta_A, \qquad B_{u,\text{aug}} = \hat{B}_u + \Delta_B.$$

These recovered perturbations should be interpreted as representative worst-case directions induced by the convex relaxation, rather than as unique physical identification errors.

J.4 Post-optimization recovery of Δ_A and Δ_B

Inputs. After solving the program, suppose we have optimal quantities

$$P^\star \succ 0, \quad X^\star = X^{\star\top}, \quad Y^\star = Y^{\star\top}, \quad M^\star,\ N^\star,\ s_A^\star > 0,\ s_B^\star > 0,\ \tau_A^\star,\ \tau_B^\star,$$

together with the nominal identified maps $\hat{A},\ \hat{B}_u,\ \hat{C}_y$ and data-driven radii $\beta_A,\ \beta_B$ that induce the lifted bounds

$$\|E_A\|_F \le \beta_A := \sqrt{1 + \|X^\star\|_2^2 + \|Y^\star\|_2^2}\,\beta_A, \qquad \|E_B\|_F \le \beta_B := \sqrt{\|M^\star\|_2^2 + \|N^\star\|_2^2\|\hat{C}_y\|_2^2}\,\beta_B.$$

Step 1: Adversarial directions for the Young bounds. Let the nominal lifted block (built with $\hat{A}, \hat{B}_u$) be

$$\mathbf{A}^\star := \mathbf{A}(\hat{A}, \hat{B}_u; X^\star, Y^\star, M^\star, N^\star), \qquad Z_{A,i}^\star := P^\star \mathbf{A}^\star, \quad i \in \{A, B\}.$$

Choose worst-case aligned perturbations that saturate the Young relaxation:

$$E_{A,i}^\star = \begin{cases} \beta_{A,i}\dfrac{Z_{A,i}^\star}{\|Z_{A,i}^\star\|_F}, & Z_{A,i}^\star \neq 0, \\ 0, & Z_{A,i}^\star = 0, \end{cases} \qquad \beta_{A,A} := \beta_A, \quad \beta_{A,B} := \beta_B. \tag{J.2}$$

This choice attains equality in Young's inequality when $s_{A,i}^\star = \|Z_{A,i}^\star\|_F/\beta_{A,i}$.

Step 2: Block structure of $E_A^\star$ and $E_B^\star$. Recall the definitions

$$E_A = \begin{bmatrix} \Delta_A Y^\star & \Delta_A \\ 0 & X^\star \Delta_A \end{bmatrix}, \qquad E_B = \begin{bmatrix} \Delta_B M^\star & \Delta_B N^\star \hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

After choosing $E_A^\star, E_B^\star$ via (J.2), extract the nonzero sub-blocks

$$A_1 := E_A^\star(1{:}n,\ 1{:}n), \quad A_2 := E_A^\star(1{:}n,\ n{+}1{:}2n), \quad A_3 := E_A^\star(n{+}1{:}2n,\ n{+}1{:}2n),$$

$$B_1 := E_B^\star(1{:}n,\ 1{:}m), \qquad B_2 := E_B^\star(1{:}n,\ m{+}1{:}m{+}p),$$

Step 3: Recovery of Δ_A (Sylvester equation). Solve the least-squares normal equation

$$\boxed{(X^{\star\top}X^\star)\,\Delta_A \;+\; \Delta_A\,(Y^\star Y^{\star\top} + I) \;=\; A_2 + A_1 Y^{\star\top} + X^{\star\top} A_3\ ,} \tag{J.3}$$

which is a Sylvester equation in Δ_A. Under mild conditions on the spectra of $X^{\star\top}X^\star$ and $Y^\star Y^{\star\top} + I$, (J.3) has a unique solution.

Special cases. If $X^\star = 0$, then $\Delta_A(Y^\star Y^{\star\top}+I) = A_2+A_1Y^{\star\top} \Rightarrow \Delta_A = (A_2+A_1Y^{\star\top})(Y^\star Y^{\star\top}+I)^{-1}$. If $Y^\star = 0$, then $(X^{\star\top}X^\star + I)\Delta_A = A_2 + X^{\star\top}A_3 \Rightarrow \Delta_A = (X^{\star\top}X^\star + I)^{-1}(A_2 + X^{\star\top}A_3)$.

Step 4: Recovery of Δ_B (right-side Gram solve). Solve

$$\boxed{\Delta_B\, G_B \;=\; B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top}, \qquad G_B := M^\star M^{\star\top} + N^\star \hat{C}_y \hat{C}_y^\top N^{\star\top}\ ,} \tag{J.4}$$

so that, if G_B is invertible, $\Delta_B = (B_1 M^{\star\top} + B_2\hat{C}_y^\top N^{\star\top})\,G_B^{-1}$. If G_B is singular or ill-conditioned, use the Moore–Penrose pseudoinverse G_B^+ or a Tikhonov-regularised inverse $(G_B + \eta I)^{-1}$ with small $\eta > 0$.

Step 5: Augmented nominal maps. Finally set

$$A_{\text{aug}} = \hat{A} + \Delta_A, \qquad B_{u,\text{aug}} = \hat{B}_u + \Delta_B,$$

and, if desired, verify feasibility with the identification radii: $\|\Delta_A\|_F \le \beta_A,\ \|\Delta_B\|_F \le \beta_B$ and with the lifted bounds $\|E_A(\Delta_A)\|_F \le \beta_A,\ \|E_B(\Delta_B)\|_F \le \beta_B$.

Remarks. (i) The recovered (Δ_A, Δ_B) are *representative* worst-case directions induced by the Young relaxation, not the unique identification errors. (ii) Numerically, (J.3) is solved with a Sylvester solver; (J.4) is a right-multiplication by a Gram factor and should use a robust linear solver or SVD-based pseudoinverse. (iii) If a single uncertainty block on $[A\ B_u]$ was used in synthesis, replace β_A, β_B by a common β and set $E_x^\star = \beta\, Z_x^\star / \|Z_x^\star\|_F$ with $Z_x^\star = P^\star[\hat{A}\ \hat{B}_u]$, then project $E_x^\star$ onto the E_A, E_B structures above before solving (J.3)–(J.4).

K Derivations for the Willems-Fundamental-Lemma-Based Data-Driven DRO-LMI Formulation

K.1 Plant, Controller, and Augmented Closed-Loop Representation

Consider the discrete-time linear time-invariant plant

$$\begin{aligned} x_{k+1} &= A_x x_k + B_u u_k + B_w w_k, \\ z_k &= C_z x_k + D_{zu} u_k + \underbrace{D_{zw}}_{\to \mathbf{0}_{n_z \times n_w}} w_k, \\ y_k &= C_y x_k + \underbrace{D_{yu}}_{\to \mathbf{0}_{n_y \times n_u}} u_k + \underbrace{D_{yw}}_{\to \mathbf{0}_{n_y \times n_w}} w_k, \end{aligned}$$

thus

$$\begin{aligned} \begin{bmatrix} x_{k+1} \\ y_k \end{bmatrix} &= \begin{bmatrix} A_x & B_u \\ C_y & \mathbf{0}_{n_y \times n_u} \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \begin{bmatrix} B_w \\ \mathbf{0}_{n_y \times n_w} \end{bmatrix} w_k \\ &= \mathcal{M}_P \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \bar{B}_w w_k, \\ z_k &= C_z x_k + D_{zu} u_k. \end{aligned}$$

The dynamic output-feedback controller is

$$\begin{aligned} x_{k+1}^c &= A_c x_k^c + B_c y_k, \\ u_k &= C_c x_k^c + D_c y_k, \end{aligned}$$

or in compact form

$$\begin{bmatrix} x_{k+1}^c \\ u_k \end{bmatrix} = \mathcal{M}_C \begin{bmatrix} x_k^c \\ y_k \end{bmatrix}, \qquad \mathcal{M}_C := \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix}.$$

Defining the augmented state

$$\chi_k := \begin{bmatrix} x_k \\ x_k^c \end{bmatrix},$$

the closed-loop system becomes

$$\begin{aligned} \chi_{k+1} &= \mathcal{A}\chi_k + \mathcal{B} w_k, \\ z_k &= \mathcal{C}\chi_k + \mathcal{D} w_k, \end{aligned}$$

or equivalently

$$\begin{bmatrix} \chi_{k+1} \\ z_k \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \chi_k \\ w_k \end{bmatrix} = \mathcal{M}_{CL} \begin{bmatrix} \chi_k \\ w_k \end{bmatrix},$$

with $\mathcal{M}_{CL} = f(\mathcal{M}_P, \mathcal{M}_C)$.

K.2 Lifting and Factorization

Let $P \succ 0$ be a symmetric weighting matrix. For $\mathcal{T} = I$, the identity

$$P = \mathcal{T}^\top \mathcal{P} \mathcal{T} = \mathcal{P} = P^\top$$

holds trivially.

Using this lifting, the augmented system matrix admits the factorization

$$\mathcal{M}_{var} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = \begin{bmatrix} P\mathcal{A} & P\mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} = \begin{bmatrix} P & \mathbf{0}_{2n_x \times n_z} \\ \mathbf{0}_{n_z \times 2n_x} & \mathbf{I}_{n_z} \end{bmatrix} \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} = \mathbf{P}\, \mathcal{M}_{CL},$$

which is convenient for subsequent LMI-based analysis and synthesis.

A sufficient condition for closed-loop stability and performance is then

$$\begin{bmatrix} -\mathbf{P}(\mathbf{I}_{2n_x+n_w} + (\lambda-1)\Psi_w \Psi_w^\top) & \lambda \Psi_w & (\mathbf{P}\mathcal{M}_{CL})^\top \\ \lambda \Psi_w^\top & -Q - \lambda \mathbf{I}_{n_w \times n_w} & \mathbf{0}_{n_w \times (2n_x+n_z)} \\ \mathbf{P}\mathcal{M}_{CL} & \mathbf{0}_{n_z \times n_w} & -\mathbf{P} \end{bmatrix} \prec 0$$

where $Q \succ 0$ and $\lambda > 0$ are design parameters, with

$$\Psi_w = \begin{bmatrix} \mathbf{0}_{2n_x \times n_w} \\ \mathbf{I}_{n_w} \end{bmatrix},$$

and

$$\mathbf{I}_{2n_x+n_w} + (\lambda-1)\Psi_w \Psi_w^\top = \begin{bmatrix} \mathbf{I}_{2n_x} & \mathbf{0}_{2n_x \times n_w} \\ \mathbf{0}_{n_w \times 2n_x} & \lambda \mathbf{I}_{n_w} \end{bmatrix}. \tag{K.1}$$

This condition follows from a Schur-complement-based dissipation inequality and ensures quadratic stability of the closed-loop system under bounded disturbances.

The state and output blocks can be rewritten as

$$\begin{bmatrix} \mathbf{A} \\ \mathbf{C} \end{bmatrix} = \begin{bmatrix} P\mathcal{A} \\ \mathcal{C} \end{bmatrix} = \mathbf{P}\, \mathcal{M}_{CL}\, \Psi_{2n_x}, \qquad \begin{bmatrix} \mathbf{B} \\ \mathbf{D} \end{bmatrix} = \begin{bmatrix} P\mathcal{B} \\ \mathcal{D} \end{bmatrix} = \mathbf{P}\, \mathcal{M}_{CL}\, \Psi_w, \qquad \Psi_{2n_x} = \begin{bmatrix} \mathbf{I}_{2n_x} \\ \mathbf{0}_{n_w \times 2n_x} \end{bmatrix}.$$

Thus, the independent constraints become

$$\begin{bmatrix} -\Psi_z^\top \mathbf{P} \Psi_z & (\mathbf{P}\mathcal{M}_{CL}\Psi_{2n_x})^\top \\ \mathbf{P}\mathcal{M}_{CL}\Psi_{2n_x} & -\mathbf{P} \end{bmatrix} \prec 0, \qquad \Psi_z = \begin{bmatrix} \mathbf{I}_{2n_x} \\ \mathbf{0}_{n_z \times 2n_x} \end{bmatrix}$$

and

$$\begin{bmatrix} -\lambda \mathbf{I}_{n_w} & \lambda \mathbf{I}_{n_w} & (\mathbf{P}\mathcal{M}_{CL}\Psi_{n_w})^\top \\ \lambda \mathbf{I}_{n_w} & -Q - \lambda \mathbf{I}_{n_w} & \mathbf{0}_{n_w \times (2n_x+n_z)} \\ \mathbf{P}\mathcal{M}_{CL}\Psi_{n_w} & \mathbf{0}_{(2n_x+n_z) \times n_w} & -\mathbf{P} \end{bmatrix} \prec 0.$$

K.3 Disturbance Absorption via Structural Substitution

Consider the substitution

$$\bar{w}_k \mapsto B_w w_k.$$

Under this substitution, the direct feedthrough term satisfies

$$\mathcal{D} = D_{zw} + D_{zu} D_c D_{yw} = 0, \qquad D_{zw}, D_{yw} \to 0,$$

and is therefore unchanged.

The closed-loop matrices become

$$\mathcal{A} = \begin{bmatrix} A_x + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix},$$

$$\mathcal{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix} = \begin{bmatrix} B_w \\ 0 \end{bmatrix}, \qquad D_{yw} \to 0$$

and therefore

$$\bar{B}_w = \begin{bmatrix} I \\ 0 \end{bmatrix}.$$

The performance output matrix is

$$\mathcal{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}.$$

Thus, one obtains the chain

$$\mathcal{M}_{var} \;\to\; \mathcal{M}_{CL} \;\to\; \mathcal{M}_C,\ \mathcal{M}_P,$$

with $\mathcal{M}_{var}$ linear in $\mathcal{M}_{CL}$, and $\mathcal{M}_{CL}$ constructed from $\mathcal{M}_C$ and $\mathcal{M}_P$.

K.4 Controller Reconstruction from the Original Decision Variables

The block $\mathcal{M}_P$ can be treated as a decision variable. The controller matrix $\mathcal{M}_C$ may either be left directly in terms of (A_c, B_c, C_c, D_c) or reconstructed from the original decision variables (X, Y, K, L, M, N).

Define

$$S := I_{n_x} - XY \in \mathbb{R}^{n_x \times n_x}. \tag{K.2}$$

Compute the singular value decomposition

$$S = U \Sigma V^\top, \tag{K.3}$$

where $U, V \in \mathbb{R}^{n_x \times n_x}$ are orthogonal and $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_{n_x}) \succeq 0$.

For numerical robustness, introduce a small $\varepsilon > 0$ and define

$$\Sigma_\varepsilon := \mathrm{diag}\big(\max(\sigma_1, \varepsilon), \ldots, \max(\sigma_{n_x}, \varepsilon)\big). \tag{K.4}$$

Then define

$$\Sigma_\varepsilon^{1/2} := \mathrm{diag}\big(\sqrt{\Sigma_{\varepsilon,11}}, \ldots, \sqrt{\Sigma_{\varepsilon,n_x n_x}}\big), \tag{K.5}$$

$$\Sigma_\varepsilon^{-1/2} := \mathrm{diag}\big(1/\sqrt{\Sigma_{\varepsilon,11}}, \ldots, 1/\sqrt{\Sigma_{\varepsilon,n_x n_x}}\big). \tag{K.6}$$

Using these, construct

$$U_{\text{new}} := U\, \Sigma_\varepsilon^{1/2}, \qquad V_{\text{new}} := V\, \Sigma_\varepsilon^{1/2}, \tag{K.7}$$

$$U_{\text{new}}^{-1} := \Sigma_\varepsilon^{-1/2} U^\top, \qquad V_{\text{new}}^{-\top} := V\, \Sigma_\varepsilon^{-1/2}. \tag{K.8}$$

Note that, in exact arithmetic,

$$U_{\text{new}}^{-1} U_{\text{new}} = I_{n_x}, \qquad V_{\text{new}}^{-\top} V_{\text{new}}^\top = I_{n_x},$$

while

$$U_{\text{new}} V_{\text{new}}^\top = S$$

up to the regularisation ε.

The controller realization (A_c, B_c, C_c, D_c) is then

$$D_c := N, \tag{K.9}$$

$$C_c := (M - NC_yY)\, V_{\text{new}}^{-\top}, \tag{K.10}$$

$$B_c := U_{\text{new}}^{-1}(L - XB_uN), \tag{K.11}$$

$$A_c := U_{\text{new}}^{-1}\Big(K - XA_xY - LC_yY - XB_u(M - NC_yY)\Big)\, V_{\text{new}}^{-\top}. \tag{K.12}$$

These equations make explicit the dependence of the controller realization on (X, Y, K, L, M, N) and on the known plant matrices (A_x, B_u, C_y).

K.5 Selector-Based Plant Reconstruction

Define the selectors

$$\psi_1 = \begin{bmatrix} \mathbf{I}_{n_x} & \mathbf{0}_{n_x \times n_y} \end{bmatrix}, \qquad \psi_2 = \begin{bmatrix} \mathbf{I}_{n_x} \\ \mathbf{0}_{n_u \times n_x} \end{bmatrix},$$

$$\psi_3 = \begin{bmatrix} \mathbf{0}_{n_x \times n_u} \\ \mathbf{I}_{n_u} \end{bmatrix}, \qquad \psi_4 = \begin{bmatrix} \mathbf{0}_{n_y \times n_x} & \mathbf{I}_{n_y} \end{bmatrix}.$$

Then the plant matrices can be written as linear functions of $\mathcal{M}_P$, namely

$$A_x = \psi_1 \mathcal{M}_P \psi_2, \qquad B_u = \psi_1 \mathcal{M}_P \psi_3, \qquad C_y = \psi_4 \mathcal{M}_P \psi_2.$$

Therefore,

$$\mathcal{M}_{CL} = \left[\begin{array}{c|c} \begin{bmatrix} \psi_1\mathcal{M}_P\psi_2 + \psi_1\mathcal{M}_P\psi_3 D_c \psi_4 \mathcal{M}_P \psi_2 & \psi_1 \mathcal{M}_P \psi_3 C_c \\ B_c \psi_4 \mathcal{M}_P \psi_2 & A_c \end{bmatrix} & \bar{B}_w \\ \hline \begin{bmatrix} C_z + D_{zu} D_c \psi_4 \mathcal{M}_P \psi_2 & D_{zu} C_c \end{bmatrix} & \mathbf{0}_{n_z \times n_w} \end{array}\right].$$

Moreover,

$$\mathcal{M}_C = \left[\begin{array}{c|c} A_c & B_c \\ \hline C_c & D_c \end{array}\right] = \left[\begin{array}{c|c} \begin{array}{c} U_{\text{new}}^{-1}\Big(K - X\psi_1\mathcal{M}_P\psi_2 Y - L\psi_4\mathcal{M}_P\psi_2 Y + \\ -X\psi_1\mathcal{M}_P\psi_3(M - N\psi_4\mathcal{M}_P\psi_2 Y)\Big) V_{\text{new}}^{-\top} \end{array} & U_{\text{new}}^{-1}(L - X\psi_1\mathcal{M}_P\psi_3 N) \\ \hline (M - N\psi_4\mathcal{M}_P\psi_2 Y) V_{\text{new}}^{-\top} & N \end{array}\right].$$

Hence,

$$\mathcal{M}_{var} = \mathbf{P}\, \mathcal{M}_{CL}, \qquad \mathcal{M}_P = \begin{bmatrix} A_x & B_u \\ C_y & \mathbf{0}_{n_y \times n_u} \end{bmatrix}_{\mathbf{WFL}}.$$

K.6 Relaxations of the Controller Reconstruction that Preserve the Main Guarantee

The realization map $(X, Y, K, L, M, N) \mapsto (A_c, B_c, C_c, D_c)$ is exact only if the factorization of

$$S := I - XY \succ 0$$

is enforced. In the original construction, one introduces factors such that $S = U_1 V_1^\top$ and defines the controller through the reconstruction formulas above.

(R1) Young–Schur (or equivalent) relaxation of the factorization. A standard convex relaxation replaces the explicit inversion/factorization of S with auxiliary variables and Schur complements, eliminating the need to compute U_1^{-1} and $V_1^{-\top}$ explicitly. In this case, the controller is not reconstructed by a closed-form formula, but the feasibility set remains a convex outer-approximation of the original one. The resulting guarantee is conservative but remains meaningful: any feasible solution still certifies robust stability/performance for the relaxed problem.

(R2) Structured X, Y (diagonal / block-diagonal). Restricting X and Y to diagonal or block-diagonal forms reduces degrees of freedom, often improving numerical conditioning. The original proof structure remains unchanged, but feasibility becomes more conservative and the best achievable performance can degrade.

(R3) Sequential factorization (majorization / SCP). One may fix a factorization $S \approx U_1 V_1^\top$ from a previous iterate and solve a convex subproblem for (X, Y, K, L, M, N), then update the factorization and repeat. This becomes a sequential convex programming heuristic. The original global guarantees no longer hold, but each subproblem remains well-posed and often yields practical controllers when the full nonconvex formulation is numerically unstable.

(R4) Replace strict definiteness with a margin. Enforcing $S \succeq \delta I$ with $\delta > 0$ improves numerical stability and makes the factorization well-conditioned. This does not change the theoretical structure; it only strengthens the assumptions.

K.7 Deliberate Deviations from the Original Construction

(B1) Treating $\mathcal{M}_C$ as a free decision variable. If (A_c, B_c, C_c, D_c) are optimized directly while (X, Y, K, L, M, N) are also present, then the change-of-variables argument connecting the LMI solution to a realizable controller is broken. The feasible set becomes strictly larger, and feasibility no longer implies that the controller corresponds to the intended convexification. The gain is numerical simplicity and the ability to search a larger space; the loss is the proof that the LMI encodes the same controller class as the original formulation.

(B2) Dropping the constraint $I - XY \succ 0$. If the positivity of $S = I - XY$ is not enforced, then the factorization $S = U_1 V_1^\top$ and the reconstruction formulas are not well-defined. The gain is avoiding strict matrix inequalities; the loss is that the controller map becomes undefined or ill-conditioned, and the resulting controller, if extracted by an ad-hoc procedure, is not covered by the original guarantee.

(B3) Freezing the factorization (one-shot linearization). A pragmatic compromise is to compute a factorization of S from a nominal (X, Y) or from an earlier solution and keep U_1, V_1 fixed while optimizing the remaining variables. This converts the reconstruction into an affine map but changes the feasible set. The gain is improved numerical robustness and speed; the loss is that the solution is only guaranteed to satisfy the modified linearized constraints, not the original ones.

(B4) Replacing the reconstruction by post-processing. Another common choice is to solve a convex problem that yields (X, Y, K, L, M, N) but then obtain (A_c, B_c, C_c, D_c) by a separate identification step. The gain is decoupling the optimization from the realization extraction; the loss is that feasibility/performance are no longer jointly certified by a single optimization problem, and post-processing can destroy the certified margins.

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